

Repository Intelligence Report

brody

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Statements are backed by repository evidence in the form path:line-range. Findings are labelled static analyzer (deterministic) or AI-inferred, and needs verification where evidence is incomplete.

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1. Executive Summary

brody: Repository intelligence, code review and code mapping. (Stated in the project's own documentation.) (package.json)

The repository contains 118 source files (29,317 lines across 178 files) organised into 23 functional areas: API Layer, Content Ingestion, Application Core, Infrastructure & Deployment, User Interface, Export.

Execution begins at src/app/layout.tsx (Next.js root layout (package.json script "dev")), scripts/worker.mts (package.json script "worker"), Dockerfile (CMD node -e "require('http').get({host:'127.0.0.1',port:process.env.PORT||3003,path:}). (src/app/layout.tsx, scripts/worker.mts:8, Dockerfile:38)

Data enters through 25 API route(s) and 9 page(s) built with Next.js, React. Data is persisted in SQLite through Drizzle ORM, modelled by 12 entities such as projects, credentials, blobs, files. (src/app/page.tsx:15, src/app/api/github/validate/route.ts:10)

It integrates with Anthropic, OpenAI, SQLite, GitHub API. (package.json, src/lib/config.ts:39, package.json)

Engineering review found 21 finding(s): 0 critical, 0 high, 8 medium, 13 lower-priority. 21 came from deterministic analyzers and 0 from AI review. 34 test file(s) exist.

2. System at a Glance

Area	Description
Primary Purpose	Repository intelligence, code review and code mapping
Application Type	full-stack monolith (Next.js) (high confidence)
Primary Languages	TypeScript (143 files), SQL (2 files), CSS (1 files), JavaScript (2 files), Python (1 files)
Frontend	Next.js, React, Tailwind CSS
Backend	custom HTTP handlers
Data Layer	Database: SQLite; Access: Drizzle ORM; 12 modelled entities
Authentication	no authentication mechanism detected
External Services	Anthropic, OpenAI, SQLite, GitHub API
Infrastructure	GitHub Actions, Docker, Docker Compose
Testing	34 test files; Vitest, Playwright

3. Architecture Overview

The codebase is best described as a full-stack monolith (Next.js) (high confidence). Both frontend and backend frameworks are present in one codebase.

Files divide into these layers: Entry Points (5), Presentation / UI (8), API / Transport (33), Domain / Services (63), Background Processing (2), AI (6), Data (3), Infrastructure & Config (11), Tests & Docs (39).

The largest functional areas are Testing (34 files), API Layer (21 files), Content Ingestion (13 files), Documentation (12 files), Application Core (11 files).

Layers: Entry Points (5 files) · Presentation / UI (8 files) · API / Transport (33 files) · Domain / Services (63 files) · Background Processing (2 files) · AI (6 files) · Data (3 files) · Infrastructure & Config (11 files) · Tests & Docs (39 files)

4. Primary Runtime Flow

GET /api/projects/:id/files/content is handled by GET /api/projects/:id/files/content, which calls guard, getReadyProject, AppError, getDb, getFileContent; along the way it reads files, symbols, relationships, findings. (src/app/api/projects/[id]/files/content/route.ts:13)

1. **GET /api/projects/:id/files/content** — function at src/app/api/projects/[id]/files/content/route.ts:13 (src/app/api/projects/[id]/files/content/route.ts:13)
2. **guard** — function at src/lib/api.ts:20 (src/lib/api.ts:20)
3. **getReadyProject** — function at src/lib/api.ts:34 (src/lib/api.ts:34)
4. **AppError** — class at src/lib/util/errors.ts:2 (src/lib/util/errors.ts:2)
5. **getDb** — function at src/lib/db/client.ts:48 (src/lib/db/client.ts:48)
6. **getFileContent** — function at src/lib/ingest/store.ts:86 (src/lib/ingest/store.ts:86)
7. **changelImpact** — function at src/lib/map/index.ts:364 (src/lib/map/index.ts:364)
8. **json** — function at src/lib/api.ts:8 (src/lib/api.ts:8)
9. **reads files** — model at src/lib/db/schema.ts:42 (src/lib/db/schema.ts:42)
10. **reads symbols** — model at src/lib/db/schema.ts:82 (src/lib/db/schema.ts:82)
11. **reads relationships** — model at src/lib/db/schema.ts:110 (src/lib/db/schema.ts:110)
12. **reads findings** — model at src/lib/db/schema.ts:128 (src/lib/db/schema.ts:128)
13. **UsageMeter.add** — method at src/lib/ai/index.ts:51 (src/lib/ai/index.ts:51)
14. **fail** — function at src/lib/api.ts:12 (src/lib/api.ts:12)

5. Major Functional Areas

Testing

- **Purpose:** Automated tests, fixtures and mocks.
- **Business function:** Supports the full-stack monolith (Next.js) through 34 file(s).
- **Primary components:** Order (model, fixtures/sample-shop/src/db/models.ts:13); User (model, fixtures/sample-shop/src/db/models.ts:7); MockProvider (service, tests/helpers/index.ts:50); authRouter (service, fixtures/sample-shop/src/routes/auth.ts:4); ordersRouter (service, fixtures/sample-shop/src/routes/orders.ts:5); app (service, fixtures/sample-shop/worker/tasks.py:7); send_receipt (job, fixtures/sample-shop/worker/tasks.py:10); EmbedProvider (service, tests/analyzers.test.ts:60)
- **Inputs:** Startup and configuration
- **Processing:** 34 file(s), key components: Order, User, MockProvider, authRouter, ordersRouter.
- **Outputs:** Results consumed via Content Ingestion, Background Jobs, Map and 17 more
- **Dependencies:** Content Ingestion, Background Jobs, Map, Data Layer, Analysis, AI Processing, Export, Review, API Layer, Shared Utilities, Infrastructure & Deployment, Search & Retrieval, Documentation, Parse, Graph, Ask, Discover, File & Media Storage, Util, Reporting & Analytics
- **Failure modes:** none identified
- **Important relationships:** No other area depends on it directly.
- **Source of explanation:** deterministic, derived from the repository graph (fixtures/sample-shop/src/db/models.ts:13, fixtures/sample-shop/src/db/models.ts:7, tests/helpers/index.ts:50)

API Layer

- **Purpose:** HTTP/RPC endpoints and request handling.
- **Business function:** Serves GET /, POST /api/github/validate, GET /api/projects, DELETE /api/projects/:id and 19 more to callers.
- **Primary components:** POST (function, src/app/api/github/validate/route.ts:10); POST (function, src/app/api/projects/[id]/analyze/route.ts:17); GET (function, src/app/api/projects/[id]/architecture/route.ts:8); GET

(function, src/app/api/projects/[id]/ask/route.ts:11); POST (function, src/app/api/projects/[id]/ask/route.ts:19); GET (function, src/app/api/projects/[id]/export/route.ts:13); GET (function, src/app/api/projects/[id]/findings/route.ts:9); GET (function, src/app/api/projects/[id]/graph/route.ts:9)

- **Inputs:** Requests to GET /, POST /api/github/validate, GET /api/projects, DELETE /api/projects/:id and 19 more
- **Processing:** 21 file(s), key components: POST, POST, GET, GET, POST.
- **Outputs:** Results consumed via Shared Utilities, User Interface, Map and 10 more
- **Dependencies:** Shared Utilities, User Interface, Map, Content Ingestion, Data Layer, Util, Documentation, Background Jobs, Discover, AI Processing, Ask, Export, Infrastructure & Deployment
- **Failure modes:** none identified
- **Important relationships:** Used by Testing.
- **Source of explanation:** deterministic, derived from the repository graph (src/app/api/github/validate/route.ts:10, src/app/api/projects/[id]/analyze/route.ts:17, src/app/api/projects/[id]/architecture/route.ts:8)

Content Ingestion

- **Purpose:** Collecting and importing data from external sources.
- **Business function:** Supports the full-stack monolith (Next.js) through 13 file(s).
- **Primary components:** scrubToken (function, src/lib/ingest/credentials.ts:36); redactSecrets (function, src/lib/ingest/secrets.ts:65); getFileContents (function, src/lib/ingest/store.ts:91); detectLanguage (function, src/lib/ingest/languages.ts:84); sanitizeRelativePath (function, src/lib/ingest/paths.ts:9); CreateProjectResult (interface, src/lib/ingest/store.ts:73); normalizeFiles (function, src/lib/ingest/normalize.ts:15); encryptSecret (function, src/lib/ingest/credentials.ts:16)
- **Inputs:** Calls from Testing, API Layer, Infrastructure & Deployment, Background Jobs and 8 more
- **Processing:** 13 file(s), key components: scrubToken, redactSecrets, getFileContents, detectLanguage, sanitizeRelativePath.
- **Outputs:** Results consumed via Util, Shared Utilities, Data Layer and 3 more
- **Dependencies:** Util, Shared Utilities, Data Layer, Infrastructure & Deployment, AI Processing, Parse
- **Failure modes:** none identified
- **Important relationships:** Used by Testing, API Layer, Infrastructure & Deployment, Background Jobs, File & Media Storage and 7 more.
- **Source of explanation:** deterministic, derived from the repository graph (src/lib/ingest/credentials.ts:36, src/lib/ingest/secrets.ts:65, src/lib/ingest/store.ts:91)

Application Core

- **Purpose:** Top-level application code.
- **Business function:** Supports the full-stack monolith (Next.js) through 11 file(s).
- **Primary components:** RootLayout (component, src/app/layout.tsx:9); register (function, src/instrumentation.ts:2); proxy (function, src/proxy.ts:8); nextConfig.headers (method, next.config.ts:15); PORT (constant, playwright.config.ts:5)
- **Inputs:** Startup and configuration
- **Processing:** 11 file(s), key components: RootLayout, register, proxy, nextConfig.headers, PORT.
- **Outputs:** Results consumed via Background Jobs
- **Dependencies:** Background Jobs
- **Failure modes:** none identified
- **Important relationships:** No other area depends on it directly.
- **Source of explanation:** deterministic, derived from the repository graph (src/app/layout.tsx:9, src/instrumentation.ts:2, src/proxy.ts:8)

Infrastructure & Deployment

- **Purpose:** Build, deployment and operations.
- **Business function:** Supports the full-stack monolith (Next.js) through 9 file(s).
- **Primary components:** `jobs.push` (job, `.github/workflows/ci.yml:3`); `jobs.verify` (job, `.github/workflows/ci.yml:8`); `jobs.pull_request` (job, `.github/workflows/ci.yml:5`); `services.app` (service, `docker-compose.yml:3`); `services.brody-data` (service, `docker-compose.yml:28`); `opt` (function, `scripts/analyze.mts:20`); `synth` (function, `scripts/benchmark.mts:19`); `ms` (function, `scripts/benchmark.mts:39`)
- **Inputs:** Calls from Parse, Export, Discover, Map and 14 more
- **Processing:** 9 file(s), key components: `jobs.push`, `jobs.verify`, `jobs.pull_request`, `services.app`, `services.brody-data`.
- **Outputs:** Results consumed via Background Jobs, Content Ingestion, Export and 6 more
- **Dependencies:** Background Jobs, Content Ingestion, Export, Data Layer, Map, AI Processing, Ask, Search & Retrieval, Documentation
- **Failure modes:** none identified
- **Important relationships:** Used by Parse, Export, Discover, Map, Testing and 13 more.
- **Source of explanation:** deterministic, derived from the repository graph (`.github/workflows/ci.yml:3`, `.github/workflows/ci.yml:8`, `.github/workflows/ci.yml:5`)

User Interface

- **Purpose:** Pages, components and client-side state.
- **Business function:** Supports the full-stack monolith (Next.js) through 7 file(s).
- **Primary components:** `Loading` (component, `src/components/ui.tsx:37`); `ErrorBox` (component, `src/components/ui.tsx:41`); `Chip` (component, `src/components/ui.tsx:23`); `DownloadMenu` (component, `src/components/download.tsx:63`); `layout` (function, `src/components/graph.tsx:43`); `FormatRow` (component, `src/components/download.tsx:41`); `Empty` (component, `src/components/ui.tsx:51`); `SourceLink` (component, `src/components/ui.tsx:70`)
- **Inputs:** Calls from API Layer, File & Media Storage, Reporting & Analytics
- **Processing:** 7 file(s), key components: `Loading`, `ErrorBox`, `Chip`, `DownloadMenu`, `layout`.
- **Outputs:** Results consumed via Shared Utilities, Map, Export and 2 more
- **Dependencies:** Shared Utilities, Map, Export, Infrastructure & Deployment, Data Layer
- **Failure modes:** none identified
- **Important relationships:** Used by API Layer, File & Media Storage, Reporting & Analytics.
- **Source of explanation:** deterministic, derived from the repository graph (`src/components/ui.tsx:37`, `src/components/ui.tsx:41`, `src/components/ui.tsx:23`)

Export

- **Purpose:** Code under `src/lib/export/`.
- **Business function:** Supports the full-stack monolith (Next.js) through 7 file(s).
- **Primary components:** `plain` (function, `src/lib/export/document.ts:117`); `parseReport` (function, `src/lib/export/document.ts:100`); `outline` (function, `src/lib/export/document.ts:120`); `loadReportData` (function, `src/lib/export/markdown.ts:16`); `buildMarkdown` (function, `src/lib/export/markdown.ts:35`); `markdownToHtml` (function, `src/lib/export/html.ts:12`); `ReportScope` (type, `src/lib/export/scopes.ts:2`); `SCOPES` (constant, `src/lib/export/scopes.ts:4`)
- **Inputs:** Calls from Infrastructure & Deployment, Testing, API Layer, User Interface and 1 more
- **Processing:** 7 file(s), key components: `plain`, `parseReport`, `outline`, `loadReportData`, `buildMarkdown`.
- **Outputs:** Results consumed via Map, Infrastructure & Deployment, Data Layer and 4 more
- **Dependencies:** Map, Infrastructure & Deployment, Data Layer, Util, Parse, Documentation, Discover
- **Failure modes:** none identified

- **Important relationships:** Used by Infrastructure & Deployment, Testing, API Layer, User Interface, Reporting & Analytics.
- **Source of explanation:** deterministic, derived from the repository graph (src/lib/export/document.ts:117, src/lib/export/document.ts:100, src/lib/export/document.ts:120)

AI Processing

- **Purpose:** Language-model prompts, agents and inference.
- **Business function:** Supports the full-stack monolith (Next.js) through 6 file(s).
- **Primary components:** UsageMeter.add (method, src/lib/ai/index.ts:51); AIProvider (interface, src/lib/ai/types.ts:30); AnthropicProvider (service, src/lib/ai/anthropic.ts:21); OpenAICompatibleProvider (service, src/lib/ai/openai.ts:6); getAIProvider (function, src/lib/ai/index.ts:26); AIResponseError (class, src/lib/ai/types.ts:45); neutralizeDelimiters (function, src/lib/ai/prompt.ts:21); untrusted (function, src/lib/ai/prompt.ts:29)
- **Inputs:** Calls from Testing, Review, Documentation, API Layer and 11 more
- **Processing:** 6 file(s), key components: UsageMeter.add, AIProvider, AnthropicProvider, OpenAICompatibleProvider, getAIProvider.
- **Outputs:** Results consumed via Shared Utilities, Content Ingestion, Infrastructure & Deployment and 1 more
- **Dependencies:** Shared Utilities, Content Ingestion, Infrastructure & Deployment, Parse, Anthropic
- **Failure modes:** Anthropic unavailable: AI-generated features stop producing output
- **Important relationships:** Used by Testing, Review, Documentation, API Layer, Search & Retrieval and 10 more.
- **Source of explanation:** deterministic, derived from the repository graph (src/lib/ai/index.ts:51, src/lib/ai/types.ts:30, src/lib/ai/anthropic.ts:21)

Parse

- **Purpose:** Code under src/lib/parse/.
- **Business function:** Serves GET /x to callers.
- **Primary components:** text (function, src/lib/parse/extract.ts:20); emptyParse (function, src/lib/parse/types.ts:70); loadLanguage (function, src/lib/parse/treesitter.ts:56); ParsedFile (interface, src/lib/parse/types.ts:53); parseFile (function, src/lib/parse/index.ts:14); serializeParsed (function, src/lib/parse/cache.ts:19); SymbolKind (type, src/lib/parse/types.ts:1); extractFromTree (function, src/lib/parse/extract.ts:778)
- **Inputs:** Requests to GET /x
- **Processing:** 6 file(s), key components: text, emptyParse, loadLanguage, ParsedFile, parseFile.
- **Outputs:** Results consumed via Infrastructure & Deployment, AI Processing, Data Layer and 1 more
- **Dependencies:** Infrastructure & Deployment, AI Processing, Data Layer, Search & Retrieval
- **Failure modes:** none identified
- **Important relationships:** Used by Data Layer, Graph, Testing, Export, AI Processing and 3 more.
- **Source of explanation:** deterministic, derived from the repository graph (src/lib/parse/extract.ts:20, src/lib/parse/types.ts:70, src/lib/parse/treesitter.ts:56)

Configuration

- **Purpose:** Runtime configuration and tooling settings.
- **Business function:** Supports the full-stack monolith (Next.js) through 5 file(s).
- **Primary components:** none extracted
- **Inputs:** Calls from Documentation
- **Processing:** 5 file(s), key components: none extracted.
- **Outputs:** Return values to callers

- **Dependencies:** Anthropic, OpenAI, SQLite
- **Failure modes:** Anthropic unavailable: AI-generated features stop producing output; OpenAI unavailable: AI-generated features stop producing output; SQLite unavailable: persisted reads and writes fail
- **Important relationships:** Used by Documentation.
- **Source of explanation:** deterministic, derived from the repository graph

Public

- **Purpose:** Code under public/.
- **Business function:** Supports the full-stack monolith (Next.js) through 5 file(s).
- **Primary components:** none extracted
- **Inputs:** Startup and configuration
- **Processing:** 5 file(s), key components: none extracted.
- **Outputs:** Return values to callers
- **Dependencies:** none
- **Failure modes:** none identified
- **Important relationships:** No other area depends on it directly.
- **Source of explanation:** deterministic, derived from the repository graph

Background Jobs

- **Purpose:** Asynchronous and scheduled work.
- **Business function:** Serves GET /api/jobs/:id, POST /api/jobs/:id/cancel to callers.
- **Primary components:** Tracker.flush (method, src/lib/jobs/index.ts:70); getJob (function, src/lib/jobs/index.ts:36); enqueueAnalysis (function, src/lib/jobs/index.ts:24); initialStages (function, src/lib/jobs/stages.ts:18); claimNextJob (function, src/lib/jobs/index.ts:253); processJob (function, src/lib/jobs/index.ts:122); startWorker (function, src/lib/jobs/worker.ts:7); recoverStaleJobs (function, src/lib/jobs/index.ts:264)
- **Inputs:** Requests to GET /api/jobs/:id, POST /api/jobs/:id/cancel
- **Processing:** 5 file(s), key components: Tracker.flush, getJob, enqueueAnalysis, initialStages, claimNextJob.
- **Outputs:** Results consumed via Data Layer, Content Ingestion, Map and 9 more
- **Dependencies:** Data Layer, Content Ingestion, Map, Shared Utilities, Util, AI Processing, Discover, Documentation, Graph, Search & Retrieval, Review, Infrastructure & Deployment
- **Failure modes:** none identified
- **Important relationships:** Used by Testing, Infrastructure & Deployment, API Layer, File & Media Storage, Application Core.
- **Source of explanation:** deterministic, derived from the repository graph (src/lib/jobs/index.ts:70, src/lib/jobs/index.ts:36, src/lib/jobs/index.ts:24)

Review

- **Purpose:** Code under src/lib/review/.
- **Business function:** Supports the full-stack monolith (Next.js) through 5 file(s).
- **Primary components:** FindingDraft (interface, src/lib/review/types.ts:11); Severity (type, src/lib/review/types.ts:1); Confidence (type, src/lib/review/types.ts:2); Category (type, src/lib/review/types.ts:3); applyUnifiedDiff (function, src/lib/review/patch.ts:37); dedupe (function, src/lib/review/verify.ts:182); verifyDeterministic (function, src/lib/review/verify.ts:25); assignCodes (function, src/lib/review/verify.ts:208)
- **Inputs:** Calls from Testing, Analysis, Background Jobs, Ask
- **Processing:** 5 file(s), key components: FindingDraft, Severity, Confidence, Category, applyUnifiedDiff.
- **Outputs:** Results consumed via Analysis, AI Processing, Util and 5 more

- **Dependencies:** Analysis, AI Processing, Util, Data Layer, Graph, Infrastructure & Deployment, Shared Utilities, Discover
- **Failure modes:** none identified
- **Important relationships:** Used by Testing, Analysis, Background Jobs, Ask.
- **Source of explanation:** deterministic, derived from the repository graph (src/lib/review/types.ts:11, src/lib/review/types.ts:1, src/lib/review/types.ts:2)

File & Media Storage

- **Purpose:** Uploads and stored media.
- **Business function:** Serves GET /api/projects/:id/files, GET /api/projects/:id/files/content, POST /api/projects/upload, GET /p/:id/files to callers.
- **Primary components:** GET (function, src/app/api/projects/[id]/files/content/route.ts:13); GET (function, src/app/api/projects/[id]/files/route.ts:9); POST (function, src/app/api/projects/upload/route.ts:14); FilesPage (component, src/app/p/[id]/files/page.tsx:115); usePersisted (hook, src/app/p/[id]/files/page.tsx:29); Ctx (type, src/app/api/projects/[id]/files/content/route.ts:10); MAX_RETURN (constant, src/app/api/projects/[id]/files/content/route.ts:11); Ctx (type, src/app/api/projects/[id]/files/route.ts:7)
- **Inputs:** Requests to GET /api/projects/:id/files, GET /api/projects/:id/files/content, POST /api/projects/upload, GET /p/:id/files
- **Processing:** 4 file(s), key components: GET, GET, POST, FilesPage, usePersisted.
- **Outputs:** Results consumed via Shared Utilities, User Interface, Map and 7 more
- **Dependencies:** Shared Utilities, User Interface, Map, Data Layer, Content Ingestion, Documentation, Util, Background Jobs, AI Processing, Infrastructure & Deployment
- **Failure modes:** none identified
- **Important relationships:** Used by Testing.
- **Source of explanation:** deterministic, derived from the repository graph (src/app/api/projects/[id]/files/content/route.ts:13, src/app/api/projects/[id]/files/route.ts:9, src/app/api/projects/upload/route.ts:14)

Discover

- **Purpose:** Code under src/lib/discover/.
- **Business function:** Serves GET /path to callers.
- **Primary components:** Architecture (interface, src/lib/discover/types.ts:129); callArgs (function, src/lib/discover/detect.ts:218); AREA_HINTS (constant, src/lib/discover/catalog.ts:211); discoverArchitecture (function, src/lib/discover/index.ts:205); detectDependencies (function, src/lib/discover/detect.ts:15); detectExternalServices (function, src/lib/discover/detect.ts:80); detectEnvVars (function, src/lib/discover/detect.ts:144); detectRoutes (function, src/lib/discover/detect.ts:250)
- **Inputs:** Requests to GET /path
- **Processing:** 4 file(s), key components: Architecture, callArgs, AREA_HINTS, discoverArchitecture, detectDependencies.
- **Outputs:** Results consumed via Infrastructure & Deployment, Data Layer, Graph and 4 more
- **Dependencies:** Infrastructure & Deployment, Data Layer, Graph, AI Processing, Util, Content Ingestion, Search & Retrieval
- **Failure modes:** none identified
- **Important relationships:** Used by API Layer, Documentation, Map, Review, Testing and 4 more.
- **Source of explanation:** deterministic, derived from the repository graph (src/lib/discover/types.ts:129, src/lib/discover/detect.ts:218, src/lib/discover/catalog.ts:211)

Util

- **Purpose:** Code under src/lib/util/.
- **Business function:** Supports the full-stack monolith (Next.js) through 4 file(s).
- **Primary components:** `AppError` (class, src/lib/util/errors.ts:2); `escapeHtml` (function, src/lib/util/text.ts:39); `sha256` (function, src/lib/util/ids.ts:8); `newId` (function, src/lib/util/ids.ts:3); `isAppError` (function, src/lib/util/errors.ts:14); `sliceLines` (function, src/lib/util/text.ts:8); `tokenize` (function, src/lib/util/text.ts:19); `mapLimit` (function, src/lib/util/concurrency.ts:2)
- **Inputs:** Calls from Content Ingestion, API Layer, Review, Documentation and 9 more
- **Processing:** 4 file(s), key components: `AppError`, `escapeHtml`, `sha256`, `newId`, `isAppError`.
- **Outputs:** Results consumed via Infrastructure & Deployment
- **Dependencies:** Infrastructure & Deployment
- **Failure modes:** none identified
- **Important relationships:** Used by Content Ingestion, API Layer, Review, Documentation, Background Jobs and 8 more.
- **Source of explanation:** deterministic, derived from the repository graph (src/lib/util/errors.ts:2, src/lib/util/text.ts:39, src/lib/util/ids.ts:8)

Data Layer

- **Purpose:** Schemas, models, migrations and data access.
- **Business function:** Persists and retrieves projects, credentials, blobs, files, symbols and 7 more.
- **Primary components:** `getDb` (function, src/lib/db/client.ts:48); `openDatabase` (function, src/lib/db/client.ts:32); `migrationsFolder` (function, src/lib/db/client.ts:20); `files` (model, src/lib/db/schema.ts:42); `symbols` (model, src/lib/db/schema.ts:82); `jobs` (model, src/lib/db/schema.ts:159); `findings` (model, src/lib/db/schema.ts:128); `projects` (model, src/lib/db/schema.ts:4)
- **Inputs:** Calls from Testing, API Layer, Background Jobs, Infrastructure & Deployment and 14 more
- **Processing:** 3 file(s), key components: `getDb`, `openDatabase`, `migrationsFolder`, `files`, `symbols`.
- **Outputs:** Records for projects, credentials, blobs, files and 8 more
- **Dependencies:** Parse, Shared Utilities, SQLite
- **Failure modes:** SQLite unavailable: persisted reads and writes fail
- **Important relationships:** Used by Testing, API Layer, Background Jobs, Infrastructure & Deployment, Documentation and 13 more.
- **Source of explanation:** deterministic, derived from the repository graph (src/lib/db/client.ts:48, src/lib/db/client.ts:32, src/lib/db/client.ts:20)

Analysis

- **Purpose:** Code under src/lib/analysis/.
- **Business function:** Supports the full-stack monolith (Next.js) through 3 file(s).
- **Primary components:** `scanText` (function, src/lib/analysis/rules.ts:162); `blankLiterals` (function, src/lib/analysis/rules.ts:114); `runTypeScriptSyntax` (function, src/lib/analysis/analyzers.ts:73); `runEslint` (function, src/lib/analysis/analyzers.ts:99); `runPython` (function, src/lib/analysis/analyzers.ts:160); `runGo` (function, src/lib/analysis/analyzers.ts:213); `structuralFindings` (function, src/lib/analysis/structural.ts:9); `AnalyzerStatus` (interface, src/lib/analysis/analyzers.ts:11)
- **Inputs:** Calls from Testing, Review
- **Processing:** 3 file(s), key components: `scanText`, `blankLiterals`, `runTypeScriptSyntax`, `runEslint`, `runPython`.
- **Outputs:** Results consumed via Review, Infrastructure & Deployment, Graph and 4 more
- **Dependencies:** Review, Infrastructure & Deployment, Graph, Shared Utilities, Data Layer, Discover, AI Processing
- **Failure modes:** none identified

- **Important relationships:** Used by Testing, Review.
- **Source of explanation:** deterministic, derived from the repository graph (src/lib/analysis/rules.ts:162, src/lib/analysis/rules.ts:114, src/lib/analysis/analyzers.ts:73)

Shared Utilities

- **Purpose:** Reusable helpers and shared code.
- **Business function:** Supports the full-stack monolith (Next.js) through 3 file(s).
- **Primary components:** json (function, src/lib/api.ts:8); fail (function, src/lib/api.ts:12); guard (function, src/lib/api.ts:20); useApi (hook, src/lib/client.ts:35); getReadyProject (function, src/lib/api.ts:34); parseCite (function, src/lib/client.ts:84); getProject (function, src/lib/api.ts:28); keyOf (function, src/lib/client.ts:33)
- **Inputs:** Calls from API Layer, File & Media Storage, User Interface, Content Ingestion and 9 more
- **Processing:** 3 file(s), key components: json, fail, guard, useApi, getReadyProject.
- **Outputs:** Results consumed via Data Layer, Util, Content Ingestion and 1 more
- **Dependencies:** Data Layer, Util, Content Ingestion, Parse, Anthropic, OpenAI, GitHub API
- **Failure modes:** Anthropic unavailable: AI-generated features stop producing output; OpenAI unavailable: AI-generated features stop producing output; GitHub API unavailable: GitHub-backed features fail
- **Important relationships:** Used by API Layer, File & Media Storage, User Interface, Content Ingestion, Background Jobs and 8 more.
- **Source of explanation:** deterministic, derived from the repository graph (src/lib/api.ts:8, src/lib/api.ts:12, src/lib/api.ts:20)

Drizzle

- **Purpose:** Code under drizzle/.
- **Business function:** Supports the full-stack monolith (Next.js) through 2 file(s).
- **Primary components:** none extracted
- **Inputs:** Startup and configuration
- **Processing:** 2 file(s), key components: none extracted.
- **Outputs:** Return values to callers
- **Dependencies:** none
- **Failure modes:** none identified
- **Important relationships:** No other area depends on it directly.
- **Source of explanation:** deterministic, derived from the repository graph

Reporting & Analytics

- **Purpose:** Aggregations, reporting and analytics.
- **Business function:** Serves GET /api/projects/:id/report, GET /p/:id/reports to callers.
- **Primary components:** GET (function, src/app/api/projects/[id]/report/route.ts:8); ReportsPage (component, src/app/p/[id]/reports/page.tsx:16); Ctx (type, src/app/api/projects/[id]/report/route.ts:5); ORDER (constant, src/app/p/[id]/reports/page.tsx:6); FOR (constant, src/app/p/[id]/reports/page.tsx:7)
- **Inputs:** Requests to GET /api/projects/:id/report, GET /p/:id/reports
- **Processing:** 2 file(s), key components: GET, ReportsPage, Ctx, ORDER, FOR.
- **Outputs:** Results consumed via Shared Utilities, User Interface, Export and 1 more
- **Dependencies:** Shared Utilities, User Interface, Export, Documentation
- **Failure modes:** none identified
- **Important relationships:** Used by Testing.
- **Source of explanation:** deterministic, derived from the repository graph (src/app/api/projects/[id]/report/route.ts:8, src/app/p/[id]/reports/page.tsx:16, src/app/api/projects/[id]/report/route.ts:5)

Search & Retrieval

- **Purpose:** Indexing and querying content.
- **Business function:** Serves GET /api/projects/:id/search to callers.
- **Primary components:** search (function, src/lib/retrieval/index.ts:197); SearchFilters (interface, src/lib/retrieval/index.ts:24); queryTerms (function, src/lib/retrieval/index.ts:164); invalidateIndex (function, src/lib/retrieval/index.ts:63); retrieveContext (function, src/lib/retrieval/index.ts:326); searchCode (function, src/lib/retrieval/index.ts:291); buildSearchIndex (function, src/lib/retrieval/index.ts:68); indexFindingsAndDocs (function, src/lib/retrieval/index.ts:107)
- **Inputs:** Requests to GET /api/projects/:id/search
- **Processing:** 2 file(s), key components: search, SearchFilters, queryTerms, invalidateIndex, retrieveContext.
- **Outputs:** Results consumed via Data Layer, Shared Utilities, AI Processing and 3 more
- **Dependencies:** Data Layer, Shared Utilities, AI Processing, Util, Infrastructure & Deployment, Content Ingestion
- **Failure modes:** none identified
- **Important relationships:** Used by Testing, Ask, Documentation, Infrastructure & Deployment, Background Jobs and 2 more.
- **Source of explanation:** deterministic, derived from the repository graph (src/lib/retrieval/index.ts:197, src/lib/retrieval/index.ts:24, src/lib/retrieval/index.ts:164)

Graph

- **Purpose:** Code under src/lib/graph/.
- **Business function:** Supports the full-stack monolith (Next.js) through 2 file(s).
- **Primary components:** LoadedFile (interface, src/lib/graph/build.ts:22); loadProjectFiles (function, src/lib/graph/build.ts:54); resolveImport (function, src/lib/graph/resolve.ts:7); readPathAliases (function, src/lib/graph/resolve.ts:118); externalPackageName (function, src/lib/graph/resolve.ts:139); buildGraph (function, src/lib/graph/build.ts:62); computeImportance (function, src/lib/graph/build.ts:325); BuildResult (interface, src/lib/graph/build.ts:10)
- **Inputs:** Calls from Documentation, Review, Discover, Testing and 2 more
- **Processing:** 2 file(s), key components: LoadedFile, loadProjectFiles, resolveImport, readPathAliases, externalPackageName.
- **Outputs:** Results consumed via Parse, Data Layer, Content Ingestion and 3 more
- **Dependencies:** Parse, Data Layer, Content Ingestion, Util, Infrastructure & Deployment, AI Processing
- **Failure modes:** none identified
- **Important relationships:** Used by Documentation, Review, Discover, Testing, Analysis and 1 more.
- **Source of explanation:** deterministic, derived from the repository graph (src/lib/graph/build.ts:22, src/lib/graph/build.ts:54, src/lib/graph/resolve.ts:7)

Map

- **Purpose:** Code under src/lib/map/.
- **Business function:** Supports the full-stack monolith (Next.js) through 2 file(s).
- **Primary components:** loadModel (function, src/lib/map/index.ts:55); Graph (interface, src/lib/map/index.ts:29); nodeTypeForFile (function, src/lib/map/index.ts:106); riskGlyph (function, src/lib/map/legend.ts:39); glyphFor (function, src/lib/map/legend.ts:35); graphToMermaid (function, src/lib/map/index.ts:493); areaGraph (function, src/lib/map/index.ts:214); nodeTypeForSymbol (function, src/lib/map/index.ts:121)
- **Inputs:** Calls from API Layer, Testing, Export, Background Jobs and 4 more
- **Processing:** 2 file(s), key components: loadModel, Graph, nodeTypeForFile, riskGlyph, glyphFor.
- **Outputs:** Results consumed via Infrastructure & Deployment, Data Layer, Discover and 3 more

- **Dependencies:** Infrastructure & Deployment, Data Layer, Discover, AI Processing, Parse, Content Ingestion
- **Failure modes:** none identified
- **Important relationships:** Used by API Layer, Testing, Export, Background Jobs, User Interface and 3 more.
- **Source of explanation:** deterministic, derived from the repository graph (src/lib/map/index.ts:55, src/lib/map/index.ts:29, src/lib/map/index.ts:106)

Ask

- **Purpose:** Code under src/lib/ask/.
- **Business function:** Supports the full-stack monolith (Next.js) through 1 file(s).
- **Primary components:** askRepository (function, src/lib/ask/index.ts:138); relevantHits (function, src/lib/ask/index.ts:183); recentQuestions (function, src/lib/ask/index.ts:198); Answer (interface, src/lib/ask/index.ts:20); Citation (interface, src/lib/ask/index.ts:12); cite (function, src/lib/ask/index.ts:39); withSnippets (function, src/lib/ask/index.ts:44); findSymbolByName (function, src/lib/ask/index.ts:70)
- **Inputs:** Calls from API Layer, Infrastructure & Deployment, Testing
- **Processing:** 1 file(s), key components: askRepository, relevantHits, recentQuestions, Answer, Citation.
- **Outputs:** Results consumed via AI Processing, Search & Retrieval, Data Layer and 6 more
- **Dependencies:** AI Processing, Search & Retrieval, Data Layer, Map, Util, Content Ingestion, Discover, Infrastructure & Deployment, Review
- **Failure modes:** none identified
- **Important relationships:** Used by API Layer, Infrastructure & Deployment, Testing.
- **Source of explanation:** deterministic, derived from the repository graph (src/lib/ask/index.ts:138, src/lib/ask/index.ts:183, src/lib/ask/index.ts:198)

6. Data Flow

GET /path

```

GET /path
  ↓
lineAt
  ↓
callArgs
  ↓
isInlineFn
  ↓
reads symbols
  ↓
jobs.push

```

GET /path is handled by GET /path, which calls lineAt, callArgs, isInlineFn, jobs.push; along the way it reads symbols. (src/lib/discover/detect.ts:315)

Data touched: symbols; external: none.

POST /api/github/validate

```

POST /api/github/validate
  ↓
guard
  ↓
json
  ↓
AppError
  ↓
parseGitHubUrl
  ↓
fetchGitHubMetadata

```

```
↓  
fail  
↓  
isAppError  
↓  
scrubToken
```

POST /api/github/validate is handled by POST /api/github/validate, which calls guard, json, AppError, parseGitHubUrl, fetchGitHubMetadata. (src/app/api/github/validate/route.ts:10)

GET /api/jobs/:id

```
GET /api/jobs/:id  
↓  
guard  
↓  
getJob  
↓  
AppError  
↓  
json  
↓  
fail  
↓  
getDb  
↓  
reads jobs  
↓  
isAppError  
↓  
scrubToken  
↓  
openDatabase
```

GET /api/jobs/:id is handled by GET /api/jobs/:id, which calls guard, getJob, AppError, json, fail; along the way it reads jobs. (src/app/api/jobs/[id]/route.ts:8)

Data touched: jobs; external: none.

POST /api/jobs/:id/cancel

```
POST /api/jobs/:id/cancel  
↓  
guard  
↓  
requestCancel  
↓  
AppError  
↓  
json  
↓  
fail  
↓  
getDb  
↓  
getJob  
↓  
isAppError  
↓  
scrubToken  
↓  
openDatabase  
↓  
reads jobs
```

POST /api/jobs/:id/cancel is handled by POST /api/jobs/:id/cancel, which calls guard, requestCancel, AppError, json, fail; along the way it reads jobs. (src/app/api/jobs/[id]/cancel/route.ts:8)

Data touched: jobs; external: none.

GET /api/projects

```
GET /api/projects
↓
guard
↓
getDb
↓
json
↓
reads projects
↓
fail
↓
openDatabase
↓
isAppError
↓
scrubToken
↓
migrationsFolder
```

GET /api/projects is handled by GET /api/projects, which calls guard, getDb, json, fail, openDatabase; along the way it reads projects. (src/app/api/projects/route.ts:7)

Data touched: projects; external: none.

DELETE /api/projects/:id

```
DELETE /api/projects/:id
↓
guard
↓
getProject
↓
listJobs
↓
requestCancel
↓
deleteProject
↓
json
↓
fail
↓
getDb
↓
AppError
↓
reads projects
↓
reads jobs
↓
getJob
↓
isAppError
```

DELETE /api/projects/:id is handled by DELETE /api/projects/:id, which calls guard, getProject, listJobs, requestCancel, deleteProject; along the way it reads projects, jobs. (src/app/api/projects/[id]/route.ts:15)

Data touched: projects, jobs; external: none.

GET /api/projects/:id

```
GET /api/projects/:id
↓
guard
↓
json
↓
projectSummary
↓
getProject
↓
fail
↓
getDb
↓
reads jobs
↓
AppError
↓
reads projects
↓
isAppError
↓
scrubToken
↓
openDatabase
```

GET /api/projects/:id is handled by GET /api/projects/:id, which calls guard, json, projectSummary, getProject, fail; along the way it reads jobs, projects. (src/app/api/projects/[id]/route.ts:8)

Data touched: jobs, projects; external: none.

POST /api/projects/:id/analyze

```
POST /api/projects/:id/analyze
↓
guard
↓
getProject
↓
getDb
↓
decryptSecret
↓
parseGitHubUrl
↓
fetchGitHubMetadata
↓
createPendingProject
↓
enqueueAnalysis
↓
json
↓
projectSummary
↓
reads credentials
↓
reads projects
↓
fail
```

POST /api/projects/:id/analyze is handled by POST /api/projects/:id/analyze, which calls guard, getProject, getDb, decryptSecret, parseGitHubUrl; along the way it reads credentials, projects. (src/app/api/projects/[id]/analyze/route.ts:17)

Data touched: credentials, projects; external: none.

GET /api/projects/:id/architecture

```
GET /api/projects/:id/architecture
↓
guard
↓
getReadyProject
↓
json
↓
architectureDiagramText
↓
architectureMermaid
↓
erMermaid
↓
legendText
↓
fail
↓
AppError
↓
getProject
↓
glyphFor
↓
loadModel
↓
box
```

GET /api/projects/:id/architecture is handled by GET /api/projects/:id/architecture, which calls guard, getReadyProject, json, architectureDiagramText, architectureMermaid. (src/app/api/projects/[id]/architecture/route.ts:8)

GET /api/projects/:id/ask

```
GET /api/projects/:id/ask
↓
guard
↓
getReadyProject
↓
json
↓
recentQuestions
↓
fail
↓
AppError
↓
getProject
↓
getDb
↓
reads questions
↓
isAppError
↓
scrubToken
↓
reads projects
↓
openDatabase
```

GET /api/projects/:id/ask is handled by GET /api/projects/:id/ask, which calls guard, getReadyProject, json,

recentQuestions, fail; along the way it reads questions, projects. (src/app/api/projects/[id]/ask/route.ts:11)

Data touched: questions, projects; external: none.

POST /api/projects/:id/ask

```
POST /api/projects/:id/ask
  ↓
guard
  ↓
getReadyProject
  ↓
json
  ↓
AppError
  ↓
askRepository
  ↓
fail
  ↓
getProject
  ↓
getDb
  ↓
getAIProvider
  ↓
retrieveContext
  ↓
queryTerms
  ↓
UsageMeter
  ↓
tryAnalyze
```

POST /api/projects/:id/ask is handled by POST /api/projects/:id/ask, which calls guard, getReadyProject, json, AppError, askRepository. (src/app/api/projects/[id]/ask/route.ts:19)

GET /api/projects/:id/export

```
GET /api/projects/:id/export
  ↓
guard
  ↓
getReadyProject
  ↓
isFormat
  ↓
AppError
  ↓
exportFile
  ↓
isScope
  ↓
fail
  ↓
getProject
  ↓
loadReportData
  ↓
buildMarkdown
  ↓
parseReport
  ↓
renderPdf
  ↓
```

GET /api/projects/:id/export is handled by GET /api/projects/:id/export, which calls guard, getReadyProject, isFormat, AppError, exportFile. (src/app/api/projects/[id]/export/route.ts:13)

GET /api/projects/:id/files

```

GET /api/projects/:id/files
  ↓
guard
  ↓
getReadyProject
  ↓
getDb
  ↓
json
  ↓
repositoryTree
  ↓
treeToText
  ↓
reads files
  ↓
fail
  ↓
AppError
  ↓
getProject
  ↓
openDatabase
  ↓
loadModel
  ↓
jobs.push

```

GET /api/projects/:id/files is handled by GET /api/projects/:id/files, which calls guard, getReadyProject, getDb, json, repositoryTree; along the way it reads files. (src/app/api/projects/[id]/files/route.ts:9)

Data touched: files; external: none.

GET /api/projects/:id/files/content

```

GET /api/projects/:id/files/content
  ↓
guard
  ↓
getReadyProject
  ↓
AppError
  ↓
getDb
  ↓
getFileContent
  ↓
changeImpact
  ↓
json
  ↓
reads files
  ↓
reads symbols
  ↓
reads relationships
  ↓
reads findings
  ↓

```

```
UsageMeter.add
↓
fail
```

GET /api/projects/:id/files/content is handled by GET /api/projects/:id/files/content, which calls guard, getReadyProject, AppError, getDb, getFileContent; along the way it reads files, symbols, relationships, findings. (src/app/api/projects/[id]/files/content/route.ts:13)

Data touched: files, symbols, relationships, findings; external: none.

GET /api/projects/:id/findings

```
GET /api/projects/:id/findings
↓
guard
↓
getReadyProject
↓
getDb
↓
intParam
↓
json
↓
reads findings
↓
list
↓
fail
↓
AppError
↓
getProject
↓
openDatabase
↓
isAppError
↓
scrubToken
```

GET /api/projects/:id/findings is handled by GET /api/projects/:id/findings, which calls guard, getReadyProject, getDb, intParam, json; along the way it reads findings. (src/app/api/projects/[id]/findings/route.ts:9)

Data touched: findings; external: none.

7. API Architecture

34 route(s) were detected across Next.js, Express-style: 25 API and 9 page routes. (src/app/page.tsx:15, src/app/api/github/validate/route.ts:10, src/app/api/jobs/[id]/route.ts:8)

0 route(s) show an authentication or authorization check near their definition; 25 API route(s) show none, which means either the check lives in shared middleware or the route is open.

Method	Route	Handler	Authentication	Kind	Location
GET	/	uploadWithProgress	Not visible	page · Next.js	src/app/page.tsx:15
POST	/api/github/validate	POST (route handler)	Not visible	api · Next.js	src/app/api/github/validate/route.ts:10
GET	/api/jobs/:id	GET (route handler)	Not visible	api · Next.js	src/app/api/jobs/[id]/route.ts:8
POST	/api/jobs/:id/cancel	POST (route handler)	Not visible	api · Next.js	src/app/api/jobs/[id]/cancel/route.ts:8

Method	Route	Handler	Authentication	Kind	Location
GET	/api/projects	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/route.ts:7
DELETE	/api/projects/:id	DELETE (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/route.ts:15
GET	/api/projects/:id	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/route.ts:8
POST	/api/projects/:id/analyze	POST (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/analyze/route.ts:17
GET	/api/projects/:id/architecture	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/architecture/route.ts:8
GET	/api/projects/:id/ask	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/ask/route.ts:11
POST	/api/projects/:id/ask	POST (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/ask/route.ts:19
GET	/api/projects/:id/export	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/export/route.ts:13
GET	/api/projects/:id/files	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/files/route.ts:9
GET	/api/projects/:id/files/content	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/files/content/route.ts:13
GET	/api/projects/:id/findings	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/findings/route.ts:9
GET	/api/projects/:id/graph	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/graph/route.ts:9
GET	/api/projects/:id/impact	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/impact/route.ts:8
GET	/api/projects/:id/overview	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/overview/route.ts:11
GET	/api/projects/:id/report	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/report/route.ts:8
GET	/api/projects/:id/search	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/search/route.ts:8
GET	/api/projects/:id/symbols/:symbolId	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/symbols/[symbolId]/route.ts:13
POST	/api/projects/github	POST (route handler)	Not visible	api · Next.js	src/app/api/projects/github/route.ts:16
POST	/api/projects/upload	POST (route handler)	Not visible	api · Next.js	src/app/api/projects/upload/route.ts:14
GET	/api/status	GET (route handler)	Not visible	api · Next.js	src/app/api/status/route.ts:30
GET	/p/:id	OverviewPage	Not visible	page · Next.js	src/app/p/[id]/page.tsx:26
GET	/p/:id/architecture	ArchitecturePage	Not visible	page · Next.js	src/app/p/[id]/architecture/page.tsx:16
GET	/p/:id/ask	Cite	Not visible	page · Next.js	src/app/p/[id]/ask/page.tsx:20
GET	/p/:id/explain	Origin	Not visible	page · Next.js	src/app/p/[id]/explain/page.tsx:11

Method	Route	Handler	Authentication	Kind	Location
GET	/p/:id/files	Splitter	Not visible	page · Next.js	src/app/p/[id]/files/page.tsx:36
GET	/p/:id/map	ImpactPanel	Not visible	page · Next.js	src/app/p/[id]/map/page.tsx:18
GET	/p/:id/reports	ReportsPage	Not visible	page · Next.js	src/app/p/[id]/reports/page.tsx:16
GET	/p/:id/review	Diff	Not visible	page · Next.js	src/app/p/[id]/review/page.tsx:14
GET	/path	handler	Not visible	api · Express-style	src/lib/discover/detect.ts:315
GET	/x	GET /x (inline)	Not visible	api · Express-style	src/lib/parse/extract.ts:306

8. Data Architecture

12 data model(s) were found using Drizzle + SQL table projects (drizzle/0000_robust_hawkeye.sql:117), Drizzle + SQL table credentials (drizzle/0000_robust_hawkeye.sql:7), Drizzle + SQL table blobs (drizzle/0000_robust_hawkeye.sql:1), Drizzle + SQL table files (drizzle/0000_robust_hawkeye.sql:20) and 8 more. (src/lib/db/schema.ts:4-26, src/lib/db/schema.ts:29-33, src/lib/db/schema.ts:36-40)

9. External Integrations

Service	Category	Where used	Why	If unavailable
Anthropic	ai	package.json, src/lib/ai/anthropic.ts:1, src/lib/config.ts:35	language model inference	AI-generated features stop producing output
OpenAI	ai	src/lib/config.ts:39, src/lib/config.ts:38, .env.example:12	language model inference and embeddings	AI-generated features stop producing output
SQLite	database	package.json, src/lib/db/client.ts:1	embedded relational storage	persisted reads and writes fail
GitHub API	developer-platform	src/lib/config.ts:57, src/lib/config.ts:56	repository and identity integration	GitHub-backed features fail

10. Infrastructure & Deployment

Deployment and operations assets: GitHub Actions, Docker, Docker Compose. (.github/workflows/ci.yml, Dockerfile, docker-compose.yml)

Network ports referenced: 3003, 3211. (Dockerfile:36, playwright.config.ts:5)

41 environment variable(s) configure the application, 5 of which look sensitive (names only: ANTHROPIC_API_KEY, CREDENTIAL_SECRET, GITHUB_TOKEN, OPENAI_API_KEY, OPENAI_BASE_URL).

Configuration (variable names only; values are never exported):

Variable	Sensitive	Read in	Declared in
AI_CONCURRENCY	no		.env.example
AI_EMBEDDING_MODEL	no	src/lib/config.ts:40	.env.example
AI_MAX_FILES_EXPLAINED	no		.env.example
AI_MAX_FILES_REVIEWED	no		.env.example
AI_MAX_SYMBOLS_EXPLAINED	no		.env.example

Variable	Sensitive	Read in	Declared in
AI_MODEL	no	src/lib/config.ts:34	.env.example
AI_PROVIDER	no	src/lib/config.ts:33	.env.example
AI_REFUSAL_FALLBACKS	no	src/lib/ai/anthropic.ts:31	.env.example
ALLOWED_HOSTS	no	src/proxy.ts:9	.env.example, docker-compose.yml
ANTHROPIC_API_KEY	yes	src/lib/config.ts:35	.env.example
ANTHROPIC_WORKSPACE_ID	no	src/lib/config.ts:37	.env.example
CLOSE_TAG	no	src/lib/ai/prompt.ts:30	
CREDENTIAL_SECRET	yes	src/lib/config.ts:66	.env.example
DATABASE_PATH	no	drizzle.config.ts:7, src/lib/config.ts:15	.env.example, docker-compose.yml
E2E_PORT	no	playwright.config.ts:5	
EMBEDDED_WORKER	no	src/instrumentation.ts:3	.env.example
GITHUB_API_BASE	no	src/lib/config.ts:57	
GITHUB_TOKEN	yes	src/lib/config.ts:56	.env.example
GO_PATH	no	src/lib/config.ts:63	
MAX_FILE_BYTES	no		.env.example
MAX_FILES	no		.env.example
MAX_TOTAL_BYTES	no		.env.example
MAX_UPLOAD_BYTES	no		.env.example
MAX_ZIP_ENTRIES	no		.env.example
MAX_ZIP_RATIO	no		.env.example
MERMAID_ONLOAD	no	src/lib/export/html.ts:86	
NEXT_BUILD_CPUS	no	next.config.ts:14	
NEXT_RUNTIME	no	src/instrumentation.ts:3	
NEXT_TELEMETRY_DISABLED	no		Dockerfile
NODE_ENV	no		Dockerfile
OPEN_TAG	no	src/lib/ai/prompt.ts:30	
OPENAI_API_KEY	yes	src/lib/config.ts:39	.env.example
OPENAI_BASE_URL	yes	src/lib/config.ts:38	.env.example
PARSER_VERSION	no	src/lib/parse/cache.ts:10	
PATH	no	src/lib/analysis/analyzers.ts:34	
PORT	no	Dockerfile:38, docker-compose.yml:7	.env.example, docker-compose.yml
PROMPT_VERSION	no	src/lib/docs/cache.ts:13	
PW_CHANNEL	no	playwright.config.ts:17	
PYTHON_PATH	no	src/lib/config.ts:62	.env.example
RUFF_PATH	no	src/lib/config.ts:61	.env.example
STATIC_ANALYSIS	no	src/lib/config.ts:60	.env.example

11. Testing Strategy

34 test file(s): 3 e2e, 17 fixture, 13 unit, 1 integration; frameworks: Vitest, Playwright. (e2e/performance.spec.ts, e2e/styling.spec.ts)

Important files with no test referencing them include src/lib/client.ts, src/lib/util/text.ts, src/components/ui.tsx. (src/lib/client.ts, src/lib/util/text.ts, src/components/ui.tsx)

12. Security Model

Authentication: no mechanism detected. 25 of 25 API route(s) have no visible access check.

Secrets: 8 likely secret value(s) were found in source (redacted in this report). (fixtures/sample-shop/src/config.ts:7, tests/ai.test.ts:150)

6 security finding(s) were recorded: 0 critical/high.

13. Code Review

21 finding(s): 0 critical, 0 high, 8 medium, 13 low, 0 informational. 21 from static analyzers, 0 AI-inferred (0 need verification).

By category: Maintainability 14 · Security 6 · Testing 1

Analyzers run:

Analyzer	Status	Findings	Detail
brody-rules	ran	0	Built-in deterministic pattern rules (security, reliability, performance, operations).
brody-structure	ran	21	Structural checks over the repository model (size, complexity, tests, operations, dependencies, secrets).
typescript	ran	0	Syntactic diagnostics via the TypeScript compiler API (no type resolution, no code execution).
eslint	ran	0	ESLint Linter API with a fixed embedded rule set; repository ESLint configs are not executed.
python-ast	ran	0	ast.parse validation in isolated mode (code is parsed, never executed).
ruff	ran	0	Ruff in --isolated mode with a fixed rule selection; repository Ruff configuration is ignored.
mypy	skipped	0	Skipped for safety: mypy loads plugins named in repository configuration, which would execute untrusted code outside a sandbox.

AI review did not run because no provider was configured; only deterministic findings are shown.

MNT-001 — High cyclomatic complexity in FilesPage (88)

Severity	Confidence	Category	Source	Status	Location
Medium	High	Maintainability	Static analyzer (brody-structure)	Verified	src/app/p/[id]/files/page.tsx:115-251

Evidence

```
function FilesPage() { – 88 independent branches counted by the parser.
```

What happens. The function contains about 88 decision points (if/loops/case/catch/logical operators).

Why it matters. Each decision path needs its own test; high branching correlates with defects and makes behaviour hard to reason about.

Recommended remediation. Simplify with early returns, lookup tables or smaller functions, and add tests covering each branch.

Related components: FilesPage

Verification: Computed from the syntax tree.

MNT-002 — High cyclomatic complexity in Home (88)

Severity	Confidence	Category	Source	Status	Location
Medium	High	Maintainability	Static analyzer (brody-structure)	Verified	src/app/page.tsx:47-241

Evidence

```
function Home() { – 88 independent branches counted by the parser.
```

What happens. The function contains about 88 decision points (if/loops/case/catch/logical operators).

Why it matters. Each decision path needs its own test; high branching correlates with defects and makes behaviour hard to reason about.

Recommended remediation. Simplify with early returns, lookup tables or smaller functions, and add tests covering each branch.

Related components: Home

Verification: Computed from the syntax tree.

MNT-003 — High cyclomatic complexity in detectRoutes (190)

Severity	Confidence	Category	Source	Status	Location
Medium	High	Maintainability	Static analyzer (brody-structure)	Verified	src/lib/discover/detect.ts:250-439

Evidence

```
function detectRoutes(files: LoadedFile[], symbols: SymbolRow[]): RouteInfo[] { – 190 independent branches counted by the parser.
```

What happens. The function contains about 190 decision points (if/loops/case/catch/logical operators).

Why it matters. Each decision path needs its own test; high branching correlates with defects and makes behaviour hard to reason about.

Recommended remediation. Simplify with early returns, lookup tables or smaller functions, and add tests covering each branch.

Related components: detectRoutes

Verification: Computed from the syntax tree.

MNT-004 — High cyclomatic complexity in detectEntryPoints (100)

Severity	Confidence	Category	Source	Status	Location
Medium	High	Maintainability	Static analyzer (brody-structure)	Verified	src/lib/discover/detect.ts:492-566

Evidence

```
function detectEntryPoints(files: LoadedFile[], symbols: SymbolRow[], deps: DependencyInfo[]): EntryPoint[] { – 100 independent branches counted by the parser.
```

What happens. The function contains about 100 decision points (if/loops/case/catch/logical operators).

Why it matters. Each decision path needs its own test; high branching correlates with defects and makes behaviour hard to reason about.

Recommended remediation. Simplify with early returns, lookup tables or smaller functions, and add tests covering each branch.

Related components: detectEntryPoints

Verification: Computed from the syntax tree.

MNT-005 — High cyclomatic complexity in enhanceWithAI (100)

Severity	Confidence	Category	Source	Status	Location
Medium	High	Maintainability	Static analyzer (brody-structure)	Verified	src/lib/docs/ai.ts:69-288

Evidence

```
async function enhanceWithAI(opts: AiDocOptions): Promise<DocReport> { – 100 independent branches counted by the parser
  .
```

What happens. The function contains about 100 decision points (if/loops/case/catch/logical operators).

Why it matters. Each decision path needs its own test; high branching correlates with defects and makes behaviour hard to reason about.

Recommended remediation. Simplify with early returns, lookup tables or smaller functions, and add tests covering each branch.

Related components: enhanceWithAI

Verification: Computed from the syntax tree.

MNT-006 — High cyclomatic complexity in buildGraph (151)

Severity	Confidence	Category	Source	Status	Location
Medium	High	Maintainability	Static analyzer (brody-structure)	Verified	src/lib/graph/build.ts:62-322

Evidence

```
async function buildGraph(projectId: string, onProgress?: (done: number, total: number) => void, onPhase?: (phase: "parse" | "graph") => void)
: Promise<BuildResult> { – 151 independent branches counted by the parser.
```

What happens. The function contains about 151 decision points (if/loops/case/catch/logical operators).

Why it matters. Each decision path needs its own test; high branching correlates with defects and makes behaviour hard to reason about.

Recommended remediation. Simplify with early returns, lookup tables or smaller functions, and add tests covering each branch.

Related components: buildGraph

Verification: Computed from the syntax tree.

MNT-007 — High cyclomatic complexity in extractTsJs (166)

Severity	Confidence	Category	Source	Status	Location
Medium	High	Maintainability	Static analyzer (brody-structure)	Verified	src/lib/parse/extract.ts:146-341

Evidence

```
function extractTsJs(ctx: Ctx, tree: Tree): void { – 166 independent branches counted by the parser.
```

What happens. The function contains about 166 decision points (if/loops/case/catch/logical operators).

Why it matters. Each decision path needs its own test; high branching correlates with defects and makes behaviour hard to reason about.

Recommended remediation. Simplify with early returns, lookup tables or smaller functions, and add tests covering each branch.

Related components: extractTsJs

Verification: Computed from the syntax tree.

MNT-008 — High cyclomatic complexity in extractPython (83)

Severity	Confidence	Category	Source	Status	Location
Medium	High	Maintainability	Static analyzer (brody-structure)	Verified	src/lib/parse/extract.ts:346-430

Evidence

```
function extractPython(ctx: Ctx, tree: Tree): void { – 83 independent branches counted by the parser.
```

What happens. The function contains about 83 decision points (if/loops/case/catch/logical operators).

Why it matters. Each decision path needs its own test; high branching correlates with defects and makes behaviour hard to reason about.

Recommended remediation. Simplify with early returns, lookup tables or smaller functions, and add tests covering each branch.

Related components: extractPython

Verification: Computed from the syntax tree.

MNT-009 — Very long component: Home (195 lines)

Severity	Confidence	Category	Source	Status	Location
Low	High	Maintainability	Static analyzer (brody-structure)	Verified	src/app/page.tsx:47-241

Evidence

```
function Home() {
```

What happens. Home spans 195 lines.

Why it matters. Long functions mix several steps, which hides bugs and makes unit testing difficult.

Recommended remediation. Extract cohesive steps into named helper functions.

Related components: Home

Verification: Computed from parsed symbol ranges.

MNT-010 — Very long function: detectRoutes (190 lines)

Severity	Confidence	Category	Source	Status	Location
Low	High	Maintainability	Static analyzer (brody-structure)	Verified	src/lib/discover/detect.ts:250-439

Evidence

```
function detectRoutes(files: LoadedFile[], symbols: SymbolRow[]): RouteInfo[] {
```

What happens. detectRoutes spans 190 lines.

Why it matters. Long functions mix several steps, which hides bugs and makes unit testing difficult.

Recommended remediation. Extract cohesive steps into named helper functions.

Related components: detectRoutes

Verification: Computed from parsed symbol ranges.

MNT-011 — Very long function: discoverArchitecture (155 lines)

Severity	Confidence	Category	Source	Status	Location
Low	High	Maintainability	Static analyzer (brody-structure)	Verified	src/lib/discover/index.ts:205-359

Evidence

```
async function discoverArchitecture(projectId: string): Promise<Architecture> {
```

What happens. discoverArchitecture spans 155 lines.

Why it matters. Long functions mix several steps, which hides bugs and makes unit testing difficult.

Recommended remediation. Extract cohesive steps into named helper functions.

Related components: discoverArchitecture

Verification: Computed from parsed symbol ranges.

MNT-012 — Very long function: enhanceWithAI (220 lines)

Severity	Confidence	Category	Source	Status	Location
Low	High	Maintainability	Static analyzer (brody-structure)	Verified	src/lib/docs/ai.ts:69-288

Evidence

```
async function enhanceWithAI(opts: AiDocOptions): Promise<DocReport> {
```

What happens. enhanceWithAI spans 220 lines.

Why it matters. Long functions mix several steps, which hides bugs and makes unit testing difficult.

Recommended remediation. Extract cohesive steps into named helper functions.

Related components: enhanceWithAI

Verification: Computed from parsed symbol ranges.

MNT-013 — Very long function: buildGraph (261 lines)

Severity	Confidence	Category	Source	Status	Location
Low	High	Maintainability	Static analyzer (brody-structure)	Verified	src/lib/graph/build.ts:62-322

Evidence

```
async function buildGraph(projectId: string, onProgress?: (done: number, total: number) => void, onPhase?: (phase: "parse" | "graph") => void)  
: Promise<BuildResult> {
```

What happens. buildGraph spans 261 lines.

Why it matters. Long functions mix several steps, which hides bugs and makes unit testing difficult.

Recommended remediation. Extract cohesive steps into named helper functions.

Related components: buildGraph

Verification: Computed from parsed symbol ranges.

MNT-014 — Very long function: extractTsJs (196 lines)

Severity	Confidence	Category	Source	Status	Location
Low	High	Maintainability	Static analyzer (brody-structure)	Verified	src/lib/parse/extract.ts:146-341

Evidence

```
function extractTsJs(ctx: Ctx, tree: Tree): void {
```

What happens. extractTsJs spans 196 lines.

Why it matters. Long functions mix several steps, which hides bugs and makes unit testing difficult.

Recommended remediation. Extract cohesive steps into named helper functions.

Related components: extractTsJs

Verification: Computed from parsed symbol ranges.

SEC-001 — Secret-like value in test or fixture file fixtures/sample-shop/src/config.ts

Severity	Confidence	Category	Source	Status	Location
Low	Low	Security	Static analyzer (brody-structure)	Verified	fixtures/sample-shop/src/config.ts:7

Evidence

```
line 7: AWS Access Key AKIA...*****TB
```

What happens. Pattern matching found 1 value(s) that look like credentials (AWS Access Key). Values are redacted in this report.

Why it matters. Credentials committed to a repository are visible to everyone with access and persist in git history even after deletion.

Business impact. Unauthorised access to the connected service and potential data exposure until the credential is rotated.

Recommended remediation. Rotate the credential immediately, remove it from history, and load it from environment configuration or a secrets manager.

Verification: Deterministic pattern match; entropy and context may still make it a false positive (for example test data).

SEC-002 — None of the 25 API routes has an authentication check

Severity	Confidence	Category	Source	Status	Location
Low	Low	Security	Static analyzer (brody-structure)	Verified	src/app/api/github/validate/route.ts:10

Evidence

```
POST /api/github/validate (src/app/api/github/validate/route.ts:10)
GET /api/jobs/:id (src/app/api/jobs/[id]/route.ts:8)
POST /api/jobs/:id/cancel (src/app/api/jobs/[id]/cancel/route.ts:8)
GET /api/projects (src/app/api/projects/route.ts:7)
DELETE /api/projects/:id (src/app/api/projects/[id]/route.ts:15)
GET /api/projects/:id (src/app/api/projects/[id]/route.ts:8)
```

What happens. No route has authentication middleware or guard, and no authentication library or provider was detected.

Why it matters. Anyone who can reach the service can call every route. That is acceptable for a local single-user tool but not for a shared or internet-facing deployment.

Recommended remediation. Put the service behind an authenticating reverse proxy, or add authentication and authorization to the routes before exposing it.

Verification: Checked middleware visible at each route definition and imports of common authentication libraries.

SEC-003 — Secret-like value in test or fixture file tests/ai.test.ts

Severity	Confidence	Category	Source	Status	Location
Low	Low	Security	Static analyzer (brody-structure)	Verified	tests/ai.test.ts:150

Evidence

```
line 150: AWS Access Key AKIA...*****TB
```

What happens. Pattern matching found 1 value(s) that look like credentials (AWS Access Key). Values are redacted in this report.

Why it matters. Credentials committed to a repository are visible to everyone with access and persist in git history even after deletion.

Business impact. Unauthorised access to the connected service and potential data exposure until the credential

is rotated.

Recommended remediation. Rotate the credential immediately, remove it from history, and load it from environment configuration or a secrets manager.

Verification: Deterministic pattern match; entropy and context may still make it a false positive (for example test data).

SEC-004 — Secret-like value in test or fixture file tests/api.test.ts

Severity	Confidence	Category	Source	Status	Location
Low	Low	Security	Static analyzer (brody-structure)	Verified	tests/api.test.ts:117

Evidence

```
line 117: AWS Access Key AKIA...*****TB
line 120: AWS Access Key AKIA...*****TB
```

What happens. Pattern matching found 2 value(s) that look like credentials (AWS Access Key). Values are redacted in this report.

Why it matters. Credentials committed to a repository are visible to everyone with access and persist in git history even after deletion.

Business impact. Unauthorised access to the connected service and potential data exposure until the credential is rotated.

Recommended remediation. Rotate the credential immediately, remove it from history, and load it from environment configuration or a secrets manager.

Verification: Deterministic pattern match; entropy and context may still make it a false positive (for example test data).

SEC-005 — Secret-like value in test or fixture file tests/pipeline.test.ts

Severity	Confidence	Category	Source	Status	Location
Low	Low	Security	Static analyzer (brody-structure)	Verified	tests/pipeline.test.ts:87

Evidence

```
line 87: AWS Access Key AKIA...*****TB
```

What happens. Pattern matching found 1 value(s) that look like credentials (AWS Access Key). Values are redacted in this report.

Why it matters. Credentials committed to a repository are visible to everyone with access and persist in git history even after deletion.

Business impact. Unauthorised access to the connected service and potential data exposure until the credential is rotated.

Recommended remediation. Rotate the credential immediately, remove it from history, and load it from environment configuration or a secrets manager.

Verification: Deterministic pattern match; entropy and context may still make it a false positive (for example test data).

SEC-006 — Secret-like value in test or fixture file tests/secrets.test.ts

Severity	Confidence	Category	Source	Status	Location
Low	Low	Security	Static analyzer (brody-structure)	Verified	tests/secrets.test.ts:5

Evidence

```
line 5: AWS Access Key AKIA...*****TB
line 33: Bearer Token Bear...*****56
```

What happens. Pattern matching found 2 value(s) that look like credentials (AWS Access Key, Bearer Token). Values are redacted in this report.

Why it matters. Credentials committed to a repository are visible to everyone with access and persist in git history even after deletion.

Business impact. Unauthorised access to the connected service and potential data exposure until the credential is rotated.

Recommended remediation. Rotate the credential immediately, remove it from history, and load it from environment configuration or a secrets manager.

Verification: Deterministic pattern match; entropy and context may still make it a false positive (for example test data).

TST-001 — 3 important files have no tests

Severity	Confidence	Category	Source	Status	Location
Low	Medium	Testing	Static analyzer (brody-structure)	Verified	src/lib/client.ts:1

Evidence

```
src/lib/client.ts
src/lib/util/text.ts
src/components/ui.tsx
```

What happens. These files rank highest by how many other files depend on them, but no test imports them or calls their symbols.

Why it matters. Failures in highly depended-upon code propagate to many callers, and regressions here are not caught by the build.

Recommended remediation. Add unit tests for the exported functions and failure paths of these files, starting with the request handlers and payment or data-access code.

Related components: src/lib/client.ts, src/lib/util/text.ts, src/components/ui.tsx

Verification: Derived from TESTS relationships in the dependency graph; coverage through end-to-end tests would not be visible.

14. Engineering Risks

No critical or high-severity findings were recorded.

15. File & Component Explanations

src/lib/db/client.ts

Purpose. src/lib/db/client.ts is a data-layer TypeScript file in the Data Layer area. It declares 4 functions, 1 type, 1 interface, 1 variable. Its documentation says: "Return the open database, opening the configured one on first use."

Key symbols: getDb(), openDatabase(), migrationsFolder(), closeDatabase(), Db(), DbHolder(), holder()

How it operates. Exposes 4 exported symbol(s) that other modules call; internal helpers are 3.

Called by: scripts/analyze.mts, scripts/benchmark.mts, scripts/verify-ai.mts, src/app/api/projects/[id]/analyze/route.ts, src/app/api/projects/[id]/files/content/route.ts, src/app/api/projects/[id]/files/route.ts, src/app/api/projects/[id]/findings/route.ts, src/app/api/projects/[id]/overview/route.ts, src/app/api/projects/[id]/symbols/[symbolId]/route.ts, src/app/api/projects/github/route.ts, src/app/api/projects/route.ts, src/app/api/projects/upload/route.ts ·

Depends on: src/lib/db/schema.ts, src/lib/config.ts, better-sqlite3, drizzle-orm

Data in: Arguments from scripts/analyze.mts, scripts/benchmark.mts, scripts/verify-ai.mts and 30 more · **Data out:** Return values to callers

src/lib/db/schema.ts

Purpose. src/lib/db/schema.ts is a TypeScript file in the Data Layer area. It defines the data models projects, credentials, blobs, files and 8 more. It declares 12 models, 7 types, 1 interface. Its documentation says: "Projects: one row per ingested codebase (a repository or an upload)."

Responsibilities: Define model projects; Define model credentials; Define model blobs; Define model files; Define model symbols

Key symbols: files, symbols, jobs, findings, projects, blobs, relationships, questions

How it operates. Exposes 20 exported symbol(s) that other modules call; internal helpers are 0.

Called by: src/app/p/[id]/files/page.tsx, src/app/p/[id]/review/page.tsx, src/components/project.tsx, src/lib/analysis/structural.ts, src/lib/api.ts, src/lib/db/client.ts, src/lib/discover/detect.ts, src/lib/discover/index.ts, src/lib/docs/ai.ts, src/lib/docs/deterministic.ts, src/lib/export/markdown.ts, src/lib/graph/build.ts · **Depends on:** drizzle-orm

Data in: Arguments from src/app/p/[id]/files/page.tsx, src/app/p/[id]/review/page.tsx, src/components/project.tsx and 16 more · **Data out:** Return values to callers

src/lib/api.ts

Purpose. src/lib/api.ts is an API layer TypeScript file in the Shared Utilities area. It declares 7 functions. Its documentation says: "Wrap a handler so AppErrors become structured JSON with actionable hints."

Key symbols: json(), fail(), guard(), getReadyProject(), getProject(), projectSummary(), intParam()

How it operates. Exposes 7 exported symbol(s) that other modules call; internal helpers are 0.

Called by: src/app/api/github/validate/route.ts, src/app/api/jobs/[id]/cancel/route.ts, src/app/api/jobs/[id]/route.ts, src/app/api/projects/[id]/analyze/route.ts, src/app/api/projects/[id]/architecture/route.ts, src/app/api/projects/[id]/ask/route.ts, src/app/api/projects/[id]/export/route.ts, src/app/api/projects/[id]/files/content/route.ts, src/app/api/projects/[id]/files/route.ts, src/app/api/projects/[id]/findings/route.ts, src/app/api/projects/[id]/graph/route.ts, src/app/api/projects/[id]/impact/route.ts · **Depends on:** src/lib/db/client.ts, src/lib/db/schema.ts, src/lib/ingest/credentials.ts, src/lib/util/errors.ts, drizzle-orm, next

Data in: Records read from projects, jobs · **Data out:** Return values to callers

src/lib/config.ts

Purpose. src/lib/config.ts is a service-layer TypeScript file in the Shared Utilities area. It declares 1 variable, 1 function, 1 type.

Key symbols: config(), int(), AppConfig()

How it operates. Exposes 2 exported symbol(s) that other modules call; internal helpers are 1.

Called by: src/app/api/projects/github/route.ts, src/app/api/projects/upload/route.ts, src/app/api/status/route.ts, src/lib/ai/anthropic.ts, src/lib/ai/index.ts, src/lib/ai/openai.ts, src/lib/analysis/analyzers.ts, src/lib/db/client.ts, src/lib/docs/ai.ts, src/lib/ingest/credentials.ts, src/lib/ingest/fs.ts, src/lib/ingest/github.ts · **Depends on:** nothing indexed

Data in: Arguments from src/app/api/projects/github/route.ts, src/app/api/projects/upload/route.ts, src/app/api/status/route.ts and 17 more · **Data out:** Return values to callers

src/lib/util/errors.ts

Purpose. src/lib/util/errors.ts is a TypeScript file in the Util area. It declares 1 class, 2 functions, 3 propertys, 1 method. Its documentation says: "Application error with an HTTP status and a user-facing, actionable message."

Key symbols: AppError, isAppError(), errorMessage(), AppError.status(), AppError.code(), AppError.hint(), AppError.constructor()

How it operates. Exposes 3 exported symbol(s) that other modules call; internal helpers are 4.

Called by: src/app/api/github/validate/route.ts, src/app/api/jobs/[id]/cancel/route.ts, src/app/api/jobs/[id]/route.ts, src/app/api/projects/[id]/ask/route.ts, src/app/api/projects/[id]/export/route.ts, src/app/api/projects/[id]/files/content/route.ts, src/app/api/projects/[id]/graph/route.ts, src/app/api/projects/[id]/impact/route.ts, src/app/api/projects/[id]/symbols/[symbolId]/route.ts, src/app/api/projects/github/route.ts, src/app/api/projects/upload/route.ts, src/lib/api.ts · **Depends on:** nothing indexed

Data in: Arguments from src/app/api/github/validate/route.ts, src/app/api/jobs/[id]/cancel/route.ts, src/app/api/jobs/[id]/route.ts and 16 more · **Data out:** Return values to callers

src/lib/parse/extract.ts

Purpose. src/lib/parse/extract.ts is an API layer TypeScript file in the Parse area. It handles GET /x. It declares 27 functions, 2 constants, 1 interface. Its documentation says: "Determine the enclosing symbol index for a node given recorded symbol ranges."

Responsibilities: Serve GET /x

Key symbols: text(), extractFromTree(), cleanComment(), walk(), lineOf(), endLineOf(), firstLine(), addSymbol()

How it operates. Requests matching its routes are dispatched to GET /x (inline).

Called by: src/lib/parse/index.ts · **Depends on:** src/lib/parse/treesitter.ts, src/lib/parse/types.ts

Data in: HTTP request parameters, headers and bodies · **Data out:** Persists to symbols

Engineering notes: Large file (799 lines). Contains a function with cyclomatic complexity 166.

.github/workflows/ci.yml

Purpose. .github/workflows/ci.yml is an infrastructure YAML file in the Infrastructure & Deployment area. It declares 3 jobs.

Responsibilities: job push; job verify; job pull_request

Key symbols: jobs.push(), jobs.verify(), jobs.pull_request()

How it operates. Exposes 3 exported symbol(s) that other modules call; internal helpers are 0.

Called by: no indexed importers · **Depends on:** nothing indexed

Data in: n/a · **Data out:** Return values to callers

Engineering notes: Parsed with text-based heuristics, so symbol data is approximate.

src/lib/ai/index.ts

Purpose. src/lib/ai/index.ts is a TypeScript file in the AI Processing area. It declares 1 method, 6 functions, 2 interfaces, 1 class, 2 variables, 2 propertys. Its documentation says: "Run a structured request and swallow provider failures into the meter so one failed pass never aborts the whole job."

Key symbols: UsageMeter.add(), getAIProvider(), providerStatus(), ProviderStatus(), tryAnalyze(), setAIProvider(), UsageMeter, checkProvider()

How it operates. Exposes 9 exported symbol(s) that other modules call; internal helpers are 5.

Called by: scripts/analyze.mts, scripts/verify-ai.mts, src/app/api/projects/[id]/overview/route.ts, src/app/api/status/route.ts, src/lib/ask/index.ts, src/lib/docs/ai.ts, src/lib/docs/index.ts, src/lib/jobs/index.ts, src/lib/retrieval/index.ts, src/lib/review/ai.ts, src/lib/review/index.ts, src/lib/review/verify.ts · **Depends on:** src/lib/config.ts, src/lib/ai/anthropic.ts, src/lib/ai/errors.ts, src/lib/ai/openai.ts, src/lib/ai/types.ts, src/lib/ai/prompt.ts, zod

Data in: Arguments from scripts/analyze.mts, scripts/verify-ai.mts, src/app/api/projects/[id]/overview/route.ts and 12 more · **Data out:** Return values to callers

src/lib/ingest/store.ts

Purpose. src/lib/ingest/store.ts is a service-layer TypeScript file in the Content Ingestion area. It declares 6 functions, 1 interface, 1 type. Its documentation says: "Insert an empty project row (used when file contents arrive later, e.g. a GitHub download inside a job)."

Key symbols: `getFileContents()`, `CreateProjectResult()`, `createPendingProject()`, `createProject()`, `attachFiles()`, `getFileContent()`, `deleteProject()`, `IncrementalStats()`

How it operates. Exposes 7 exported symbol(s) that other modules call; internal helpers are 1.

Called by: `scripts/analyze.mts`, `scripts/benchmark.mts`, `scripts/verify-ai.mts`, `src/app/api/projects/[id]/analyze/route.ts`, `src/app/api/projects/[id]/files/content/route.ts`, `src/app/api/projects/[id]/route.ts`, `src/app/api/projects/[id]/symbols/[symbolId]/route.ts`, `src/app/api/projects/github/route.ts`, `src/lib/ask/index.ts`, `src/lib/graph/build.ts`, `src/lib/ingest/index.ts`, `src/lib/ingest/upload.ts` · **Depends on:** `src/lib/db/client.ts`, `src/lib/util/ids.ts`, `src/lib/ingest/types.ts`, `src/lib/ingest/credentials.ts`, `drizzle-orm`

Data in: Records read from projects, files, blobs · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 17.

src/lib/discover/types.ts

Purpose. `src/lib/discover/types.ts` is a TypeScript file in the Discover area. It declares 15 interfaces.

Key symbols: `Architecture()`, `Evidence()`, `EntryPoint()`, `RouteInfo()`, `ModelInfo()`, `ExternalService()`, `DependencyInfo()`, `EnvVar()`

How it operates. Exposes 15 exported symbol(s) that other modules call; internal helpers are 0.

Called by: `src/app/api/projects/[id]/architecture/route.ts`, `src/app/api/projects/[id]/overview/route.ts`, `src/app/p/[id]/architecture/page.tsx`, `src/app/p/[id]/explain/page.tsx`, `src/app/p/[id]/page.tsx`, `src/lib/analysis/structural.ts`, `src/lib/ask/index.ts`, `src/lib/discover/detect.ts`, `src/lib/discover/index.ts`, `src/lib/docs/deterministic.ts`, `src/lib/docs/index.ts`, `src/lib/export/markdown.ts` · **Depends on:** nothing indexed

Data in: Arguments from `src/app/api/projects/[id]/architecture/route.ts`, `src/app/api/projects/[id]/overview/route.ts`, `src/app/p/[id]/architecture/page.tsx` and 14 more · **Data out:** Return values to callers

src/lib/jobs/index.ts

Purpose. `src/lib/jobs/index.ts` is a background-processing TypeScript file in the Background Jobs area. It declares 12 methods, 8 functions, 2 classes, 5 propertyts, 1 constant. Its documentation says: "Atomically claim the oldest queued job."

Key symbols: `Tracker.flush()`, `getJob()`, `enqueueAnalysis()`, `claimNextJob()`, `processJob()`, `recoverStaleJobs()`, `requestCancel()`, `listJobs()`

How it operates. Exposes 8 exported symbol(s) that other modules call; internal helpers are 20.

Called by: `scripts/analyze.mts`, `scripts/benchmark.mts`, `scripts/verify-ai.mts`, `src/app/api/jobs/[id]/cancel/route.ts`, `src/app/api/jobs/[id]/route.ts`, `src/app/api/projects/[id]/analyze/route.ts`, `src/app/api/projects/[id]/route.ts`, `src/app/api/projects/github/route.ts`, `src/app/api/projects/upload/route.ts`, `src/lib/jobs/worker.ts`, `tests/api.test.ts`, `tests/helpers/index.ts` · **Depends on:** `src/lib/db/client.ts`, `src/lib/db/schema.ts`, `src/lib/ai/index.ts`, `src/lib/config.ts`, `src/lib/discover/index.ts`, `src/lib/docs/index.ts`, `src/lib/graph/build.ts`, `src/lib/ingest/credentials.ts`, `src/lib/ingest/github.ts`, `src/lib/ingest/normalize.ts`, `src/lib/ingest/store.ts`, `src/lib/map/index.ts`, `src/lib/retrieval/index.ts`, `src/lib/review/index.ts`, `src/lib/util/errors.ts`, `src/lib/util/ids.ts`

Data in: Records read from projects, jobs, credentials, files · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 67.

src/lib/map/index.ts

Purpose. `src/lib/map/index.ts` is a service-layer TypeScript file in the Map area. It declares 20 functions, 8 interfaces, 2 variables, 2 constants. Its documentation says:

"-----"

Key symbols: `loadModel()`, `Graph()`, `nodeTypeForFile()`, `graphToMermaid()`, `areaGraph()`, `nodeTypeForSymbol()`, `changeImpact()`, `repositoryTree()`

How it operates. Exposes 22 exported symbol(s) that other modules call; internal helpers are 10.

Called by: `scripts/benchmark.mts`, `src/app/api/projects/[id]/architecture/route.ts`, `src/app/api/projects/[id]/files/content/route.ts`, `src/app/api/projects/[id]/files/route.ts`, `src/app/api/projects/[id]/graph/route.ts`, `src/app/api/`

projects/[id]/impact/route.ts, src/app/api/projects/[id]/symbols/[symbolId]/route.ts, src/app/p/[id]/architecture/page.tsx, src/app/p/[id]/files/page.tsx, src/app/p/[id]/map/page.tsx, src/components/graph.tsx, src/lib/ask/index.ts · **Depends on:** src/lib/db/client.ts, src/lib/db/schema.ts, src/lib/discover/types.ts, src/lib/discover/catalog.ts, src/lib/map/legend.ts, drizzle-orm

Data in: Records read from projects, files, symbols, relationships, findings · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 51.

src/lib/graph/build.ts

Purpose. src/lib/graph/build.ts is a service-layer TypeScript file in the Graph area. It declares 4 interfaces, 3 functions, 3 constants. Its documentation says: "Load included, text-bearing files for a project along with content."

Key symbols: LoadedFile(), loadProjectFiles(), buildGraph(), computeImportance(), BuildResult(), SymRef(), RelDraft(), CALLABLE_KINDS()

How it operates. Exposes 5 exported symbol(s) that other modules call; internal helpers are 5.

Called by: src/lib/analysis/analyzers.ts, src/lib/analysis/structural.ts, src/lib/discover/detect.ts, src/lib/discover/index.ts, src/lib/docs/ai.ts, src/lib/docs/cache.ts, src/lib/docs/deterministic.ts, src/lib/docs/index.ts, src/lib/jobs/index.ts, src/lib/review/ai.ts, src/lib/review/index.ts, src/lib/review/verify.ts · **Depends on:** src/lib/db/client.ts, src/lib/db/schema.ts, src/lib/ingest/store.ts, src/lib/parse/index.ts, src/lib/parse/cache.ts, src/lib/util/ids.ts, src/lib/graph/resolve.ts, drizzle-orm

Data in: Records read from files · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 151.

src/lib/docs/types.ts

Purpose. src/lib/docs/types.ts is a TypeScript file in the Documentation area. It declares 8 interfaces. Its documentation says: "A single claim with the repository evidence that supports it."

Key symbols: DocReport(), FileDoc(), SymbolDoc(), Statement(), AreaDoc(), DocSection(), FlowDoc(), ConflictRecord()

How it operates. Exposes 8 exported symbol(s) that other modules call; internal helpers are 0.

Called by: scripts/verify-ai.mts, src/app/api/projects/[id]/files/content/route.ts, src/app/api/projects/[id]/overview/route.ts, src/app/api/projects/[id]/report/route.ts, src/app/api/projects/[id]/symbols/[symbolId]/route.ts, src/app/p/[id]/explain/page.tsx, src/app/p/[id]/files/page.tsx, src/app/p/[id]/page.tsx, src/lib/docs/ai.ts, src/lib/docs/cache.ts, src/lib/docs/deterministic.ts, src/lib/docs/index.ts · **Depends on:** nothing indexed

Data in: Arguments from scripts/verify-ai.mts, src/app/api/projects/[id]/files/content/route.ts, src/app/api/projects/[id]/overview/route.ts and 12 more · **Data out:** Return values to callers

src/lib/ingest/normalize.ts

Purpose. src/lib/ingest/normalize.ts is a service-layer TypeScript file in the Content Ingestion area. It declares 1 function. Its documentation says: "Turn raw bytes into the normalized file model: gitignore filtering, binary detection, hashing, classification, duplicate detection, size flags. Files excluded by default are kept as metadata so users "

Key symbols: normalizeFiles()

How it operates. Exposes 1 exported symbol(s) that other modules call; internal helpers are 0.

Called by: scripts/analyze.mts, scripts/benchmark.mts, scripts/verify-ai.mts, src/lib/ingest/index.ts, src/lib/ingest/upload.ts, src/lib/jobs/index.ts, tests/api.test.ts, tests/helpers/index.ts, tests/ingest.test.ts, tests/pipeline.test.ts ·

Depends on: src/lib/config.ts, src/lib/util/ids.ts, src/lib/util/text.ts, src/lib/ingest/classify.ts, src/lib/ingest/languages.ts, src/lib/ingest/types.ts, ignore, isbinaryfile

Data in: Arguments from scripts/analyze.mts, scripts/benchmark.mts, scripts/verify-ai.mts and 7 more · **Data out:** Persists to files

Engineering notes: Contains a function with cyclomatic complexity 39.

src/lib/client.ts

Purpose. src/lib/client.ts is a user-interface TypeScript file in the Shared Utilities area. It declares 1 hook, 5 functions, 1 class, 3 propertys, 1 method. Its documentation says: "Parse "path:12-30" citations into a navigable link target."

Key symbols: useApi(), parseCite(), keyOf(), ApiError, api(), fmtDuration(), fmtBytes(), ApiError.code()

How it operates. Exposes 6 exported symbol(s) that other modules call; internal helpers are 5.

Called by: src/app/p/[id]/architecture/page.tsx, src/app/p/[id]/ask/page.tsx, src/app/p/[id]/explain/page.tsx, src/app/p/[id]/files/page.tsx, src/app/p/[id]/layout.tsx, src/app/p/[id]/map/page.tsx, src/app/p/[id]/page.tsx, src/app/p/[id]/review/page.tsx, src/app/page.tsx, src/components/project.tsx, src/components/ui.tsx · **Depends on:** react

Data in: Arguments from src/app/p/[id]/architecture/page.tsx, src/app/p/[id]/ask/page.tsx, src/app/p/[id]/explain/page.tsx and 8 more · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 18.

src/lib/util/text.ts

Purpose. src/lib/util/text.ts is a TypeScript file in the Util area. It declares 6 functions. Its documentation says: "Split identifiers into searchable terms: camelCase, snake_case, kebab-case, paths."

Key symbols: escapeHtml(), sliceLines(), tokenize(), countLines(), formatBytes(), truncate()

How it operates. Exposes 6 exported symbol(s) that other modules call; internal helpers are 0.

Called by: src/app/api/projects/[id]/symbols/[symbolId]/route.ts, src/lib/ask/index.ts, src/lib/docs/ai.ts, src/lib/docs/deterministic.ts, src/lib/export/html.ts, src/lib/export/markdown.ts, src/lib/ingest/normalize.ts, src/lib/retrieval/index.ts, src/lib/review/ai.ts, src/lib/review/verify.ts · **Depends on:** nothing indexed

Data in: Arguments from src/app/api/projects/[id]/symbols/[symbolId]/route.ts, src/lib/ask/index.ts, src/lib/docs/ai.ts and 7 more · **Data out:** Return values to callers

src/components/ui.tsx

Purpose. src/components/ui.tsx is a user-interface TypeScript file in the User Interface area. It declares 11 components, 1 constant. Its documentation says: "Link to a source location in the code explorer."

Responsibilities: component Loading; component ErrorBox; component Chip; component Empty; component SourceLink

Key symbols: Loading(), ErrorBox(), Chip(), Empty(), SourceLink(), SeverityBadge(), OriginBadge(), Evidence()

How it operates. Exposes 12 exported symbol(s) that other modules call; internal helpers are 0.

Called by: src/app/p/[id]/architecture/page.tsx, src/app/p/[id]/ask/page.tsx, src/app/p/[id]/explain/page.tsx, src/app/p/[id]/files/page.tsx, src/app/p/[id]/layout.tsx, src/app/p/[id]/map/page.tsx, src/app/p/[id]/page.tsx, src/app/p/[id]/review/page.tsx, src/app/page.tsx, src/components/project.tsx · **Depends on:** src/lib/client.ts, next, react

Data in: Arguments from src/app/p/[id]/architecture/page.tsx, src/app/p/[id]/ask/page.tsx, src/app/p/[id]/explain/page.tsx and 7 more · **Data out:** Return values to callers

src/lib/jobs/worker.ts

Purpose. src/lib/jobs/worker.ts is an entry-point TypeScript file in the Background Jobs area. It declares 3 functions, 1 interface, 1 variable. Its documentation says: "Start the in-process job loop. Safe to call more than once."

Key symbols: startWorker(), drainQueue(), stopWorker(), WorkerState(), state()

How it operates. Exposes 3 exported symbol(s) that other modules call; internal helpers are 2.

Called by: scripts/analyze.mts, scripts/benchmark.mts, scripts/verify-ai.mts, scripts/worker.mts, src/instrumentation.ts, tests/helpers/index.ts, tests/pipeline.test.ts · **Depends on:** src/lib/jobs/index.ts

Data in: Arguments from scripts/analyze.mts, scripts/benchmark.mts, scripts/verify-ai.mts and 4 more · **Data out:** Return values to callers

src/lib/retrieval/index.ts

Purpose. src/lib/retrieval/index.ts is a service-layer TypeScript file in the Search & Retrieval area. It declares 10 functions, 6 interfaces, 1 type, 1 constant, 1 variable. Its documentation says: "Hybrid retrieval: lexical BM25, structural graph expansion, importance prior, optional semantic similarity."

Key symbols: search(), SearchFilters(), queryTerms(), invalidateIndex(), retrieveContext(), searchCode(), buildSearchIndex(), indexFindingsAndDocs()

How it operates. Exposes 12 exported symbol(s) that other modules call; internal helpers are 7.

Called by: scripts/benchmark.mts, src/app/api/projects/[id]/search/route.ts, src/lib/ask/index.ts, src/lib/docs/ai.ts, src/lib/docs/index.ts, src/lib/jobs/index.ts, tests/analyzers.test.ts, tests/map.test.ts · **Depends on:** src/lib/db/client.ts, src/lib/db/schema.ts, src/lib/ingest/store.ts, src/lib/ai/index.ts, src/lib/ai/prompt.ts, src/lib/util/text.ts, drizzle-orm

Data in: Records read from files, symbols, findings, projects, indexEntries and 1 more · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 53.

src/components/download.tsx

Purpose. src/components/download.tsx is a user-interface TypeScript file in the User Interface area. It declares 2 components, 2 functions, 1 constant, 1 type. Its documentation says: "One control on every result view: view or download that result as PDF, Word or Markdown."

Responsibilities: component DownloadMenu; component FormatRow

Key symbols: DownloadMenu(), FormatRow(), exportUrl(), saveReport(), FORMAT_INFO(), Format()

How it operates. Exposes 5 exported symbol(s) that other modules call; internal helpers are 1.

Called by: src/app/p/[id]/architecture/page.tsx, src/app/p/[id]/ask/page.tsx, src/app/p/[id]/explain/page.tsx, src/app/p/[id]/layout.tsx, src/app/p/[id]/map/page.tsx, src/app/p/[id]/page.tsx, src/app/p/[id]/reports/page.tsx, src/app/p/[id]/review/page.tsx, src/app/page.tsx · **Depends on:** src/lib/export/scopes.ts, react

Data in: Arguments from src/app/p/[id]/architecture/page.tsx, src/app/p/[id]/ask/page.tsx, src/app/p/[id]/explain/page.tsx and 6 more · **Data out:** Return values to callers

src/lib/util/ids.ts

Purpose. src/lib/util/ids.ts is a TypeScript file in the Util area. It declares 3 functions.

Key symbols: sha256(), newId(), shortHash()

How it operates. Exposes 3 exported symbol(s) that other modules call; internal helpers are 0.

Called by: src/lib/ask/index.ts, src/lib/discover/index.ts, src/lib/docs/cache.ts, src/lib/graph/build.ts, src/lib/ingest/normalize.ts, src/lib/ingest/store.ts, src/lib/jobs/index.ts, src/lib/review/index.ts · **Depends on:** nothing indexed

Data in: Arguments from src/lib/ask/index.ts, src/lib/discover/index.ts, src/lib/docs/cache.ts and 5 more · **Data out:** Return values to callers

src/lib/ingest/credentials.ts

Purpose. src/lib/ingest/credentials.ts is a service-layer TypeScript file in the Content Ingestion area. It declares 4 functions, 1 variable. Its documentation says: "Scrub anything that looks like a token from a string before it is logged."

Key symbols: scrubToken(), encryptSecret(), key(), decryptSecret(), processKey()

How it operates. Exposes 3 exported symbol(s) that other modules call; internal helpers are 2.

Called by: src/app/api/projects/[id]/analyze/route.ts, src/lib/api.ts, src/lib/ingest/github.ts, src/lib/ingest/store.ts, src/lib/jobs/index.ts, tests/secrets.test.ts · **Depends on:** src/lib/config.ts

Data in: Arguments from `src/app/api/projects/[id]/analyze/route.ts`, `src/lib/api.ts`, `src/lib/ingest/github.ts` and 3 more · **Data out:** Return values to callers

`src/lib/ingest/zip.ts`

Purpose. `src/lib/ingest/zip.ts` is a service-layer TypeScript file in the Content Ingestion area. It declares 1 function, 1 interface. Its documentation says: "Extract a ZIP buffer defensively: entry count limits, per-file and total size limits, compression ratio checks (zip bombs), symlink rejection and path traversal rejection. Nothing is written to disk."

Key symbols: `extractZip()`, `ZipExtractResult()`

How it operates. Exposes 2 exported symbol(s) that other modules call; internal helpers are 0.

Called by: `src/lib/ingest/github.ts`, `src/lib/ingest/index.ts`, `src/lib/ingest/upload.ts`, `tests/exports.test.ts`, `tests/ingest.test.ts` · **Depends on:** `src/lib/config.ts`, `src/lib/util/errors.ts`, `src/lib/ingest/paths.ts`, `src/lib/ingest/types.ts`, `yauzl`

Data in: Arguments from `src/lib/ingest/github.ts`, `src/lib/ingest/index.ts`, `src/lib/ingest/upload.ts` and 2 more ·

Data out: Persists to files

Engineering notes: Contains a function with cyclomatic complexity 21.

`src/lib/ask/index.ts`

Purpose. `src/lib/ask/index.ts` is a service-layer TypeScript file in the Ask area. It declares 9 functions, 2 interfaces, 1 variable. Its documentation says: "-----"

Key symbols: `askRepository()`, `relevantHits()`, `recentQuestions()`, `Answer()`, `Citation()`, `cite()`, `withSnippets()`, `findSymbolByName()`

How it operates. Exposes 5 exported symbol(s) that other modules call; internal helpers are 7.

Called by: `scripts/benchmark.mts`, `scripts/verify-ai.mts`, `src/app/api/projects/[id]/ask/route.ts`, `src/app/p/[id]/ask/page.tsx`, `tests/exports.test.ts`, `tests/map.test.ts` · **Depends on:** `src/lib/db/client.ts`, `src/lib/discover/types.ts`, `src/lib/ai/index.ts`, `src/lib/ingest/store.ts`, `src/lib/map/index.ts`, `src/lib/retrieval/index.ts`, `src/lib/util/ids.ts`, `src/lib/util/text.ts`, `drizzle-orm`, `zod`

Data in: Records read from files, symbols, projects, questions · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 42.

`src/lib/review/types.ts`

Purpose. `src/lib/review/types.ts` is a TypeScript file in the Review area. It declares 1 interface, 4 types, 2 constants.

Key symbols: `FindingDraft()`, `Severity()`, `Confidence()`, `Category()`, `CATEGORY_PREFIX()`, `SEVERITY_ORDER()`, `Verification()`

How it operates. Exposes 7 exported symbol(s) that other modules call; internal helpers are 0.

Called by: `src/lib/analysis/analyzers.ts`, `src/lib/analysis/rules.ts`, `src/lib/analysis/structural.ts`, `src/lib/review/ai.ts`, `src/lib/review/index.ts`, `src/lib/review/verify.ts`, `tests/review.test.ts` · **Depends on:** nothing indexed

Data in: Arguments from `src/lib/analysis/analyzers.ts`, `src/lib/analysis/rules.ts`, `src/lib/analysis/structural.ts` and 4 more · **Data out:** Return values to callers

`src/lib/ingest/github.ts`

Purpose. `src/lib/ingest/github.ts` is a service-layer TypeScript file in the Content Ingestion area. It declares 5 functions, 5 interfaces. Its documentation says: "Download the repository snapshot as a zipball and extract it in memory."

Key symbols: `parseGitHubUrl()`, `fetchGitHubMetadata()`, `downloadGitHubSnapshot()`, `GitHubMetadata()`, `GitHubRef()`, `headers()`, `gh()`, `RepoResponse()`

How it operates. Exposes 5 exported symbol(s) that other modules call; internal helpers are 5.

Called by: src/app/api/github/validate/route.ts, src/app/api/projects/[id]/analyze/route.ts, src/app/api/projects/github/route.ts, src/app/page.tsx, src/lib/ingest/index.ts, src/lib/jobs/index.ts, tests/ingest.test.ts · **Depends on:** src/lib/config.ts, src/lib/util/errors.ts, src/lib/ingest/credentials.ts, src/lib/ingest/zip.ts, src/lib/ingest/types.ts

Data in: Arguments from src/app/api/github/validate/route.ts, src/app/api/projects/[id]/analyze/route.ts, src/app/api/projects/github/route.ts and 4 more · **Data out:** Return values to callers

src/lib/export/document.ts

Purpose. src/lib/export/document.ts is a service-layer TypeScript file in the Export area. It declares 6 functions, 2 interfaces, 1 type. Its documentation says: "Parse the report Markdown into a format-neutral document that the PDF and DOCX renderers share."

Key symbols: plain(), parseReport(), outline(), Run(), Block(), ReportDocument(), blocks(), runs()

How it operates. Exposes 6 exported symbol(s) that other modules call; internal helpers are 3.

Called by: scripts/convert.mts, src/lib/export/docx.ts, src/lib/export/index.ts, src/lib/export/pdf.ts, tests/exports.test.ts · **Depends on:** marked

Data in: Arguments from scripts/convert.mts, src/lib/export/docx.ts, src/lib/export/index.ts and 2 more · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 23.

src/lib/ingest/types.ts

Purpose. src/lib/ingest/types.ts is a TypeScript file in the Content Ingestion area. It declares 4 interfaces.

Key symbols: RawFile(), NormalizedFile(), IngestStats(), IngestSource()

How it operates. Exposes 4 exported symbol(s) that other modules call; internal helpers are 0.

Called by: src/lib/ingest/fs.ts, src/lib/ingest/github.ts, src/lib/ingest/index.ts, src/lib/ingest/normalize.ts, src/lib/ingest/store.ts, src/lib/ingest/upload.ts, src/lib/ingest/zip.ts · **Depends on:** nothing indexed

Data in: Arguments from src/lib/ingest/fs.ts, src/lib/ingest/github.ts, src/lib/ingest/index.ts and 4 more · **Data out:** Return values to callers

src/lib/ai/types.ts

Purpose. src/lib/ai/types.ts is a TypeScript file in the AI Processing area. It declares 4 interfaces, 2 classes, 4 methods. Its documentation says: "Provider abstraction. Repository intelligence code depends only on this interface, never on a specific vendor SDK."

Key symbols: AIProvider(), AIResponseError, AnalysisRequest(), AnalysisResult(), AIUnavailableError, AnalysisUsage(), AIProvider.analyze(), AIProvider.embed()

How it operates. Exposes 6 exported symbol(s) that other modules call; internal helpers are 4.

Called by: src/lib/ai/anthropic.ts, src/lib/ai/index.ts, src/lib/ai/openai.ts · **Depends on:** zod

Data in: Arguments from src/lib/ai/anthropic.ts, src/lib/ai/index.ts, src/lib/ai/openai.ts · **Data out:** Return values to callers

src/lib/ingest/secrets.ts

Purpose. src/lib/ingest/secrets.ts is a service-layer TypeScript file in the Content Ingestion area. It declares 4 functions, 1 interface, 4 constants. Its documentation says: "Replace likely secret values so they never reach a model or a log."

Key symbols: redactSecrets(), detectSecrets(), SecretMatch(), scan(), redact(), PATTERNS(), PLACEHOLDER(), TEMPLATE_FILE()

How it operates. Exposes 3 exported symbol(s) that other modules call; internal helpers are 6.

Called by: src/lib/ai/prompt.ts, src/lib/discover/index.ts, src/lib/ingest/index.ts, tests/secrets.test.ts · **Depends on:** nothing indexed

Data in: Arguments from src/lib/ai/prompt.ts, src/lib/discover/index.ts, src/lib/ingest/index.ts and 1 more · **Data**

out: Return values to callers

src/lib/export/index.ts

Purpose. src/lib/export/index.ts is a service-layer TypeScript file in the Export area. It declares 5 functions, 3 constants, 1 type, 1 interface, 1 variable. Its documentation says: "Produce one report file. PDF and Word are generated from the same Markdown the app shows, so all formats agree."

Key symbols: exportFile(), isFormat(), buildHtml(), buildJson(), FORMATS(), ExportFormat(), CONTENT_TYPES(), ExportResult()

How it operates. Exposes 8 exported symbol(s) that other modules call; internal helpers are 3.

Called by: scripts/analyze.mts, scripts/benchmark.mts, src/app/api/projects/[id]/export/route.ts, tests/exports.test.ts, tests/map.test.ts · **Depends on:** src/lib/db/client.ts, src/lib/export/document.ts, src/lib/export/docx.ts, src/lib/export/html.ts, src/lib/export/markdown.ts, src/lib/export/pdf.ts, drizzle-orm

Data in: Records read from files, symbols, relationships · **Data out:** Return values to callers

src/lib/ai/anthropic.ts

Purpose. src/lib/ai/anthropic.ts is a TypeScript file in the AI Processing area. It declares 1 service, 4 methods, 1 interface. Its documentation says: "Anthropic provider using structured outputs. The response is parsed against the request's Zod schema by the SDK; a failed parse or truncated response is retried once, never more. On Fable/Opus 5 tier "

Responsibilities: service AnthropicProvider

Key symbols: AnthropicProvider, AnthropicProvider.call(), AnthropicProvider.analyze(), AnthropicProvider.ping(), ParsedMessage(), AnthropicProvider.constructor()

How it operates. Exposes 1 exported symbol(s) that other modules call; internal helpers are 5.

Called by: src/lib/ai/index.ts, tests/provider.test.ts · **Depends on:** src/lib/config.ts, src/lib/ai/types.ts, @anthropic-ai/sdk

Data in: Arguments from src/lib/ai/index.ts, tests/provider.test.ts · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 20.

src/lib/ai/openai.ts

Purpose. src/lib/ai/openai.ts is a TypeScript file in the AI Processing area. It declares 1 service, 4 methods. Its documentation says: "Provider for any OpenAI-compatible chat completions endpoint (OpenAI, Azure gateways, vLLM, Ollama, ...)."

Responsibilities: service OpenAICompatibleProvider

Key symbols: OpenAICompatibleProvider, OpenAICompatibleProvider.post(), OpenAICompatibleProvider.constructor(), OpenAICompatibleProvider.analyze(), OpenAICompatibleProvider.embed()

How it operates. Exposes 1 exported symbol(s) that other modules call; internal helpers are 4.

Called by: src/lib/ai/index.ts, tests/provider.test.ts · **Depends on:** src/lib/config.ts, src/lib/ai/types.ts, zod

Data in: Arguments from src/lib/ai/index.ts, tests/provider.test.ts · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 17.

src/lib/ingest/paths.ts

Purpose. src/lib/ingest/paths.ts is a service-layer TypeScript file in the Content Ingestion area. It declares 2 functions. Its documentation says: "Normalize an archive/upload path and reject anything that could escape the project root. Returns undefined for entries that should simply be skipped (directories, empty names)."

Key symbols: sanitizeRelativePath(), stripCommonRoot()

How it operates. Exposes 2 exported symbol(s) that other modules call; internal helpers are 0.

Called by: src/lib/ingest/fs.ts, src/lib/ingest/index.ts, src/lib/ingest/upload.ts, src/lib/ingest/zip.ts, tests/ingest.test.ts · **Depends on:** src/lib/util/errors.ts, src/lib/ingest/classify.ts

Data in: Arguments from src/lib/ingest/fs.ts, src/lib/ingest/index.ts, src/lib/ingest/upload.ts and 2 more · **Data out:** Return values to callers

src/lib/parse/types.ts

Purpose. src/lib/parse/types.ts is a service-layer TypeScript file in the Parse area. It declares 1 function, 5 interfaces, 1 type.

Key symbols: emptyParse(), ParsedFile(), SymbolKind(), ParsedSymbol(), ParsedImport(), ParsedCall(), ParsedInheritance()

How it operates. Exposes 7 exported symbol(s) that other modules call; internal helpers are 0.

Called by: src/lib/parse/cache.ts, src/lib/parse/extract.ts, src/lib/parse/index.ts, src/lib/parse/textfallback.ts ·

Depends on: nothing indexed

Data in: Arguments from src/lib/parse/cache.ts, src/lib/parse/extract.ts, src/lib/parse/index.ts and 1 more · **Data**

out: Return values to callers

src/lib/ingest/classify.ts

Purpose. src/lib/ingest/classify.ts is a service-layer TypeScript file in the Content Ingestion area. It declares 2 functions, 13 constants, 1 type, 1 interface.

Key symbols: isDefaultExcludedPath(), DEFAULT_EXCLUDED_DIRS(), classifyPath(), FileClassification(), Classified(), MANIFEST_NAMES(), CONFIG_NAMES(), SCHEMA_HINTS()

How it operates. Exposes 5 exported symbol(s) that other modules call; internal helpers are 12.

Called by: src/app/page.tsx, src/lib/ingest/fs.ts, src/lib/ingest/index.ts, src/lib/ingest/normalize.ts, src/lib/ingest/paths.ts · **Depends on:** src/lib/ingest/languages.ts

Data in: Arguments from src/app/page.tsx, src/lib/ingest/fs.ts, src/lib/ingest/index.ts and 2 more · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 31.

src/lib/analysis/rules.ts

Purpose. src/lib/analysis/rules.ts is a service-layer TypeScript file in the Analysis area. It declares 4 functions, 1 interface, 2 constants. Its documentation says: "Blank the contents of string, template and regex literals and drop trailing // comments so code rules only see code. Line based and deliberately simple: an unterminated template continues on the next "

Key symbols: scanText(), blankLiterals(), Rule(), RULES(), ignored(), toFinding(), JS()

How it operates. Exposes 4 exported symbol(s) that other modules call; internal helpers are 3.

Called by: src/lib/review/index.ts, tests/precision.test.ts, tests/review.test.ts · **Depends on:** src/lib/review/types.ts

Data in: Arguments from src/lib/review/index.ts, tests/precision.test.ts, tests/review.test.ts · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 32.

src/lib/map/legend.ts

Purpose. src/lib/map/legend.ts is a service-layer TypeScript file in the Map area. It declares 3 functions, 2 types, 3 constants. Its documentation says: "Single source of truth for map symbols; used by the UI and by every export."

Key symbols: riskGlyph(), glyphFor(), legendText(), RiskLevel(), NODE_LEGEND(), EDGE_LEGEND(), RISK_LEGEND(), NodeType()

How it operates. Exposes 8 exported symbol(s) that other modules call; internal helpers are 0.

Called by: src/app/p/[id]/map/page.tsx, src/components/graph.tsx, src/components/legend.tsx, src/lib/map/

index.ts · **Depends on:** nothing indexed

Data in: Arguments from src/app/p/[id]/map/page.tsx, src/components/graph.tsx, src/components/legend.tsx and 1 more · **Data out:** Return values to callers

src/lib/ingest/languages.ts

Purpose. src/lib/ingest/languages.ts is a service-layer TypeScript file in the Content Ingestion area. It declares 2 functions, 4 constants. Its documentation says: "Languages considered program source (as opposed to config/docs/data)."

Key symbols: detectLanguage(), SOURCE_LANGUAGES(), extensionOf(), AST_LANGUAGES(), EXT_MAP(), NAME_MAP()

How it operates. Exposes 4 exported symbol(s) that other modules call; internal helpers are 2.

Called by: src/lib/ingest/classify.ts, src/lib/ingest/index.ts, src/lib/ingest/normalize.ts · **Depends on:** nothing indexed

Data in: Arguments from src/lib/ingest/classify.ts, src/lib/ingest/index.ts, src/lib/ingest/normalize.ts · **Data out:** Return values to callers

src/lib/jobs/stages.ts

Purpose. src/lib/jobs/stages.ts is a background-processing TypeScript file in the Background Jobs area. It declares 1 function, 1 constant.

Key symbols: initialStages(), STAGE_DEFS()

How it operates. Exposes 2 exported symbol(s) that other modules call; internal helpers are 0.

Called by: src/lib/jobs/index.ts, tests/pipeline.test.ts · **Depends on:** src/lib/db/schema.ts

Data in: Arguments from src/lib/jobs/index.ts, tests/pipeline.test.ts · **Data out:** Return values to callers

src/lib/export/scopes.ts

Purpose. src/lib/export/scopes.ts is a service-layer TypeScript file in the Export area. It declares 1 type, 1 constant, 1 function. Its documentation says: "Client-safe report metadata (no server imports), shared by the UI and the exporters."

Key symbols: ReportScope(), SCOPES(), isScope()

How it operates. Exposes 3 exported symbol(s) that other modules call; internal helpers are 0.

Called by: src/app/p/[id]/reports/page.tsx, src/components/download.tsx, src/lib/export/markdown.ts · **Depends on:** nothing indexed

Data in: Arguments from src/app/p/[id]/reports/page.tsx, src/components/download.tsx, src/lib/export/markdown.ts · **Data out:** Return values to callers

src/lib/ingest/fs.ts

Purpose. src/lib/ingest/fs.ts is a service-layer TypeScript file in the Content Ingestion area. It declares 2 functions, 1 interface. Its documentation says: "Dependency and build directories (node_modules, dist, .env, ...) are not read by default: they are counted and reported in pruned instead, so a normal project folder does not hit the file limit. Pa"

Key symbols: readDirectory(), PrunedDir(), countFiles()

How it operates. Exposes 2 exported symbol(s) that other modules call; internal helpers are 1.

Called by: scripts/analyze.mts, scripts/verify-ai.mts, tests/ingest.test.ts · **Depends on:** src/lib/config.ts, src/lib/util/errors.ts, src/lib/ingest/classify.ts, src/lib/ingest/paths.ts, src/lib/ingest/types.ts

Data in: Arguments from scripts/analyze.mts, scripts/verify-ai.mts, tests/ingest.test.ts · **Data out:** Persists to files

src/lib/parse/index.ts

Purpose. src/lib/parse/index.ts is a service-layer TypeScript file in the Parse area. It declares 2 functions, 1 constant. Its documentation says: "Parse one file into symbols, imports, calls and relationships. AST parsing is used when a grammar exists; otherwise a documented text fallback runs."

Key symbols: parseFile(), fallback(), MAX_AST_BYTES()

How it operates. Exposes 1 exported symbol(s) that other modules call; internal helpers are 2.

Called by: src/lib/graph/build.ts, tests/parse.test.ts · **Depends on:** src/lib/parse/extract.ts, src/lib/parse/treesitter.ts, src/lib/parse/textfallback.ts, src/lib/parse/types.ts

Data in: Arguments from src/lib/graph/build.ts, tests/parse.test.ts · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 19.

src/lib/ai/errors.ts

Purpose. src/lib/ai/errors.ts is a TypeScript file in the AI Processing area. It declares 1 function. Its documentation says: "Turn provider failures into a short summary plus an actionable hint. Never includes credentials."

Key symbols: explainAIError()

How it operates. Exposes 1 exported symbol(s) that other modules call; internal helpers are 0.

Called by: src/lib/ai/index.ts, tests/provider.test.ts · **Depends on:** nothing indexed

Data in: Arguments from src/lib/ai/index.ts, tests/provider.test.ts · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 28.

src/lib/export/html.ts

Purpose. src/lib/export/html.ts is a service-layer TypeScript file in the Export area. It declares 3 functions, 1 constant. Its documentation says: "Converts the report's own Markdown subset (headings, tables, fences, lists, paragraphs) to HTML."

Key symbols: markdownToHtml(), buildHtmlDocument(), inline(), MERMAID_ONLOAD()

How it operates. Exposes 2 exported symbol(s) that other modules call; internal helpers are 2.

Called by: src/lib/export/index.ts, tests/map.test.ts · **Depends on:** src/lib/util/text.ts

Data in: Arguments from src/lib/export/index.ts, tests/map.test.ts · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 23.

src/lib/util/concurrency.ts

Purpose. src/lib/util/concurrency.ts is a TypeScript file in the Util area. It declares 1 function. Its documentation says: "Run tasks with bounded concurrency, preserving result order."

Key symbols: mapLimit()

How it operates. Exposes 1 exported symbol(s) that other modules call; internal helpers are 0.

Called by: src/lib/docs/ai.ts, src/lib/review/ai.ts, src/lib/review/verify.ts · **Depends on:** nothing indexed

Data in: Arguments from src/lib/docs/ai.ts, src/lib/review/ai.ts, src/lib/review/verify.ts · **Data out:** Return values to callers

src/lib/discover/catalog.ts

Purpose. src/lib/discover/catalog.ts is a TypeScript file in the Discover area. It declares 4 constants, 2 interfaces. Its documentation says: "Directory / keyword hints for functional areas."

Key symbols: AREA_HINTS(), SERVICE_SIGNATURES(), SENSITIVE_ENV(), FRAMEWORK_SIGNATURES(), ServiceSignature(), FrameworkSignature()

How it operates. Exposes 6 exported symbol(s) that other modules call; internal helpers are 0.

Called by: src/lib/discover/detect.ts, src/lib/discover/index.ts, src/lib/map/index.ts · **Depends on:** nothing indexed

Data in: Arguments from `src/lib/discover/detect.ts`, `src/lib/discover/index.ts`, `src/lib/map/index.ts` · **Data out:** Return values to callers

`.env.example`

Purpose. `.env.example` is a configuration Dotenv file in the Configuration area.

How it operates. No executable symbols were extracted.

Called by: `README.md` · **Depends on:** nothing indexed

Data in: Arguments from `README.md` · **Data out:** Return values to callers

Engineering notes: Parsed with text-based heuristics, so symbol data is approximate.

`src/lib/review/patch.ts`

Purpose. `src/lib/review/patch.ts` is a service-layer TypeScript file in the Review area. It declares 2 functions, 2 interfaces. Its documentation says: "Minimal unified-diff applier used to validate AI-suggested patches against the real file before they are shown. A patch that does not apply cleanly is discarded rather than presented as fact."

Key symbols: `applyUnifiedDiff()`, `PatchResult()`, `parseHunks()`, `Hunk()`

How it operates. Exposes 2 exported symbol(s) that other modules call; internal helpers are 2.

Called by: `src/lib/review/verify.ts`, `tests/review.test.ts` · **Depends on:** nothing indexed

Data in: Arguments from `src/lib/review/verify.ts`, `tests/review.test.ts` · **Data out:** Return values to callers

`src/lib/review/verify.ts`

Purpose. `src/lib/review/verify.ts` is a service-layer TypeScript file in the Review area. It declares 8 functions, 1 interface, 1 variable, 1 constant. Its documentation says: "Merge duplicate findings. Two static findings are duplicates only when they carry the same title; an AI finding duplicates another finding when it covers the same lines or is worded very similarly nea"

Key symbols: `dedupe()`, `verifyDeterministic()`, `assignCodes()`, `verifyWithAI()`, `VerifyContext()`, `titleTokens()`, `similarity()`, `norm()`

How it operates. Exposes 5 exported symbol(s) that other modules call; internal helpers are 6.

Called by: `src/lib/review/index.ts`, `tests/review.test.ts` · **Depends on:** `src/lib/config.ts`, `src/lib/db/schema.ts`, `src/lib/graph/build.ts`, `src/lib/ai/index.ts`, `src/lib/util/concurrency.ts`, `src/lib/util/text.ts`, `src/lib/review/patch.ts`, `src/lib/review/types.ts`, `zod`

Data in: Records read from symbols · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 37.

`src/lib/ingest/upload.ts`

Purpose. `src/lib/ingest/upload.ts` is a service-layer TypeScript file in the Content Ingestion area. It declares 1 function, 1 type, 2 interfaces. Its documentation says: "Validate and store uploaded content as a new project. Throws `AppError` with actionable messages."

Key symbols: `ingestUpload()`, `UploadMode()`, `UploadedItem()`, `IngestOutcome()`

How it operates. Exposes 4 exported symbol(s) that other modules call; internal helpers are 0.

Called by: `src/app/api/projects/upload/route.ts`, `tests/ingest.test.ts` · **Depends on:** `src/lib/config.ts`, `src/lib/util/errors.ts`, `src/lib/ingest/normalize.ts`, `src/lib/ingest/paths.ts`, `src/lib/ingest/store.ts`, `src/lib/ingest/types.ts`, `src/lib/ingest/zip.ts`

Data in: Records read from files · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 19.

src/lib/analysis/analyzers.ts

Purpose. src/lib/analysis/analyzers.ts is a service-layer TypeScript file in the Analysis area. It declares 7 functions, 1 interface, 1 constant. Its documentation says: "-----"

Key symbols: runTypeScriptSyntax(), runEslint(), runPython(), runGo(), AnalyzerStatus(), run(), writeTemp(), sev()

How it operates. Exposes 5 exported symbol(s) that other modules call; internal helpers are 4.

Called by: src/lib/review/index.ts, tests/analyzers.test.ts · **Depends on:** src/lib/config.ts, src/lib/graph/build.ts, src/lib/review/types.ts, typescript, eslint, @typescript-eslint/parser

Data in: Records read from files · **Data out:** Persists to findings

Engineering notes: Contains a function with cyclomatic complexity 29.

src/lib/docs/ai.ts

Purpose. src/lib/docs/ai.ts is a service-layer TypeScript file in the Documentation area. It declares 4 functions, 1 interface, 7 variables, 1 constant. Its documentation says: "Parse "path:12-30" style citations and keep only those that point at real lines."

Key symbols: validateEvidence(), crossCheckDataClaims(), enhanceWithAI(), AiDocOptions(), wrapFacts(), Ev(), StatementSchema(), SymbolBatchSchema()

How it operates. Exposes 4 exported symbol(s) that other modules call; internal helpers are 9.

Called by: src/lib/docs/index.ts, tests/ai.test.ts · **Depends on:** src/lib/config.ts, src/lib/db/schema.ts, src/lib/graph/build.ts, src/lib/ai/index.ts, src/lib/retrieval/index.ts, src/lib/util/concurrency.ts, src/lib/docs/cache.ts, src/lib/util/text.ts, src/lib/docs/deterministic.ts, src/lib/docs/types.ts, zod

Data in: Records read from files, findings · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 100.

src/lib/review/ai.ts

Purpose. src/lib/review/ai.ts is a service-layer TypeScript file in the Review area. It declares 4 functions, 1 variable, 1 type, 2 constants, 3 interfaces. Its documentation says: "Choose the files worth an AI review, ordered by importance and risk."

Key symbols: runAiReview(), selectReviewFiles(), AiFindingSchema(), AiFinding(), REVIEW_PASSES(), ReviewUnit(), AiReviewResult(), fileBrief()

How it operates. Exposes 7 exported symbol(s) that other modules call; internal helpers are 4.

Called by: src/lib/review/index.ts, tests/review.test.ts · **Depends on:** src/lib/config.ts, src/lib/db/client.ts, src/lib/db/schema.ts, src/lib/discover/types.ts, src/lib/graph/build.ts, src/lib/ai/index.ts, src/lib/util/concurrency.ts, src/lib/util/text.ts, src/lib/review/types.ts, drizzle-orm, zod

Data in: Records read from symbols, relationships · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 26.

src/lib/ai/prompt.ts

Purpose. src/lib/ai/prompt.ts is a TypeScript file in the AI Processing area. It declares 4 functions, 1 interface, 3 constants. Its documentation says: "Remove our own delimiters from repository text so a hostile file cannot close the untrusted block early."

Key symbols: neutralizeDelimiters(), untrusted(), renderEvidence(), EvidenceItem(), SAFETY_PREAMBLE(), redactForModel(), OPEN_TAG(), CLOSE_TAG()

How it operates. Exposes 6 exported symbol(s) that other modules call; internal helpers are 2.

Called by: src/lib/ai/index.ts, src/lib/retrieval/index.ts · **Depends on:** src/lib/ingest/secrets.ts

Data in: Arguments from src/lib/ai/index.ts, src/lib/retrieval/index.ts · **Data out:** Return values to callers

docker-compose.yml

Purpose. docker-compose.yml is an infrastructure YAML file in the Infrastructure & Deployment area. It declares 2 services.

Responsibilities: service app; service brody-data

Key symbols: services.app, services.brody-data

How it operates. Exposes 2 exported symbol(s) that other modules call; internal helpers are 0.

Called by: no indexed importers · **Depends on:** nothing indexed

Data in: n/a · **Data out:** Return values to callers

Engineering notes: Parsed with text-based heuristics, so symbol data is approximate.

src/lib/parse/treesitter.ts

Purpose. src/lib/parse/treesitter.ts is a service-layer TypeScript file in the Parse area. It declares 5 functions, 1 constant, 1 interface, 1 variable, 1 method. Its documentation says: "Map of app language name -> grammar package + wasm file."

Key symbols: loadLanguage(), grammarKeyFor(), parseWithTreeSitter(), grammarPath(), ensureInit(), GRAMMARS(), TsHolder(), holder()

How it operates. Exposes 3 exported symbol(s) that other modules call; internal helpers are 6.

Called by: src/lib/parse/extract.ts, src/lib/parse/index.ts · **Depends on:** web-tree-sitter

Data in: Arguments from src/lib/parse/extract.ts, src/lib/parse/index.ts · **Data out:** Return values to callers

src/components/graph.tsx

Purpose. src/components/graph.tsx is a user-interface TypeScript file in the User Interface area. It declares 1 function, 2 components, 1 interface, 2 constants, 1 variable. Its documentation says: "A read-only graph canvas for embedding in other views. Selecting a node calls onSelect."

Responsibilities: component GraphCanvas; component MapNode

Key symbols: layout(), GraphCanvas(), NodeData(), RISK_COLOR(), nodeTypes(), TYPE_SHAPE(), MapNode()

How it operates. Exposes 5 exported symbol(s) that other modules call; internal helpers are 2.

Called by: src/app/p/[id]/architecture/page.tsx, src/app/p/[id]/map/page.tsx · **Depends on:** src/lib/map/index.ts, src/lib/map/legend.ts, @xyflow/react, @dagrejs/dagre, react

Data in: Arguments from src/app/p/[id]/architecture/page.tsx, src/app/p/[id]/map/page.tsx · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 16.

src/lib/export/docx.ts

Purpose. src/lib/export/docx.ts is a service-layer TypeScript file in the Export area. It declares 3 functions, 4 constants. Its documentation says: "Render the report as a Word document."

Key symbols: renderDocx(), render(), children(), FONT(), MONO(), LINK(), HEADINGS()

How it operates. Exposes 1 exported symbol(s) that other modules call; internal helpers are 6.

Called by: scripts/convert.mts, src/lib/export/index.ts · **Depends on:** src/lib/export/document.ts, docx

Data in: Arguments from scripts/convert.mts, src/lib/export/index.ts · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 23.

src/lib/export/pdf.ts

Purpose. src/lib/export/pdf.ts is a service-layer TypeScript file in the Export area. It declares 14 functions, 2 methods, 6 constants, 1 type, 1 variable, 1 interface. Its documentation says: "Render the report as a PDF with cover, contents with page numbers, bookmarks and a running footer."

Key symbols: `renderPdf()`, `bottom()`, `coverage.hasGlyphForCodePoint()`, `build()`, `covers()`, `withFallback()`, `writeRuns()`, `setup()`

How it operates. Exposes 1 exported symbol(s) that other modules call; internal helpers are 24.

Called by: `scripts/convert.mts`, `src/lib/export/index.ts` · **Depends on:** `src/lib/export/document.ts`, `pdfkit`, `fontkit`

Data in: Arguments from `scripts/convert.mts`, `src/lib/export/index.ts` · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 25.

`src/components/legend.tsx`

Purpose. `src/components/legend.tsx` is a user-interface TypeScript file in the User Interface area. It declares 1 component. Its documentation says: "The legend is identical everywhere: glyphs, line styles and risk marks, never colour alone."

Responsibilities: component Legend

Key symbols: `Legend()`

How it operates. Exposes 1 exported symbol(s) that other modules call; internal helpers are 0.

Called by: `src/app/p/[id]/architecture/page.tsx`, `src/app/p/[id]/map/page.tsx` · **Depends on:** `src/lib/map/legend.ts`

Data in: Arguments from `src/app/p/[id]/architecture/page.tsx`, `src/app/p/[id]/map/page.tsx` · **Data out:** Return values to callers

`src/lib/docs/deterministic.ts`

Purpose. `src/lib/docs/deterministic.ts` is a service-layer TypeScript file in the Documentation area. It declares 11 functions, 1 interface.

Key symbols: `readmeSummary()`, `buildSymbolDocs()`, `buildFileDocs()`, `buildAreaDocs()`, `buildFlowDocs()`, `buildReport()`, `DocInputs()`, `loc()`

How it operates. Exposes 7 exported symbol(s) that other modules call; internal helpers are 5.

Called by: `src/lib/docs/ai.ts`, `src/lib/docs/index.ts` · **Depends on:** `src/lib/db/schema.ts`, `src/lib/discover/types.ts`, `src/lib/graph/build.ts`, `src/lib/util/text.ts`, `src/lib/docs/types.ts`

Data in: Records read from files, symbols, findings · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 73.

`src/app/api/github/validate/route.ts`

Purpose. `src/app/api/github/validate/route.ts` is an API layer TypeScript file in the API Layer area. It handles POST `/api/github/validate`. It declares 2 variables, 1 function.

Responsibilities: Serve POST `/api/github/validate`

Key symbols: `dynamic()`, `POST()`, `Body()`

How it operates. Requests matching its routes are dispatched to POST (route handler).

Called by: `tests/api.test.ts` · **Depends on:** `src/lib/api.ts`, `src/lib/ingest/github.ts`, `src/lib/util/errors.ts`, `zod`

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

`src/app/api/jobs/[id]/cancel/route.ts`

Purpose. `src/app/api/jobs/[id]/cancel/route.ts` is an API layer TypeScript file in the Background Jobs area. It handles POST `/api/jobs/:id/cancel`. It declares 1 variable, 1 function, 1 type.

Responsibilities: Serve POST `/api/jobs/:id/cancel`

Key symbols: `dynamic()`, `POST()`, `Ctx()`

How it operates. Requests matching its routes are dispatched to POST (route handler).

Called by: `tests/api.test.ts` · **Depends on:** `src/lib/api.ts`, `src/lib/jobs/index.ts`, `src/lib/util/errors.ts`

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

src/app/api/jobs/[id]/route.ts

Purpose. src/app/api/jobs/[id]/route.ts is an API layer TypeScript file in the Background Jobs area. It handles GET /api/jobs/:id. It declares 1 variable, 1 function, 1 type.

Responsibilities: Serve GET /api/jobs/:id

Key symbols: dynamic(), GET(), Ctx()

How it operates. Requests matching its routes are dispatched to GET (route handler).

Called by: tests/api.test.ts · **Depends on:** src/lib/api.ts, src/lib/jobs/index.ts, src/lib/util/errors.ts

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

src/app/api/projects/[id]/architecture/route.ts

Purpose. src/app/api/projects/[id]/architecture/route.ts is an API layer TypeScript file in the API Layer area. It handles GET /api/projects/:id/architecture. It declares 1 variable, 1 function, 1 type.

Responsibilities: Serve GET /api/projects/:id/architecture

Key symbols: dynamic(), GET(), Ctx()

How it operates. Requests matching its routes are dispatched to GET (route handler).

Called by: tests/api.test.ts · **Depends on:** src/lib/api.ts, src/lib/discover/types.ts, src/lib/map/index.ts

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

src/app/api/projects/[id]/ask/route.ts

Purpose. src/app/api/projects/[id]/ask/route.ts is an API layer TypeScript file in the API Layer area. It handles GET /api/projects/:id/ask, POST /api/projects/:id/ask. It declares 3 variables, 2 functions, 1 type.

Responsibilities: Serve GET /api/projects/:id/ask; Serve POST /api/projects/:id/ask

Key symbols: dynamic(), maxDuration(), GET(), POST(), Ctx(), Body()

How it operates. Requests matching its routes are dispatched to GET (route handler), POST (route handler).

Called by: tests/api.test.ts · **Depends on:** src/lib/ask/index.ts, src/lib/api.ts, src/lib/util/errors.ts, zod

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

src/app/api/projects/[id]/export/route.ts

Purpose. src/app/api/projects/[id]/export/route.ts is an API layer TypeScript file in the API Layer area. It handles GET /api/projects/:id/export. It declares 2 variables, 1 function, 1 type. Its documentation says: "GET /api/projects/:id/export?format=pdf|docx|md|html|json|print&scope=full|review|explain|architecture|map|ask&download=0|1 download=0 serves the file inline so the browser can display it (PDF, HTML, "

Responsibilities: Serve GET /api/projects/:id/export

Key symbols: dynamic(), maxDuration(), GET(), Ctx()

How it operates. Requests matching its routes are dispatched to GET (route handler).

Called by: tests/api.test.ts · **Depends on:** src/lib/api.ts, src/lib/export/index.ts, src/lib/util/errors.ts

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

Engineering notes: Contains a function with cyclomatic complexity 16.

src/app/api/projects/[id]/files/content/route.ts

Purpose. src/app/api/projects/[id]/files/content/route.ts is an API layer TypeScript file in the File & Media Storage area. It handles GET /api/projects/:id/files/content. It declares 1 variable, 1 function, 1 type, 1 constant.

Responsibilities: Serve GET /api/projects/:id/files/content

Key symbols: `dynamic()`, `GET()`, `Ctx()`, `MAX_RETURN()`

How it operates. Requests matching its routes are dispatched to GET (route handler).

Called by: `tests/api.test.ts` · **Depends on:** `src/lib/api.ts`, `src/lib/db/client.ts`, `src/lib/docs/types.ts`, `src/lib/ingest/store.ts`, `src/lib/map/index.ts`, `src/lib/util/errors.ts`, `drizzle-orm`

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

Engineering notes: Contains a function with cyclomatic complexity 35.

`src/app/api/projects/[id]/files/route.ts`

Purpose. `src/app/api/projects/[id]/files/route.ts` is an API layer TypeScript file in the File & Media Storage area. It handles GET `/api/projects/:id/files`. It declares 1 variable, 1 function, 1 type.

Responsibilities: Serve GET `/api/projects/:id/files`

Key symbols: `dynamic()`, `GET()`, `Ctx()`

How it operates. Requests matching its routes are dispatched to GET (route handler).

Called by: `tests/api.test.ts` · **Depends on:** `src/lib/api.ts`, `src/lib/db/client.ts`, `src/lib/map/index.ts`, `drizzle-orm`

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

`src/app/api/projects/[id]/findings/route.ts`

Purpose. `src/app/api/projects/[id]/findings/route.ts` is an API layer TypeScript file in the API Layer area. It handles GET `/api/projects/:id/findings`. It declares 1 variable, 1 function, 1 type, 1 constant.

Responsibilities: Serve GET `/api/projects/:id/findings`

Key symbols: `dynamic()`, `GET()`, `Ctx()`, `SEV()`

How it operates. Requests matching its routes are dispatched to GET (route handler).

Called by: `tests/api.test.ts` · **Depends on:** `src/lib/api.ts`, `src/lib/db/client.ts`, `drizzle-orm`

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

Engineering notes: Contains a function with cyclomatic complexity 24.

`src/app/api/projects/[id]/graph/route.ts`

Purpose. `src/app/api/projects/[id]/graph/route.ts` is an API layer TypeScript file in the API Layer area. It handles GET `/api/projects/:id/graph`. It declares 1 variable, 1 function, 1 type. Its documentation says: "type=area (high level) | module (files, optionally filtered by area or directory) | symbol (neighbourhood)."

Responsibilities: Serve GET `/api/projects/:id/graph`

Key symbols: `dynamic()`, `GET()`, `Ctx()`

How it operates. Requests matching its routes are dispatched to GET (route handler).

Called by: `tests/api.test.ts` · **Depends on:** `src/lib/api.ts`, `src/lib/map/index.ts`, `src/lib/util/errors.ts`

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

`src/app/api/projects/[id]/impact/route.ts`

Purpose. `src/app/api/projects/[id]/impact/route.ts` is an API layer TypeScript file in the API Layer area. It handles GET `/api/projects/:id/impact`. It declares 1 variable, 1 function, 1 type.

Responsibilities: Serve GET `/api/projects/:id/impact`

Key symbols: `dynamic()`, `GET()`, `Ctx()`

How it operates. Requests matching its routes are dispatched to GET (route handler).

Called by: `tests/api.test.ts` · **Depends on:** `src/lib/api.ts`, `src/lib/map/index.ts`, `src/lib/util/errors.ts`

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

src/app/api/projects/[id]/overview/route.ts

Purpose. src/app/api/projects/[id]/overview/route.ts is an API layer TypeScript file in the API Layer area. It handles GET /api/projects/:id/overview. It declares 1 variable, 1 function, 1 type.

Responsibilities: Serve GET /api/projects/:id/overview

Key symbols: dynamic(), GET(), Ctx()

How it operates. Requests matching its routes are dispatched to GET (route handler).

Called by: tests/api.test.ts · **Depends on:** src/lib/api.ts, src/lib/ai/index.ts, src/lib/db/client.ts, src/lib/discover/types.ts, src/lib/docs/types.ts, drizzle-orm

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

Engineering notes: Contains a function with cyclomatic complexity 19.

src/app/api/projects/[id]/report/route.ts

Purpose. src/app/api/projects/[id]/report/route.ts is an API layer TypeScript file in the Reporting & Analytics area. It handles GET /api/projects/:id/report. It declares 1 variable, 1 function, 1 type. Its documentation says: "Generated documentation as structured JSON (section, area, file and symbol level)."

Responsibilities: Serve GET /api/projects/:id/report

Key symbols: dynamic(), GET(), Ctx()

How it operates. Requests matching its routes are dispatched to GET (route handler).

Called by: tests/api.test.ts · **Depends on:** src/lib/api.ts, src/lib/docs/types.ts

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

src/app/api/projects/[id]/route.ts

Purpose. src/app/api/projects/[id]/route.ts is an API layer TypeScript file in the API Layer area. It handles DELETE /api/projects/:id, GET /api/projects/:id. It declares 1 variable, 2 functions, 1 type.

Responsibilities: Serve DELETE /api/projects/:id; Serve GET /api/projects/:id

Key symbols: dynamic(), GET(), DELETE(), Ctx()

How it operates. Requests matching its routes are dispatched to DELETE (route handler), GET (route handler).

Called by: tests/api.test.ts · **Depends on:** src/lib/api.ts, src/lib/ingest/store.ts, src/lib/jobs/index.ts

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

src/app/api/projects/[id]/search/route.ts

Purpose. src/app/api/projects/[id]/search/route.ts is an API layer TypeScript file in the Search & Retrieval area. It handles GET /api/projects/:id/search. It declares 1 variable, 1 function, 1 type, 1 constant.

Responsibilities: Serve GET /api/projects/:id/search

Key symbols: dynamic(), GET(), Ctx(), KINDS()

How it operates. Requests matching its routes are dispatched to GET (route handler).

Called by: tests/api.test.ts · **Depends on:** src/lib/api.ts, src/lib/retrieval/index.ts

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

src/app/api/projects/[id]/symbols/[symbolId]/route.ts

Purpose. src/app/api/projects/[id]/symbols/[symbolId]/route.ts is an API layer TypeScript file in the API Layer area. It handles GET /api/projects/:id/symbols/:symbolId. It declares 1 variable, 1 function, 1 type.

Responsibilities: Serve GET /api/projects/:id/symbols/:symbolId

Key symbols: dynamic(), GET(), Ctx()

How it operates. Requests matching its routes are dispatched to GET (route handler).

Called by: tests/api.test.ts · **Depends on:** src/lib/api.ts, src/lib/db/client.ts, src/lib/docs/types.ts, src/lib/ingest/store.ts, src/lib/map/index.ts, src/lib/util/errors.ts, src/lib/util/text.ts, drizzle-orm

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

src/app/api/projects/github/route.ts

Purpose. src/app/api/projects/github/route.ts is an API layer TypeScript file in the API Layer area. It handles POST /api/projects/github. It declares 2 variables, 1 function. Its documentation says: "Validate the repository, create the project and queue analysis. The download happens inside the job so progress and failures persist."

Responsibilities: Serve POST /api/projects/github

Key symbols: dynamic(), POST(), Body()

How it operates. Requests matching its routes are dispatched to POST (route handler).

Called by: tests/api.test.ts · **Depends on:** src/lib/api.ts, src/lib/db/client.ts, src/lib/config.ts, src/lib/ingest/github.ts, src/lib/ingest/store.ts, src/lib/jobs/index.ts, src/lib/util/errors.ts, zod, drizzle-orm

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

src/app/api/projects/route.ts

Purpose. src/app/api/projects/route.ts is an API layer TypeScript file in the API Layer area. It handles GET /api/projects. It declares 1 variable, 1 function.

Responsibilities: Serve GET /api/projects

Key symbols: dynamic(), GET()

How it operates. Requests matching its routes are dispatched to GET (route handler).

Called by: tests/api.test.ts · **Depends on:** src/lib/api.ts, src/lib/db/client.ts, drizzle-orm

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

src/app/api/projects/upload/route.ts

Purpose. src/app/api/projects/upload/route.ts is an API layer TypeScript file in the File & Media Storage area. It handles POST /api/projects/upload. It declares 2 variables, 1 function, 1 constant.

Responsibilities: Serve POST /api/projects/upload

Key symbols: dynamic(), maxDuration(), POST(), MODES()

How it operates. Requests matching its routes are dispatched to POST (route handler).

Called by: tests/api.test.ts · **Depends on:** src/lib/api.ts, src/lib/config.ts, src/lib/db/client.ts, src/lib/ingest/upload.ts, src/lib/jobs/index.ts, src/lib/util/errors.ts, drizzle-orm

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

src/app/api/status/route.ts

Purpose. src/app/api/status/route.ts is an API layer TypeScript file in the API Layer area. It handles GET /api/status. It declares 3 variables, 3 functions. Its documentation says: "Analyzer availability is probed asynchronously and cached for a minute so the event loop is never blocked."

Responsibilities: Serve GET /api/status

Key symbols: dynamic(), GET(), run(), has(), analyzers(), cached()

How it operates. Requests matching its routes are dispatched to GET (route handler).

Called by: tests/api.test.ts · **Depends on:** src/lib/api.ts, src/lib/ai/index.ts, src/lib/config.ts

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

src/lib/export/markdown.ts

Purpose. src/lib/export/markdown.ts is a service-layer TypeScript file in the Export area. It declares 10 functions, 1 interface, 1 constant.

Key symbols: loadReportData(), buildMarkdown(), ReportData(), cell(), cite(), fence(), apiTable(), para()

How it operates. Exposes 3 exported symbol(s) that other modules call; internal helpers are 9.

Called by: src/lib/export/index.ts · **Depends on:** src/lib/db/client.ts, src/lib/db/schema.ts, src/lib/discover/types.ts, src/lib/docs/types.ts, src/lib/map/index.ts, src/lib/util/text.ts, src/lib/export/scopes.ts, drizzle-orm

Data in: Records read from projects, findings, questions, files, symbols · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 67.

src/lib/parse/cache.ts

Purpose. src/lib/parse/cache.ts is a service-layer TypeScript file in the Parse area. It declares 5 functions, 1 constant, 1 interface. Its documentation says: "Bump when extraction logic changes so stale cached parses are ignored."

Key symbols: serializeParsed(), deserializeParsed(), parseCacheKey(), getCachedParse(), putCachedParse(), PARSER_VERSION(), Serialized()

How it operates. Exposes 6 exported symbol(s) that other modules call; internal helpers are 1.

Called by: src/lib/graph/build.ts · **Depends on:** src/lib/db/client.ts, src/lib/parse/types.ts, drizzle-orm

Data in: Records read from parseCache · **Data out:** Return values to callers

src/lib/parse/textfallback.ts

Purpose. src/lib/parse/textfallback.ts is a service-layer TypeScript file in the Parse area. It declares 8 functions. Its documentation says: "SQL: tables, views, functions, indexes."

Key symbols: parseSql(), parsePrisma(), parseGraphQL(), parseMarkdown(), parseConfig(), parseGeneric(), lineNumberAt(), blockEnd()

How it operates. Exposes 6 exported symbol(s) that other modules call; internal helpers are 2.

Called by: src/lib/parse/index.ts · **Depends on:** src/lib/parse/types.ts

Data in: Records read from symbols · **Data out:** Persists to symbols

Engineering notes: Contains a function with cyclomatic complexity 34.

src/lib/graph/resolve.ts

Purpose. src/lib/graph/resolve.ts is a service-layer TypeScript file in the Graph area. It declares 3 functions. Its documentation says: "Resolve an import specifier to a repository file path. Returns undefined when the import refers to an external package or cannot be located."

Key symbols: resolveImport(), readPathAliases(), externalPackageName()

How it operates. Exposes 3 exported symbol(s) that other modules call; internal helpers are 0.

Called by: src/lib/graph/build.ts · **Depends on:** src/lib/db/schema.ts

Data in: Arguments from src/lib/graph/build.ts · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 82.

src/components/project.tsx

Purpose. src/components/project.tsx is a user-interface TypeScript file in the User Interface area. It declares 3 components, 3 interfaces, 1 variable, 1 hook, 2 constants.

Responsibilities: component AnalysisProgress; component SearchBox; component StatusChip

Key symbols: AnalysisProgress(), SearchBox(), StatusChip(), ProjectSummary(), ProjectCtx(),

useProject(), Ctx(), ICON()

How it operates. Exposes 6 exported symbol(s) that other modules call; internal helpers are 4.

Called by: src/app/p/[id]/layout.tsx · **Depends on:** src/lib/client.ts, src/lib/db/schema.ts, src/components/ui.tsx, next, react

Data in: Arguments from src/app/p/[id]/layout.tsx · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 40.

src/lib/discover/detect.ts

Purpose. src/lib/discover/detect.ts is an API layer TypeScript file in the Discover area. It handles GET /path. It declares 14 functions, 4 constants. Its documentation says: "Split the top-level arguments of a call whose "(" is at openIdx, respecting nesting and string literals."

Responsibilities: Serve GET /path

Key symbols: callArgs(), detectDependencies(), detectExternalServices(), detectEnvVars(), detectRoutes(), detectModels(), detectEntryPoints(), detectInfra()

How it operates. Requests matching its routes are dispatched to handler.

Called by: src/lib/discover/index.ts · **Depends on:** src/lib/graph/build.ts, src/lib/db/schema.ts, src/lib/discover/catalog.ts, src/lib/discover/types.ts

Data in: HTTP request parameters, headers and bodies · **Data out:** Persists to files

Engineering notes: Large file (647 lines). Contains a function with cyclomatic complexity 190.

src/components/mermaid.tsx

Purpose. src/components/mermaid.tsx is a user-interface TypeScript file in the User Interface area. It declares 1 component. Its documentation says: "Renders Mermaid source client-side, falling back to the source text if rendering fails."

Responsibilities: component MermaidView

Key symbols: MermaidView()

How it operates. Exposes 1 exported symbol(s) that other modules call; internal helpers are 0.

Called by: src/app/p/[id]/architecture/page.tsx · **Depends on:** react, mermaid

Data in: Arguments from src/app/p/[id]/architecture/page.tsx · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 19.

src/lib/discover/index.ts

Purpose. src/lib/discover/index.ts is a service-layer TypeScript file in the Discover area. It declares 7 functions, 1 constant. Its documentation says: "Run architecture discovery and persist results on the project."

Key symbols: discoverArchitecture(), assignRole(), humanizeDir(), areaFor(), detectTests(), buildFlows(), detectPattern(), ROLE_ORDER()

How it operates. Exposes 2 exported symbol(s) that other modules call; internal helpers are 6.

Called by: src/lib/jobs/index.ts · **Depends on:** src/lib/db/client.ts, src/lib/db/schema.ts, src/lib/graph/build.ts, src/lib/ingest/secrets.ts, src/lib/util/ids.ts, src/lib/discover/catalog.ts, src/lib/discover/detect.ts, src/lib/discover/types.ts, drizzle-orm

Data in: Records read from files, symbols, relationships, projects · **Data out:** Persists to files

Engineering notes: Contains a function with cyclomatic complexity 76.

src/lib/docs/index.ts

Purpose. src/lib/docs/index.ts is a service-layer TypeScript file in the Documentation area. It declares 1 function.

Key symbols: generateDocs()

How it operates. Exposes 1 exported symbol(s) that other modules call; internal helpers are 0.

Called by: src/lib/jobs/index.ts · **Depends on:** src/lib/db/client.ts, src/lib/discover/types.ts, src/lib/graph/build.ts, src/lib/ai/index.ts, src/lib/retrieval/index.ts, src/lib/docs/ai.ts, src/lib/docs/deterministic.ts, src/lib/docs/types.ts, drizzle-orm

Data in: Records read from projects, symbols, relationships, findings · **Data out:** Return values to callers

src/lib/review/index.ts

Purpose. src/lib/review/index.ts is a service-layer TypeScript file in the Review area. It declares 2 functions, 2 interfaces. Its documentation says: "Full review: static + AI candidates -> verification -> dedupe -> persisted findings with stable codes."

Key symbols: runReview(), runStaticReview(), StaticReviewResult(), FullReviewResult()

How it operates. Exposes 4 exported symbol(s) that other modules call; internal helpers are 0.

Called by: src/lib/jobs/index.ts · **Depends on:** src/lib/analysis/analyzers.ts, src/lib/analysis/rules.ts, src/lib/analysis/structural.ts, src/lib/config.ts, src/lib/db/client.ts, src/lib/discover/types.ts, src/lib/graph/build.ts, src/lib/ai/index.ts, src/lib/util/ids.ts, src/lib/review/ai.ts, src/lib/review/verify.ts, src/lib/review/types.ts, drizzle-orm

Data in: Records read from symbols, relationships · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 22.

src/lib/analysis/structural.ts

Purpose. src/lib/analysis/structural.ts is a service-layer TypeScript file in the Analysis area. It declares 1 function, 1 type. Its documentation says: "Deterministic findings derived from the repository model, not from AI."

Key symbols: structuralFindings(), Draft()

How it operates. Exposes 1 exported symbol(s) that other modules call; internal helpers are 1.

Called by: src/lib/review/index.ts · **Depends on:** src/lib/graph/build.ts, src/lib/db/schema.ts, src/lib/discover/types.ts, src/lib/review/types.ts

Data in: Records read from files, symbols · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 72.

src/lib/docs/cache.ts

Purpose. src/lib/docs/cache.ts is a service-layer TypeScript file in the Documentation area. It declares 5 functions, 3 constants. Its documentation says: "Bump when prompts or schemas for explanations change."

Key symbols: fileExplainKey(), symbolExplainKey(), getExplain(), putExplain(), pick(), SYMBOL_CACHED_FIELDS(), FILE_CACHED_FIELDS(), PROMPT_VERSION()

How it operates. Exposes 7 exported symbol(s) that other modules call; internal helpers are 1.

Called by: src/lib/docs/ai.ts · **Depends on:** src/lib/db/client.ts, src/lib/util/ids.ts, src/lib/graph/build.ts, src/lib/docs/types.ts, drizzle-orm

Data in: Records read from explainCache · **Data out:** Return values to callers

.dockerignore

Purpose. .dockerignore is an infrastructure Gitignore file in the Infrastructure & Deployment area.

How it operates. No executable symbols were extracted.

Called by: no indexed importers · **Depends on:** nothing indexed

Data in: n/a · **Data out:** Return values to callers

Engineering notes: Parsed with no structural parser, so symbol data is approximate.

Dockerfile

Purpose. Dockerfile is an infrastructure Dockerfile file in the Infrastructure & Deployment area.

How it operates. No executable symbols were extracted.

Called by: no indexed importers · **Depends on:** nothing indexed

Data in: n/a · **Data out:** Return values to callers

Engineering notes: Parsed with text-based heuristics, so symbol data is approximate.

package.json

Purpose. package.json is a configuration JSON file in the Configuration area.

How it operates. No executable symbols were extracted.

Called by: no indexed importers · **Depends on:** nothing indexed

Data in: n/a · **Data out:** Return values to callers

src/app/globals.css

Purpose. src/app/globals.css is a CSS file in the Application Core area.

How it operates. No executable symbols were extracted.

Called by: src/app/layout.tsx · **Depends on:** tailwindcss

Data in: Arguments from src/app/layout.tsx · **Data out:** Return values to callers

drizzle/0000_robust_hawkeye.sql

Purpose. drizzle/0000_robust_hawkeye.sql is a SQL file in the Data Layer area. It declares 12 tables, 13 properties.

Key symbols: blobs(), credentials(), explain_cache(), files(), files_project_idx(), files_project_path_idx(), findings(), findings_project_idx()

How it operates. Exposes 25 exported symbol(s) that other modules call; internal helpers are 0.

Called by: no indexed importers · **Depends on:** nothing indexed

Data in: n/a · **Data out:** Return values to callers

Engineering notes: Parsed with text-based heuristics, so symbol data is approximate.

src/app/api/projects/[id]/analyze/route.ts

Purpose. src/app/api/projects/[id]/analyze/route.ts is an API layer TypeScript file in the API Layer area. It handles POST /api/projects/:id/analyze. It declares 1 variable, 1 function, 1 type. Its documentation says: "Re-run analysis. For GitHub projects the latest commit of the same ref is fetched; a new commit creates a new project so files can be diffed against the previous run (incremental analysis). An unchang"

Responsibilities: Serve POST /api/projects/:id/analyze

Key symbols: dynamic(), POST(), Ctx()

How it operates. Requests matching its routes are dispatched to POST (route handler).

Called by: no indexed importers · **Depends on:** src/lib/api.ts, src/lib/db/client.ts, src/lib/ingest/credentials.ts, src/lib/ingest/github.ts, src/lib/ingest/store.ts, src/lib/jobs/index.ts, drizzle-orm

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

Engineering notes: No other indexed file imports this one; it may be an entry point, dynamically loaded, or dead code.

src/app/layout.tsx

Purpose. src/app/layout.tsx is an entry-point TypeScript file in the Application Core area. It declares 1 variable, 1 component.

Responsibilities: component RootLayout

Key symbols: metadata(), RootLayout()

How it operates. Exposes 2 exported symbol(s) that other modules call; internal helpers are 0.

Called by: no indexed importers · **Depends on:** src/app/globals.css, next

Data in: n/a · **Data out:** Return values to callers

src/app/p/[id]/architecture/page.tsx

Purpose. src/app/p/[id]/architecture/page.tsx is an API layer TypeScript file in the API Layer area. It handles GET /p/:id/architecture. It declares 1 component, 1 interface.

Responsibilities: Serve GET /p/:id/architecture; component ArchitecturePage

Key symbols: ArchitecturePage(), Resp()

How it operates. Requests matching its routes are dispatched to ArchitecturePage.

Called by: no indexed importers · **Depends on:** src/components/download.tsx, src/components/legend.tsx, src/components/graph.tsx, src/components/mermaid.tsx, src/lib/map/index.ts, src/components/ui.tsx, src/lib/client.ts, src/lib/discover/types.ts, next, react

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

Engineering notes: Contains a function with cyclomatic complexity 50. No other indexed file imports this one; it may be an entry point, dynamically loaded, or dead code.

src/app/p/[id]/ask/page.tsx

Purpose. src/app/p/[id]/ask/page.tsx is an API layer TypeScript file in the API Layer area. It handles GET /p/:id/ask. It declares 2 components, 1 interface, 1 constant.

Responsibilities: Serve GET /p/:id/ask; component AskPage; component Cite

Key symbols: AskPage(), Turn(), SUGGESTIONS(), Cite()

How it operates. Requests matching its routes are dispatched to Cite.

Called by: no indexed importers · **Depends on:** src/components/download.tsx, src/components/ui.tsx, src/lib/client.ts, src/lib/ask/index.ts, next, react

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

Engineering notes: Contains a function with cyclomatic complexity 29. No other indexed file imports this one; it may be an entry point, dynamically loaded, or dead code.

src/app/p/[id]/explain/page.tsx

Purpose. src/app/p/[id]/explain/page.tsx is an API layer TypeScript file in the API Layer area. It handles GET /p/:id/explain. It declares 8 components.

Responsibilities: Serve GET /p/:id/explain; component ExplainPage; component Origin; component Statements; component Section; component DocSectionView

Key symbols: ExplainPage(), Origin(), Statements(), Section(), DocSectionView(), AreaCard(), FileExplain(), SymbolExplain()

How it operates. Requests matching its routes are dispatched to Origin.

Called by: no indexed importers · **Depends on:** src/components/download.tsx, src/components/ui.tsx, src/lib/client.ts, src/lib/discover/types.ts, src/lib/docs/types.ts, next, react

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

Engineering notes: Contains a function with cyclomatic complexity 25. No other indexed file imports this one; it may be an entry point, dynamically loaded, or dead code.

src/app/p/[id]/files/page.tsx

Purpose. src/app/p/[id]/files/page.tsx is an API layer TypeScript file in the File & Media Storage area. It handles GET /p/:id/files. It declares 5 components, 1 hook, 3 interfaces, 4 constants. Its documentation says: "Panel widths survive reloads. This page only renders after client-side data loads, so reading storage during init cannot cause a hydration mismatch."

Responsibilities: Serve GET /p/:id/files; component FilesPage; component Splitter; component Tree; component Code; component List

Key symbols: FilesPage(), usePersisted(), FilesResp(), FileResp(), SymResp(), PRISM(), MAX_LINES(), LIGHT()

How it operates. Requests matching its routes are dispatched to Splitter.

Called by: no indexed importers · **Depends on:** src/components/ui.tsx, src/lib/client.ts, src/lib/docs/types.ts, src/lib/db/schema.ts, src/lib/map/index.ts, prism-react-renderer, next, react

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

Engineering notes: Contains a function with cyclomatic complexity 88. No other indexed file imports this one; it may be an entry point, dynamically loaded, or dead code.

src/app/p/[id]/layout.tsx

Purpose. src/app/p/[id]/layout.tsx is a user-interface TypeScript file in the User Interface area. It declares 1 component, 1 constant.

Responsibilities: component ProjectLayout

Key symbols: ProjectLayout(), NAV()

How it operates. Exposes 1 exported symbol(s) that other modules call; internal helpers are 1.

Called by: no indexed importers · **Depends on:** src/components/download.tsx, src/components/project.tsx, src/components/ui.tsx, src/lib/client.ts, next, react

Data in: n/a · **Data out:** Return values to callers

Engineering notes: Contains a function with cyclomatic complexity 34. No other indexed file imports this one; it may be an entry point, dynamically loaded, or dead code.

src/app/p/[id]/map/page.tsx

Purpose. src/app/p/[id]/map/page.tsx is an API layer TypeScript file in the API Layer area. It handles GET /p/:id/map. It declares 3 components, 1 type, 2 interfaces.

Responsibilities: Serve GET /p/:id/map; component MapPage; component ImpactPanel; component MapInner

Key symbols: MapPage(), Mode(), GraphResp(), ImpactResp(), ImpactPanel(), MapInner()

How it operates. Requests matching its routes are dispatched to ImpactPanel.

Called by: no indexed importers · **Depends on:** src/components/download.tsx, src/components/legend.tsx, src/components/ui.tsx, src/lib/client.ts, src/lib/map/index.ts, src/lib/map/legend.ts, src/components/graph.tsx, @xyflow/react, next, react

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

Engineering notes: Contains a function with cyclomatic complexity 55. No other indexed file imports this one; it may be an entry point, dynamically loaded, or dead code.

src/app/p/[id]/page.tsx

Purpose. src/app/p/[id]/page.tsx is an API layer TypeScript file in the API Layer area. It handles GET /p/:id. It declares 1 component, 1 interface, 1 constant.

Responsibilities: Serve GET /p/:id; component OverviewPage

Key symbols: OverviewPage(), Overview(), SEV()

How it operates. Requests matching its routes are dispatched to OverviewPage.

Called by: no indexed importers · **Depends on:** src/components/ui.tsx, src/components/download.tsx, src/lib/client.ts, src/lib/discover/types.ts, src/lib/docs/types.ts, next

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

Engineering notes: Contains a function with cyclomatic complexity 46. No other indexed file imports this one; it may be an entry point, dynamically loaded, or dead code.

src/app/p/[id]/reports/page.tsx

Purpose. src/app/p/[id]/reports/page.tsx is an API layer TypeScript file in the Reporting & Analytics area. It handles GET /p/:id/reports. It declares 1 component, 2 constants.

Responsibilities: Serve GET /p/:id/reports; component ReportsPage

Key symbols: ReportsPage(), ORDER(), FOR()

How it operates. Requests matching its routes are dispatched to ReportsPage.

Called by: no indexed importers · **Depends on:** src/components/download.tsx, src/lib/export/scopes.ts, next

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

Engineering notes: No other indexed file imports this one; it may be an entry point, dynamically loaded, or dead code.

src/app/p/[id]/review/page.tsx

Purpose. src/app/p/[id]/review/page.tsx is an API layer TypeScript file in the API Layer area. It handles GET /p/:id/review. It declares 3 components, 1 type, 1 interface, 1 constant.

Responsibilities: Serve GET /p/:id/review; component ReviewPage; component Diff; component Facet

Key symbols: ReviewPage(), Finding(), Resp(), SEV(), Diff(), Facet()

How it operates. Requests matching its routes are dispatched to Diff.

Called by: no indexed importers · **Depends on:** src/components/download.tsx, src/components/ui.tsx, src/lib/client.ts, src/lib/db/schema.ts, next, react

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

Engineering notes: Contains a function with cyclomatic complexity 54. No other indexed file imports this one; it may be an entry point, dynamically loaded, or dead code.

src/app/page.tsx

Purpose. src/app/page.tsx is an API layer TypeScript file in the API Layer area. It handles GET /. It declares 2 components, 1 function, 1 type, 2 interfaces.

Responsibilities: Serve GET /; component Home; component LanguageBar

Key symbols: Home(), uploadWithProgress(), Mode(), ProjectListItem(), Status(), LanguageBar()

How it operates. Requests matching its routes are dispatched to uploadWithProgress.

Called by: no indexed importers · **Depends on:** src/lib/client.ts, src/lib/ingest/classify.ts, src/components/download.tsx, src/components/ui.tsx, src/lib/ingest/github.ts, next, react

Data in: HTTP request parameters, headers and bodies · **Data out:** HTTP responses

Engineering notes: Contains a function with cyclomatic complexity 88. No other indexed file imports this one; it may be an entry point, dynamically loaded, or dead code.

src/instrumentation.ts

Purpose. src/instrumentation.ts is an entry-point TypeScript file in the Application Core area. It declares 1 function. Its documentation says: "Start the in-process analysis worker when the Node.js server boots."

Key symbols: register()

How it operates. Exposes 1 exported symbol(s) that other modules call; internal helpers are 0.

Called by: no indexed importers · **Depends on:** src/lib/jobs/worker.ts

Data in: n/a · **Data out:** Return values to callers

src/proxy.ts

Purpose. src/proxy.ts is an entry-point TypeScript file in the Application Core area. It declares 1 function. Its documentation says: "Optional host allow-list. When ALLOWED_HOSTS is set (comma separated hostnames, no port), requests addressed to any other hostname, including localhost, are refused. This pins the app to its intended "

Key symbols: proxy()

How it operates. Exposes 1 exported symbol(s) that other modules call; internal helpers are 0.

Called by: no indexed importers · **Depends on:** next

Data in: n/a · **Data out:** Return values to callers

scripts/analyze.mts

Purpose. scripts/analyze.mts is a service-layer TypeScript file in the Infrastructure & Deployment area. It declares 1 function, 7 variables.

Key symbols: opt(), args(), dir(), out(), name(), job(), done(), n()

How it operates. Exposes 0 exported symbol(s) that other modules call; internal helpers are 8.

Called by: no indexed importers · **Depends on:** src/lib/ai/index.ts, src/lib/db/client.ts, src/lib/export/index.ts, src/lib/ingest/fs.ts, src/lib/ingest/normalize.ts, src/lib/ingest/store.ts, src/lib/jobs/index.ts, src/lib/jobs/worker.ts, drizzle-orm

Data in: Records read from findings · **Data out:** Return values to callers

Engineering notes: No other indexed file imports this one; it may be an entry point, dynamically loaded, or dead code.

scripts/benchmark.mts

Purpose. scripts/benchmark.mts is a service-layer TypeScript file in the Infrastructure & Deployment area. It declares 3 functions, 1 variable.

Key symbols: synth(), ms(), rss(), sizes()

How it operates. Exposes 0 exported symbol(s) that other modules call; internal helpers are 4.

Called by: no indexed importers · **Depends on:** src/lib/db/client.ts, src/lib/export/index.ts, src/lib/ingest/normalize.ts, src/lib/ingest/store.ts, src/lib/jobs/index.ts, src/lib/jobs/worker.ts, src/lib/retrieval/index.ts, src/lib/map/index.ts, src/lib/ask/index.ts

Data in: Records read from symbols · **Data out:** Persists to files

Engineering notes: No other indexed file imports this one; it may be an entry point, dynamically loaded, or dead code.

eslint.config.mjs

Purpose. eslint.config.mjs is a JavaScript file in the Application Core area. It declares 1 variable.

Key symbols: eslintConfig()

How it operates. Exposes 0 exported symbol(s) that other modules call; internal helpers are 1.

Called by: no indexed importers · **Depends on:** eslint, eslint-config-next

Data in: n/a · **Data out:** Return values to callers

Engineering notes: No other indexed file imports this one; it may be an entry point, dynamically loaded, or dead code.

next.config.ts

Purpose. next.config.ts is a service-layer TypeScript file in the Application Core area. It declares 1 variable, 1 method.

Key symbols: nextConfig(), nextConfig.headers()

How it operates. Exposes 0 exported symbol(s) that other modules call; internal helpers are 2.

Called by: no indexed importers · **Depends on:** next

Data in: n/a · **Data out:** Return values to callers

Engineering notes: No other indexed file imports this one; it may be an entry point, dynamically loaded, or dead code.

playwright.config.ts

Purpose. playwright.config.ts is a TypeScript file in the Application Core area. It declares 1 constant, 1 variable.

Key symbols: PORT(), dbPath()

How it operates. Exposes 0 exported symbol(s) that other modules call; internal helpers are 2.

Called by: no indexed importers · **Depends on:** @playwright/test

Data in: n/a · **Data out:** Return values to callers

Engineering notes: No other indexed file imports this one; it may be an entry point, dynamically loaded, or dead code.

postcss.config.mjs

Purpose. postcss.config.mjs is a JavaScript file in the Application Core area. It declares 1 variable.

Key symbols: config()

How it operates. Exposes 0 exported symbol(s) that other modules call; internal helpers are 1.

Called by: no indexed importers · **Depends on:** nothing indexed

Data in: n/a · **Data out:** Return values to callers

Engineering notes: No other indexed file imports this one; it may be an entry point, dynamically loaded, or dead code.

16. Detailed Symbol Explanations

jobs.push (job)

.github/workflows/ci.yml:3-3 · workflow job push

- **Purpose:** Job jobs.push (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.

- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** e2e/performance.spec.ts, noHorizontalOverflow, e2e/styling.spec.ts, synth, scripts/benchmark.mts, POST, ProjectLayout, ImpactPanel, Home, AnalysisProgress
- **Important behavior:** Part of the module's public surface.

text (function)

src/lib/parse/extract.ts:20-24 · function text(n: Node | null | undefined, max = 200): string {

- **Purpose:** Function text (no doc comment; role inferred from name and relationships).
- **Inputs:** n: Node | null | undefined, max = 200
- **Process:** Contains 3 independent execution paths.
- **Outputs:** string
- **Dependencies:** none indexed
- **Used by:** e2e/styling.spec.ts, e2e/workflow.spec.ts, OpenAICompatibleProvider.post, api, projects, credentials, blobs, files, symbols, relationships
- **Important behavior:** Internal to its module.

getDb (function)

src/lib/db/client.ts:48-50 · function getDb(): Db {

- **Purpose:** Return the open database, opening the configured one on first use.
- **Inputs:** none declared
- **Process:** Contains 2 independent execution paths. Delegates to openDatabase.
- **Outputs:** Db
- **Dependencies:** openDatabase
- **Used by:** scripts/analyze.mts, scripts/verify-ai.mts, src/app/api/projects/[id]/analyze/route.ts, src/app/api/projects/[id]/files/content/route.ts, src/app/api/projects/[id]/files/route.ts, src/app/api/projects/[id]/findings/route.ts, src/app/api/projects/[id]/overview/route.ts, src/app/api/projects/[id]/symbols/[symbolId]/route.ts, src/app/api/projects/github/route.ts, src/app/api/projects/route.ts
- **Important behavior:** Part of the module's public surface.

openDatabase (function)

src/lib/db/client.ts:32-45 · function openDatabase(dbPath: string = config.databasePath): Db {

- **Purpose:** Function openDatabase (no doc comment; role inferred from name and relationships).
- **Inputs:** dbPath: string = config.databasePath
- **Process:** Contains 4 independent execution paths. Delegates to migrationsFolder.
- **Outputs:** Db
- **Dependencies:** migrationsFolder
- **Used by:** scripts/benchmark.mts, scripts/verify-ai.mts, tests/helpers/index.ts, getDb, freshDb
- **Important behavior:** Part of the module's public surface.

migrationsFolder (function)

src/lib/db/client.ts:20-30 · function migrationsFolder(): string {

- **Purpose:** Resolve the migrations folder for both source and standalone builds.
- **Inputs:** none declared
- **Process:** Contains 3 independent execution paths.
- **Outputs:** string
- **Dependencies:** none indexed
- **Used by:** openDatabase

- **Important behavior:** Internal to its module.

files (model)

src/lib/db/schema.ts:42-80 · files = sqliteTable(

- **Purpose:** Data model files.
- **Inputs:** none declared
- **Process:** Delegates to text.
- **Outputs:** not declared
- **Dependencies:** text
- **Used by:** synth, fl, GET, FilesPage, runTypeScriptSyntax, runEslint, runPython, runGo, structuralFindings, withSnippets
- **Important behavior:** Part of the module's public surface.

json (function)

src/lib/api.ts:8-10 · function json(data: unknown, status = 200): NextResponse {

- **Purpose:** Function json (no doc comment; role inferred from name and relationships).
- **Inputs:** data: unknown, status = 200
- **Process:** Contains 1 independent execution paths.
- **Outputs:** NextResponse
- **Dependencies:** none indexed
- **Used by:** src/app/api/github/validate/route.ts, src/app/api/jobs/[id]/cancel/route.ts, src/app/api/jobs/[id]/route.ts, src/app/api/projects/[id]/analyze/route.ts, src/app/api/projects/[id]/architecture/route.ts, src/app/api/projects/[id]/ask/route.ts, src/app/api/projects/[id]/files/content/route.ts, src/app/api/projects/[id]/files/route.ts, src/app/api/projects/[id]/findings/route.ts, src/app/api/projects/[id]/graph/route.ts
- **Important behavior:** Part of the module's public surface.

symbols (model)

src/lib/db/schema.ts:82-107 · symbols = sqliteTable(

- **Purpose:** Data model symbols.
- **Inputs:** none declared
- **Process:** Delegates to text.
- **Outputs:** not declared
- **Dependencies:** text
- **Used by:** scripts/benchmark.mts, sym, GET, FilesPage, structuralFindings, findSymbolByName, graphAnswer, detectRoutes, detectEntryPoints, discoverArchitecture
- **Important behavior:** Part of the module's public surface.

AppError (class)

src/lib/util/errors.ts:2-12 · class AppError extends Error {

- **Purpose:** Application error with an HTTP status and a user-facing, actionable message.
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** src/app/api/github/validate/route.ts, src/app/api/jobs/[id]/cancel/route.ts, src/app/api/jobs/[id]/route.ts, src/app/api/projects/[id]/ask/route.ts, src/app/api/projects/[id]/export/route.ts, src/app/api/projects/[id]/files/content/route.ts, src/app/api/projects/[id]/graph/route.ts, src/app/api/projects/[id]/impact/route.ts, src/app/api/projects/[id]/symbols/[symbolId]/route.ts, src/app/api/projects/github/route.ts

- **Important behavior:** Part of the module's public surface.

jobs (model)

src/lib/db/schema.ts:159-179 · jobs = sqliteTable(

- **Purpose:** Data model jobs.
- **Inputs:** none declared
- **Process:** Delegates to text.
- **Outputs:** not declared
- **Dependencies:** text
- **Used by:** projectSummary, enqueueAnalysis, getJob, latestJobForProject, Tracker.isCancelled, claimNextJob, recoverStaleJobs, listJobs, tests/ai.test.ts, tests/pipeline.test.ts
- **Important behavior:** Part of the module's public surface.

findings (model)

src/lib/db/schema.ts:128-157 · findings = sqliteTable(

- **Purpose:** Data model findings.
- **Inputs:** none declared
- **Process:** Delegates to text.
- **Outputs:** not declared
- **Dependencies:** text
- **Used by:** n, findings, scripts/verify-ai.mts, aiFindings, GET, ReviewPage, runTypeScriptSyntax, runEslint, runPython, runGo
- **Important behavior:** Part of the module's public surface.

projects (model)

src/lib/db/schema.ts:4-26 · projects = sqliteTable("projects", {

- **Purpose:** Projects: one row per ingested codebase (a repository or an upload).
- **Inputs:** none declared
- **Process:** Delegates to text.
- **Outputs:** not declared
- **Dependencies:** text
- **Used by:** docs, POST, GET, getProject, askRepository, discoverArchitecture, generateDocs, loadReportData, attachFiles, deleteProject
- **Important behavior:** Part of the module's public surface.

UsageMeter.add (method)

src/lib/ai/index.ts:51-55 · add(u: AnalysisUsage): void {

- **Purpose:** Method UsageMeter.add (no doc comment; role inferred from name and relationships).
- **Inputs:** u: AnalysisUsage
- **Process:** Contains 1 independent execution paths.
- **Outputs:** void
- **Dependencies:** none indexed
- **Used by:** GET, FilesPage, tryAnalyze, writeTemp, detectDependencies, dedupePorts, detectEntryPoints, mergeTableAndOrm, buildFlows, crossCheckDataClaims
- **Important behavior:** Internal to its module.

AIProvider (interface)

src/lib/ai/types.ts:30-36 · interface AIProvider {

- **Purpose:** Provider abstraction. Repository intelligence code depends only on this interface, never on a specific vendor SDK.
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** *src/lib/ai/anthropic.ts*, *src/lib/ai/index.ts*, *src/lib/ai/openai.ts*, AnthropicProvider, OpenAICompatibleProvider, EmbedProvider, MockProvider
- **Important behavior:** Part of the module's public surface.

AnthropicProvider (service)

src/lib/ai/anthropic.ts:21-81 · class AnthropicProvider implements AIProvider {

- **Purpose:** Anthropic provider using structured outputs. The response is parsed against the request's Zod schema by the SDK; a failed parse or truncated response is retried once, never more. On Fable/Opus 5 tier models the server-side refusal fallback is requested by default. If the API rejects those beta param
- **Inputs:** none declared
- **Process:** Contains 20 independent execution paths. Delegates to AIProvider.
- **Outputs:** not declared
- **Dependencies:** AIProvider
- **Used by:** *src/lib/ai/index.ts*, *tests/provider.test.ts*, getAIProvider
- **Important behavior:** Part of the module's public surface.

OpenAICompatibleProvider (service)

src/lib/ai/openai.ts:6-70 · class OpenAICompatibleProvider implements AIProvider {

- **Purpose:** Provider for any OpenAI-compatible chat completions endpoint (OpenAI, Azure gateways, vLLM, Ollama, ...).
- **Inputs:** none declared
- **Process:** Contains 17 independent execution paths. Delegates to AIProvider.
- **Outputs:** not declared
- **Dependencies:** AIProvider
- **Used by:** *src/lib/ai/index.ts*, *tests/provider.test.ts*, getAIProvider
- **Important behavior:** Part of the module's public surface.

blobs (model)

src/lib/db/schema.ts:36-40 · blobs = sqliteTable("blobs", {

- **Purpose:** Content-addressed blob store so identical files across projects share bytes.
- **Inputs:** none declared
- **Process:** Delegates to text.
- **Outputs:** not declared
- **Dependencies:** text
- **Used by:** getFileContent, getFileContents
- **Important behavior:** Part of the module's public surface.

relationships (model)

src/lib/db/schema.ts:110-126 · relationships = sqliteTable(

- **Purpose:** Directed relationships between symbols or files.
- **Inputs:** none declared
- **Process:** Delegates to text.
- **Outputs:** not declared
- **Dependencies:** text
- **Used by:** GET, discoverArchitecture, generateDocs, buildJson, loadModel, search, runAiReview, runReview, tests/pipeline.test.ts
- **Important behavior:** Part of the module's public surface.

questions (model)

src/lib/db/schema.ts:212-222 · questions = sqliteTable(

- **Purpose:** Data model questions.
- **Inputs:** none declared
- **Process:** Delegates to text.
- **Outputs:** not declared
- **Dependencies:** text
- **Used by:** recentQuestions, buildMarkdown
- **Important behavior:** Part of the module's public surface.

explainCache (model)

src/lib/db/schema.ts:189-194 · explainCache = sqliteTable("explain_cache", {

- **Purpose:** AI explanation reuse keyed by content hash plus the hashes of direct dependencies.
- **Inputs:** none declared
- **Process:** Delegates to text.
- **Outputs:** not declared
- **Dependencies:** text
- **Used by:** getExplain
- **Important behavior:** Part of the module's public surface.

fail (function)

src/lib/api.ts:12-17 · function fail(e: unknown): NextResponse {

- **Purpose:** Function fail (no doc comment; role inferred from name and relationships).
- **Inputs:** e: unknown
- **Process:** Contains 5 independent execution paths. Delegates to isAppError, json, scrubToken.
- **Outputs:** NextResponse
- **Dependencies:** isAppError, json, scrubToken
- **Used by:** guard
- **Important behavior:** Part of the module's public surface.

guard (function)

src/lib/api.ts:20-26 · async function guard(fn: () => Promise<Response> | Response): Promise<Response> {

- **Purpose:** Wrap a handler so AppErrors become structured JSON with actionable hints.
- **Inputs:** fn: (
- **Process:** Contains 2 independent execution paths. Delegates to fail.

- **Outputs:** Promise | Response): Promise
- **Dependencies:** fail
- **Used by:** src/app/api/github/validate/route.ts, src/app/api/jobs/[id]/cancel/route.ts, src/app/api/jobs/[id]/route.ts, src/app/api/projects/[id]/analyze/route.ts, src/app/api/projects/[id]/architecture/route.ts, src/app/api/projects/[id]/ask/route.ts, src/app/api/projects/[id]/export/route.ts, src/app/api/projects/[id]/files/content/route.ts, src/app/api/projects/[id]/files/route.ts, src/app/api/projects/[id]/findings/route.ts
- **Important behavior:** Part of the module's public surface.

parseCache (model)

src/lib/db/schema.ts:182-186 · parseCache = sqliteTable("parse_cache", {

- **Purpose:** Parse results keyed by content hash so unchanged files are never re-parsed.
- **Inputs:** none declared
- **Process:** Delegates to text.
- **Outputs:** not declared
- **Dependencies:** text
- **Used by:** getCacheParse
- **Important behavior:** Part of the module's public surface.

indexEntries (model)

src/lib/db/schema.ts:197-210 · indexEntries = sqliteTable(

- **Purpose:** Lexical + semantic index entries for retrieval.
- **Inputs:** none declared
- **Process:** Delegates to text.
- **Outputs:** not declared
- **Dependencies:** text
- **Used by:** load
- **Important behavior:** Part of the module's public surface.

jobs.verify (job)

.github/workflows/ci.yml:8-8 · workflow job verify

- **Purpose:** Job jobs.verify (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** runEslint
- **Important behavior:** Part of the module's public surface.

credentials (model)

src/lib/db/schema.ts:29-33 · credentials = sqliteTable("credentials", {

- **Purpose:** Encrypted repository credentials, session scoped and never logged.
- **Inputs:** none declared
- **Process:** Delegates to text.
- **Outputs:** not declared
- **Dependencies:** text
- **Used by:** POST, processJob
- **Important behavior:** Part of the module's public surface.

jobs.pull_request (job)

.github/workflows/ci.yml:5-5 · workflow job pull_request

- **Purpose:** Job jobs.pull_request (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** no known call sites
- **Important behavior:** Part of the module's public surface.

services.app (service)

docker-compose.yml:3-3 · service app (brody:latest)

- **Purpose:** Service services.app (no doc comment; role inferred from name and relationships).
- **Inputs:** brody:latest
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** no known call sites
- **Important behavior:** Part of the module's public surface.

services.brody-data (service)

docker-compose.yml:28-28 · service brody-data

- **Purpose:** Service services.brody-data (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** no known call sites
- **Important behavior:** Part of the module's public surface.

loadModel (function)

src/lib/map/index.ts:55-104 · function loadModel(projectId: string): Model {

- **Purpose:** Function loadModel (no doc comment; role inferred from name and relationships).
- **Inputs:** projectId: string
- **Process:** Contains 30 independent execution paths. Delegates to getDb, reads projects, reads files, reads symbols, reads relationships and 1 more.
- **Outputs:** Model
- **Dependencies:** getDb, reads projects, reads files, reads symbols, reads relationships, reads findings, jobs.push
- **Used by:** scripts/benchmark.mts, src/lib/ask/index.ts, src/lib/export/markdown.ts, src/lib/jobs/index.ts, tests/map.test.ts, findSymbolByName, graphAnswer, buildMarkdown, processJob, repositoryTree
- **Important behavior:** Part of the module's public surface.

useApi (hook)

src/lib/client.ts:35-69 · function useApi<T>(path: string | null, opts: { pollMs?: number; stop?: (d: T) => boolean; keepPrevious?: boolean } = {}) {

- **Purpose:** Hook useApi (no doc comment; role inferred from name and relationships).

- **Inputs:** path: string | null, opts: { pollMs?: number; stop?: (d: T
- **Process:** Contains 18 independent execution paths. Delegates to keyOf.
- **Outputs:** boolean; keepPrevious?: boolean }
- **Dependencies:** keyOf
- **Used by:** src/app/p/[id]/architecture/page.tsx, src/app/p/[id]/ask/page.tsx, src/app/p/[id]/explain/page.tsx, src/app/p/[id]/files/page.tsx, src/app/p/[id]/layout.tsx, src/app/p/[id]/map/page.tsx, src/app/p/[id]/page.tsx, src/app/p/[id]/review/page.tsx, src/app/page.tsx, src/components/project.tsx
- **Important behavior:** Part of the module's public surface.

plain (function)

src/lib/export/document.ts:117-117 · plain = (rs: Run[]): string => rs.map((r) => r.text).join("")

- **Purpose:** Function plain (no doc comment; role inferred from name and relationships).
- **Inputs:** rs: Run[]
- **Process:** Contains 1 independent execution paths.
- **Outputs:** string
- **Dependencies:** none indexed
- **Used by:** src/lib/export/docx.ts, src/lib/export/pdf.ts, outline, textHeight, drawTable, drawBlocks
- **Important behavior:** Part of the module's public surface.

Tracker.flush (method)

src/lib/jobs/index.ts:70-72 · private flush(extra: Partial<typeof schema.jobs.\$inferInsert> = {}) {

- **Purpose:** Method Tracker.flush (no doc comment; role inferred from name and relationships).
- **Inputs:** extra: Partial<typeof schema.jobs.\$inferInsert> = {}
- **Process:** Contains 1 independent execution paths. Delegates to getDb.
- **Outputs:** not declared
- **Dependencies:** getDb
- **Used by:** Tracker.start, Tracker.detail, Tracker.done, Tracker.skip, Tracker.warn, Tracker.fail, Tracker.note, Tracker.heartbeat
- **Important behavior:** Internal to its module.

scrubToken (function)

src/lib/ingest/credentials.ts:36-38 · function scrubToken(message: string): string {

- **Purpose:** Scrub anything that looks like a token from a string before it is logged.
- **Inputs:** message: string
- **Process:** Contains 1 independent execution paths.
- **Outputs:** string
- **Dependencies:** none indexed
- **Used by:** src/lib/api.ts, src/lib/ingest/github.ts, src/lib/jobs/index.ts, tests/secrets.test.ts, fail, gh, downloadGitHubSnapshot, Tracker.note, processJob
- **Important behavior:** Part of the module's public surface.

getJob (function)

src/lib/jobs/index.ts:36-38 · function getJob(jobId: string): JobRow | undefined {

- **Purpose:** Function getJob (no doc comment; role inferred from name and relationships).
- **Inputs:** jobId: string
- **Process:** Contains 1 independent execution paths. Delegates to reads jobs, getDb.
- **Outputs:** JobRow | undefined

- **Dependencies:** reads jobs, getDb
- **Used by:** scripts/analyze.mts, scripts/benchmark.mts, scripts/verify-ai.mts, src/app/api/jobs/[id]/route.ts, tests/helpers/index.ts, tests/pipeline.test.ts, done, j, GET, requestCancel
- **Important behavior:** Part of the module's public surface.

getAIProvider (function)

src/lib/ai/index.ts:26-39 · function getAIProvider(): AIProvider | null {

- **Purpose:** Function getAIProvider (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Contains 13 independent execution paths. Delegates to AnthropicProvider, OpenAICompatibleProvider.
- **Outputs:** AIProvider | null
- **Dependencies:** AnthropicProvider, OpenAICompatibleProvider
- **Used by:** scripts/analyze.mts, scripts/verify-ai.mts, src/lib/ask/index.ts, src/lib/jobs/index.ts, src/lib/retrieval/index.ts, providerStatus, requireProvider, checkProvider, askRepository, processJob
- **Important behavior:** Part of the module's public surface.

redactSecrets (function)

src/lib/ingest/secrets.ts:65-75 · function redactSecrets(text: string): string {

- **Purpose:** Replace likely secret values so they never reach a model or a log.
- **Inputs:** text: string
- **Process:** Contains 3 independent execution paths.
- **Outputs:** string
- **Dependencies:** none indexed
- **Used by:** src/lib/ai/prompt.ts, tests/secrets.test.ts, untrusted, renderEvidence, redactForModel
- **Important behavior:** Part of the module's public surface.

escapeHtml (function)

src/lib/util/text.ts:39-41 · function escapeHtml(s: string): string {

- **Purpose:** Function escapeHtml (no doc comment; role inferred from name and relationships).
- **Inputs:** s: string
- **Process:** Contains 1 independent execution paths.
- **Outputs:** string
- **Dependencies:** none indexed
- **Used by:** src/lib/export/html.ts, inline, markdownToHtml, buildHtmlDocument
- **Important behavior:** Part of the module's public surface.

getReadyProject (function)

src/lib/api.ts:34-39 · function getReadyProject(id: string): ProjectRow {

- **Purpose:** Function getReadyProject (no doc comment; role inferred from name and relationships).
- **Inputs:** id: string
- **Process:** Contains 3 independent execution paths. Delegates to getProject, AppError.
- **Outputs:** ProjectRow
- **Dependencies:** getProject, AppError
- **Used by:** src/app/api/projects/[id]/architecture/route.ts, src/app/api/projects/[id]/ask/route.ts, src/app/api/projects/[id]/export/route.ts, src/app/api/projects/[id]/files/content/route.ts, src/app/api/projects/[id]/files/route.ts, src/app/api/projects/[id]/findings/route.ts, src/app/api/projects/[id]/graph/route.ts, src/app/api/projects/[id]/impact/route.ts, src/app/api/projects/[id]/report/route.ts, src/app/api/projects/[id]/search/route.ts

- **Important behavior:** Part of the module's public surface.

AIResponseError (class)

src/lib/ai/types.ts:45-50 · class AIResponseError extends Error {

- **Purpose:** Class AIResponseError (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** *src/lib/ai/anthropic.ts*, *src/lib/ai/openai.ts*, *AnthropicProvider.analyze*, *OpenAICompatibleProvider.post*, *OpenAICompatibleProvider.analyze*, *OpenAICompatibleProvider.embed*
- **Important behavior:** Part of the module's public surface.

sha256 (function)

src/lib/util/ids.ts:8-10 · function sha256(input: string | Buffer): string {

- **Purpose:** Function sha256 (no doc comment; role inferred from name and relationships).
- **Inputs:** input: string | Buffer
- **Process:** Contains 1 independent execution paths.
- **Outputs:** string
- **Dependencies:** none indexed
- **Used by:** *src/lib/docs/cache.ts*, *src/lib/ingest/normalize.ts*, *src/lib/ingest/store.ts*, *fileExplainKey*, *normalizeFiles*, *attachFiles*, *shortHash*
- **Important behavior:** Part of the module's public surface.

enqueueAnalysis (function)

src/lib/jobs/index.ts:24-34 · function enqueueAnalysis(projectId: string): JobRow {

- **Purpose:** Function enqueueAnalysis (no doc comment; role inferred from name and relationships).
- **Inputs:** projectId: string
- **Process:** Contains 3 independent execution paths. Delegates to *getDb*, reads projects, *AppError*, reads jobs, *newId* and 1 more.
- **Outputs:** JobRow
- **Dependencies:** *getDb*, reads projects, *AppError*, reads jobs, *newId*, *initialStages*
- **Used by:** *scripts/analyze.mts*, *scripts/benchmark.mts*, *scripts/verify-ai.mts*, *src/app/api/projects/[id]/analyze/route.ts*, *src/app/api/projects/github/route.ts*, *src/app/api/projects/upload/route.ts*, *tests/api.test.ts*, *tests/helpers/index.ts*, *tests/pipeline.test.ts*, *job*
- **Important behavior:** Part of the module's public surface.

parseCite (function)

src/lib/client.ts:84-87 · function parseCite(c: string): { path: string; line?: number; end?: number } {

- **Purpose:** Parse "path:12-30" citations into a navigable link target.
- **Inputs:** c: string
- **Process:** Contains 3 independent execution paths.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** *src/components/ui.tsx*, *SourceLink*
- **Important behavior:** Part of the module's public surface.

getProject (function)

src/lib/api.ts:28-32 · function getProject(id: string): ProjectRow {

- **Purpose:** Function getProject (no doc comment; role inferred from name and relationships).
- **Inputs:** id: string
- **Process:** Contains 2 independent execution paths. Delegates to reads projects, getDb, AppError.
- **Outputs:** ProjectRow
- **Dependencies:** reads projects, getDb, AppError
- **Used by:** src/app/api/projects/[id]/analyze/route.ts, src/app/api/projects/[id]/overview/route.ts, src/app/api/projects/[id]/route.ts, POST, GET, DELETE, getReadyProject, tests/api.test.ts
- **Important behavior:** Part of the module's public surface.

newId (function)

src/lib/util/ids.ts:3-6 · function newId(prefix?: string): string {

- **Purpose:** Function newId (no doc comment; role inferred from name and relationships).
- **Inputs:** prefix?: string
- **Process:** Contains 2 independent execution paths.
- **Outputs:** string
- **Dependencies:** none indexed
- **Used by:** src/lib/ask/index.ts, src/lib/discover/index.ts, src/lib/graph/build.ts, src/lib/ingest/store.ts, src/lib/jobs/index.ts, src/lib/review/index.ts, askRepository, buildFlows, discoverArchitecture, buildGraph
- **Important behavior:** Part of the module's public surface.

emptyParse (function)

src/lib/parse/types.ts:70-72 · function emptyParse(status: ParsedFile["status"], parser: string, error?: string): ParsedFile {

- **Purpose:** Function emptyParse (no doc comment; role inferred from name and relationships).
- **Inputs:** status: ParsedFile["status"], parser: string, error?: string
- **Process:** Contains 1 independent execution paths.
- **Outputs:** ParsedFile
- **Dependencies:** none indexed
- **Used by:** src/lib/parse/extract.ts, src/lib/parse/index.ts, src/lib/parse/textfallback.ts, extractFromTree, parseFile, fallback, parseSql, parsePrisma, parseGraphQL, parseMarkdown
- **Important behavior:** Part of the module's public surface.

isAppError (function)

src/lib/util/errors.ts:14-16 · function isAppError(e: unknown): e is AppError {

- **Purpose:** Function isAppError (no doc comment; role inferred from name and relationships).
- **Inputs:** e: unknown
- **Process:** Contains 1 independent execution paths.
- **Outputs:** e is AppError
- **Dependencies:** none indexed
- **Used by:** src/lib/api.ts, fail
- **Important behavior:** Part of the module's public surface.

Graph (interface)

src/lib/map/index.ts:29-34 · interface Graph {

- **Purpose:** Interface Graph (no doc comment; role inferred from name and relationships).

- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** src/app/p/[id]/architecture/page.tsx, src/app/p/[id]/map/page.tsx, src/components/graph.tsx, layout
- **Important behavior:** Part of the module's public surface.

initialStages (function)

src/lib/jobs/stages.ts:18-20 · function initialStages(): JobStage[] {

- **Purpose:** Function initialStages (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Contains 1 independent execution paths.
- **Outputs:** JobStage[]
- **Dependencies:** none indexed
- **Used by:** src/lib/jobs/index.ts, enqueueAnalysis, Tracker.constructor, recoverStaleJobs
- **Important behavior:** Part of the module's public surface.

nodeTypeForFile (function)

src/lib/map/index.ts:106-119 · function nodeTypeForFile(f: FileRow): NodeType {

- **Purpose:** Function nodeTypeForFile (no doc comment; role inferred from name and relationships).
- **Inputs:** f: FileRow
- **Process:** Contains 12 independent execution paths.
- **Outputs:** NodeType
- **Dependencies:** none indexed
- **Used by:** nodeTypeForSymbol, moduleGraph, changelImpact
- **Important behavior:** Part of the module's public surface.

Loading (component)

src/components/ui.tsx:37-39 · function Loading({ label = "Loading" }: { label?: string }) {

- **Purpose:** Component Loading (no doc comment; role inferred from name and relationships).
- **Inputs:** { label = "Loading" }: { label?: string }
- **Process:** Contains 1 independent execution paths.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** src/app/p/[id]/architecture/page.tsx, src/app/p/[id]/ask/page.tsx, src/app/p/[id]/explain/page.tsx, src/app/p/[id]/files/page.tsx, src/app/p/[id]/layout.tsx, src/app/p/[id]/map/page.tsx, src/app/p/[id]/page.tsx, src/app/p/[id]/review/page.tsx, src/app/page.tsx, src/components/project.tsx
- **Important behavior:** Part of the module's public surface.

ErrorMessage (component)

src/components/ui.tsx:41-49 · function ErrorMessage({ error, onRetry }: { error: { message: string; hint?: string }; onRetry?: () => void }) {

- **Purpose:** Component ErrorMessage (no doc comment; role inferred from name and relationships).
- **Inputs:** { error, onRetry }: { error: { message: string; hint?: string }; onRetry?: () => void }
- **Process:** Contains 3 independent execution paths.
- **Outputs:** void }
- **Dependencies:** none indexed

- **Used by:** src/app/p/[id]/architecture/page.tsx, src/app/p/[id]/ask/page.tsx, src/app/p/[id]/explain/page.tsx, src/app/p/[id]/files/page.tsx, src/app/p/[id]/layout.tsx, src/app/p/[id]/map/page.tsx, src/app/p/[id]/page.tsx, src/app/p/[id]/review/page.tsx, src/app/page.tsx, src/components/project.tsx
- **Important behavior:** Part of the module's public surface.

claimNextJob (function)

src/lib/jobs/index.ts:253-261 · function claimNextJob(): JobRow | undefined {

- **Purpose:** Atomically claim the oldest queued job.
- **Inputs:** none declared
- **Process:** Contains 3 independent execution paths. Delegates to getDb, reads jobs.
- **Outputs:** JobRow | undefined
- **Dependencies:** getDb, reads jobs
- **Used by:** src/lib/jobs/worker.ts, startWorker, drainQueue, tests/pipeline.test.ts
- **Important behavior:** Part of the module's public surface.

processJob (function)

src/lib/jobs/index.ts:122-250 · async function processJob(jobId: string): Promise<void> {

- **Purpose:** Run one job to completion. Never throws: failures are persisted on the job and project.
- **Inputs:** jobId: string
- **Process:** Contains 67 independent execution paths. Delegates to getDb, getJob, Tracker, Tracker.heartbeat, UsageMeter and 1 more.
- **Outputs:** Promise
- **Dependencies:** getDb, getJob, Tracker, Tracker.heartbeat, UsageMeter, reads projects, AppError, getAuthProvider, Tracker.note, providerStatus
- **Used by:** src/lib/jobs/worker.ts, startWorker, drainQueue
- **Important behavior:** Part of the module's public surface.

Chip (component)

src/components/ui.tsx:23-26 · function Chip({ children, tone = "neutral", title }: { children: ReactNode; tone?: "neutral" | "ok" | "warn" | "info" | "danger"; title?: st

- **Purpose:** Component Chip (no doc comment; role inferred from name and relationships).
- **Inputs:** { children, tone = "neutral", title }: { children: ReactNode; tone?: "neutral" | "ok" | "warn" | "info" | "danger"; title?: string }
- **Process:** Contains 5 independent execution paths.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** src/app/p/[id]/architecture/page.tsx, src/app/p/[id]/ask/page.tsx, src/app/p/[id]/explain/page.tsx, src/app/p/[id]/files/page.tsx, src/app/p/[id]/map/page.tsx, src/app/p/[id]/page.tsx, src/app/p/[id]/review/page.tsx, src/app/page.tsx, src/components/project.tsx, ArchitecturePage
- **Important behavior:** Part of the module's public surface.

riskGlyph (function)

src/lib/map/legend.ts:39-41 · function riskGlyph(level: RiskLevel): string {

- **Purpose:** Function riskGlyph (no doc comment; role inferred from name and relationships).
- **Inputs:** level: RiskLevel
- **Process:** Contains 2 independent execution paths.
- **Outputs:** string
- **Dependencies:** none indexed

- **Used by:** src/app/p/[id]/map/page.tsx, src/components/graph.tsx, src/lib/map/index.ts, MapInner, MapNode, codeMapMarkdown, treeToText, graphToMermaid
- **Important behavior:** Part of the module's public surface.

keyOf (function)

src/lib/client.ts:33-33 · keyOf = (p: string) => p.replace(/([?&])k=\d+&?/, "\$1").replace(/[?&]\$/, "")

- **Purpose:** Function keyOf (no doc comment; role inferred from name and relationships).
- **Inputs:** p: string
- **Process:** Contains 1 independent execution paths.
- **Outputs:** p.replace(/([?&])k
- **Dependencies:** none indexed
- **Used by:** useApi
- **Important behavior:** Internal to its module.

glyphFor (function)

src/lib/map/legend.ts:35-37 · function glyphFor(type: NodeType): string {

- **Purpose:** Function glyphFor (no doc comment; role inferred from name and relationships).
- **Inputs:** type: NodeType
- **Process:** Contains 2 independent execution paths.
- **Outputs:** string
- **Dependencies:** none indexed
- **Used by:** src/app/p/[id]/map/page.tsx, src/components/graph.tsx, src/lib/map/index.ts, MapInner, MapNode, codeMapMarkdown, architectureDiagramText, graphToMermaid
- **Important behavior:** Part of the module's public surface.

startWorker (function)

src/lib/jobs/workers.ts:7-29 · function startWorker(pollMs = 1500): void {

- **Purpose:** Start the in-process job loop. Safe to call more than once.
- **Inputs:** pollMs = 1500
- **Process:** Contains 5 independent execution paths. Delegates to recoverStaleJobs, claimNextJob, processJob.
- **Outputs:** void
- **Dependencies:** recoverStaleJobs, claimNextJob, processJob
- **Used by:** scripts/worker.mts, register
- **Important behavior:** Part of the module's public surface.

neutralizeDelimiters (function)

src/lib/ai/prompts.ts:21-23 · function neutralizeDelimiters(text: string): string {

- **Purpose:** Remove our own delimiters from repository text so a hostile file cannot close the untrusted block early.
- **Inputs:** text: string
- **Process:** Contains 1 independent execution paths.
- **Outputs:** string
- **Dependencies:** none indexed
- **Used by:** untrusted, renderEvidence
- **Important behavior:** Part of the module's public surface.

getFileContents (function)

src/lib/ingest/store.ts:91-100 · function getFileContents(hashes: string[]): Map<string, string> {

- **Purpose:** Function getFileContents (no doc comment; role inferred from name and relationships).
- **Inputs:** hashes: string[]
- **Process:** Contains 3 independent execution paths. Delegates to getDb, reads blobs.
- **Outputs:** Map<string, string>
- **Dependencies:** getDb, reads blobs
- **Used by:** src/lib/ask/index.ts, src/lib/graph/build.ts, src/lib/retrieval/index.ts, withSnippets, loadProjectFiles, searchCode, retrieveContext
- **Important behavior:** Part of the module's public surface.

askRepository (function)

src/lib/ask/index.ts:138-180 · async function askRepository(projectId: string, question: string): Promise<Answer> {

- **Purpose:** -----
- **Inputs:** projectId: string, question: string
- **Process:** Contains 22 independent execution paths. Delegates to getDb, reads projects, graphAnswer, getAIProvider, retrieveContext and 1 more.
- **Outputs:** Promise
- **Dependencies:** getDb, reads projects, graphAnswer, getAIProvider, retrieveContext, queryTerms, relevantHits, UsageMeter, renderEvidence, tryAnalyze
- **Used by:** scripts/benchmark.mts, scripts/verify-ai.mts, src/app/api/projects/[id]/ask/route.ts, tests/exports.test.ts, tests/map.test.ts, a, POST
- **Important behavior:** Part of the module's public surface.

untrusted (function)

src/lib/ai/prompt.ts:29-31 · function untrusted(text: string): string {

- **Purpose:** Wrap any repository-derived text (paths, symbol names, README excerpts, earlier summaries) in the untrusted block. Every dynamic data segment of a prompt must go through this or renderEvidence.
- **Inputs:** text: string
- **Process:** Contains 1 independent execution paths. Delegates to neutralizeDelimiters, redactSecrets.
- **Outputs:** string
- **Dependencies:** neutralizeDelimiters, redactSecrets
- **Used by:** askRepository, wrapFacts, enhanceWithAI, runAiReview, verifyWithAI
- **Important behavior:** Part of the module's public surface.

parseReport (function)

src/lib/export/document.ts:100-115 · function parseReport(markdown: string): ReportDocument {

- **Purpose:** Parse the report Markdown into a format-neutral document that the PDF and DOCX renderers share.
- **Inputs:** markdown: string
- **Process:** Contains 8 independent execution paths. Delegates to blocks, jobs.push.
- **Outputs:** ReportDocument
- **Dependencies:** blocks, jobs.push
- **Used by:** scripts/convert.mts, src/lib/export/index.ts, tests/exports.test.ts, doc, exportFile
- **Important behavior:** Part of the module's public surface.

outline (function)

src/lib/export/document.ts:120-122 · function outline(doc: ReportDocument): string[] {

- **Purpose:** The level-2 headings, used for the contents list.
- **Inputs:** doc: ReportDocument
- **Process:** Contains 2 independent execution paths. Delegates to plain.
- **Outputs:** string[]
- **Dependencies:** plain
- **Used by:** *src/lib/export/docx.ts*, *src/lib/export/pdf.ts*, *renderDocx*, *build*
- **Important behavior:** Part of the module's public surface.

ApiError (class)

src/lib/client.ts:4-14 · class ApiError extends Error {

- **Purpose:** Class ApiError (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** *src/app/p/[id]/ask/page.tsx*, *src/app/page.tsx*, *src/components/project.tsx*, *uploadWithProgress*, *Home*, *api*
- **Important behavior:** Part of the module's public surface.

graphToMermaid (function)

src/lib/map/index.ts:493-506 · function graphToMermaid(g: Graph, direction: "LR" | "TD" = "LR"): string {

- **Purpose:** Function graphToMermaid (no doc comment; role inferred from name and relationships).
- **Inputs:** g: Graph, direction: "LR" | "TD" = "LR"
- **Process:** Contains 12 independent execution paths. Delegates to *glyphFor*, *mmLabel*, *riskGlyph*, *mermaidId*, *jobs.push*.
- **Outputs:** string
- **Dependencies:** *glyphFor*, *mmLabel*, *riskGlyph*, *mermaidId*, *jobs.push*
- **Used by:** *src/app/api/projects/[id]/graph/route.ts*, *src/lib/export/markdown.ts*, *tests/map.test.ts*, *GET*, *codeMapMarkdown*, *architectureMermaid*, *impactMermaid*
- **Important behavior:** Part of the module's public surface.

sliceLines (function)

src/lib/util/text.ts:8-11 · function sliceLines(text: string, start: number, end: number): string {

- **Purpose:** Function sliceLines (no doc comment; role inferred from name and relationships).
- **Inputs:** text: string, start: number, end: number
- **Process:** Contains 1 independent execution paths.
- **Outputs:** string
- **Dependencies:** none indexed
- **Used by:** *src/app/api/projects/[id]/symbols/[symbolId]/route.ts*, *src/lib/ask/index.ts*, *src/lib/docs/ai.ts*, *src/lib/retrieval/index.ts*, *src/lib/review/ai.ts*, *src/lib/review/verify.ts*, *GET*, *withSnippets*, *enhanceWithAI*, *retrieveContext*
- **Important behavior:** Part of the module's public surface.

DownloadMenu (component)

src/components/download.tsx:63-91 · function DownloadMenu({ projectId, scope = "full", label = "Download", align = "right", small = false }: { projectId: string; scope?: Report

- **Purpose:** One control on every result view: view or download that result as PDF, Word or Markdown.
- **Inputs:** none declared
- **Process:** Contains 8 independent execution paths.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** *src/app/p/[id]/architecture/page.tsx*, *src/app/p/[id]/ask/page.tsx*, *src/app/p/[id]/explain/page.tsx*, *src/app/p/[id]/layout.tsx*, *src/app/p/[id]/map/page.tsx*, *src/app/p/[id]/review/page.tsx*, *src/app/page.tsx*, *ArchitecturePage*, *Cite*, *Origin*
- **Important behavior:** Part of the module's public surface.

recoverStaleJobs (function)

src/lib/jobs/index.ts:264-269 · function recoverStaleJobs(now = Date.now()): number {

- **Purpose:** Requeue jobs whose worker died (no heartbeat), so a restart never leaves analysis stuck.
- **Inputs:** *now = Date.now()*
- **Process:** Contains 3 independent execution paths. Delegates to *getDb*, reads *jobs*, *initialStages*.
- **Outputs:** *number*
- **Dependencies:** *getDb*, reads *jobs*, *initialStages*
- **Used by:** *src/lib/jobs/worker.ts*, *startWorker*, *tests/pipeline.test.ts*
- **Important behavior:** Part of the module's public surface.

Layout (function)

src/components/graph.tsx:43-62 · function layout(g: Graph, selected: string | null, highlight: Set<string>): { nodes: Node<NodeData>[]; edges: Edge[] } {

- **Purpose:** Function layout (no doc comment; role inferred from name and relationships).
- **Inputs:** *g: Graph*, *selected: string | null*, *highlight: Set*
- **Process:** Contains 16 independent execution paths. Delegates to *Graph*.
- **Outputs:** not declared
- **Dependencies:** *Graph*
- **Used by:** *src/app/p/[id]/map/page.tsx*, *MapInner*, *GraphCanvas*
- **Important behavior:** Part of the module's public surface.

search (function)

src/lib/retrieval/index.ts:197-288 · async function search(projectId: string, query: string, opts: RetrievalOptions = {}): Promise<SearchHit[]> {

- **Purpose:** Hybrid retrieval: lexical BM25, structural graph expansion, importance prior, optional semantic similarity.
- **Inputs:** *projectId: string*, *query: string*, *opts: RetrievalOptions = {}*
- **Process:** Contains 53 independent execution paths. Delegates to *load*, *queryTerms*, *getAIProvider*, *UsageMeter.add*, *passes* and 1 more.
- **Outputs:** *Promise<SearchHit[]>*
- **Dependencies:** *load*, *queryTerms*, *getAIProvider*, *UsageMeter.add*, *passes*, *cosine*, *jobs.push*, reads *relationships*, *getDb*, *searchCode*
- **Used by:** *scripts/benchmark.mts*, *src/app/api/projects/[id]/search/route.ts*, *tests/analyzers.test.ts*, *tests/map.test.ts*, *GET*, *detectEntryPoints*, *detectAIComponents*, *parseConfig*, *retrieveContext*
- **Important behavior:** Part of the module's public surface.

scanText (function)

src/lib/analysis/rules.ts:162-202 · function scanText(path: string, language: string, text: string, isTest: boolean): FindingDraft[] {

- **Purpose:** Function scanText (no doc comment; role inferred from name and relationships).
- **Inputs:** path: string, language: string, text: string, isTest: boolean
- **Process:** Contains 30 independent execution paths. Delegates to blankLiterals, jobs.push, ignored, toFinding.
- **Outputs:** FindingDraft[]
- **Dependencies:** blankLiterals, jobs.push, ignored, toFinding
- **Used by:** src/lib/review/index.ts, tests/precision.test.ts, tests/review.test.ts, runStaticReview, ids
- **Important behavior:** Part of the module's public surface.

FormatRow (component)

src/components/download.tsx:41-60 · function FormatRow({ projectId, scope, format, compact = false, inMenu = false, onDone }: { projectId: string; scope: ReportScope; format: F

- **Purpose:** Component FormatRow (no doc comment; role inferred from name and relationships).
- **Inputs:** { projectId, scope, format, compact = false, inMenu = false, onDone }: { projectId: string; scope: ReportScope; format: Format; compact?: boolean; inMenu?: boolean; onDone?: (
- **Process:** Contains 10 independent execution paths. Delegates to saveReport, exportUrl.
- **Outputs:** v...
- **Dependencies:** saveReport, exportUrl
- **Used by:** src/app/p/[id]/page.tsx, src/app/p/[id]/reports/page.tsx, OverviewPage, ReportsPage
- **Important behavior:** Part of the module's public surface.

Empty (component)

src/components/ui.tsx:51-58 · function Empty({ title, children }: { title: string; children?: ReactNode }) {

- **Purpose:** Component Empty (no doc comment; role inferred from name and relationships).
- **Inputs:** { title, children }: { title: string; children?: ReactNode }
- **Process:** Contains 2 independent execution paths.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** src/app/p/[id]/architecture/page.tsx, src/app/p/[id]/explain/page.tsx, src/app/p/[id]/files/page.tsx, src/app/p/[id]/map/page.tsx, src/app/p/[id]/page.tsx, src/app/p/[id]/review/page.tsx, src/app/page.tsx, ArchitecturePage, Origin, usePersisted
- **Important behavior:** Part of the module's public surface.

LoadLanguage (function)

src/lib/parse/treesitter.ts:56-67 · async function loadLanguage(key: string): Promise<Language | null> {

- **Purpose:** Function loadLanguage (no doc comment; role inferred from name and relationships).
- **Inputs:** key: string
- **Process:** Contains 4 independent execution paths. Delegates to ensureInit, grammarPath, load.
- **Outputs:** Promise<Language | null>
- **Dependencies:** ensureInit, grammarPath, load
- **Used by:** parseWithTreeSitter
- **Important behavior:** Part of the module's public surface.

providerStatus (function)

src/lib/ai/index.ts:41-45 · function providerStatus(): ProviderStatus {

- **Purpose:** Function providerStatus (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Contains 2 independent execution paths. Delegates to getAIProvider.
- **Outputs:** ProviderStatus
- **Dependencies:** getAIProvider
- **Used by:** scripts/verify-ai.mts, src/app/api/projects/[id]/overview/route.ts, src/app/api/status/route.ts, src/lib/jobs/index.ts, GET, requireProvider, checkProvider, processJob
- **Important behavior:** Part of the module's public surface.

tokenize (function)

src/lib/util/text.ts:19-30 · function tokenize(input: string): string[] {

- **Purpose:** Split identifiers into searchable terms: camelCase, snake_case, kebab-case, paths.
- **Inputs:** input: string
- **Process:** Contains 3 independent execution paths. Delegates to jobs.push.
- **Outputs:** string[]
- **Dependencies:** jobs.push
- **Used by:** src/lib/retrieval/index.ts, buildSearchIndex, indexFindingsAndDocs, queryTerms
- **Important behavior:** Part of the module's public surface.

SourceLink (component)

src/components/ui.tsx:70-74 · function SourceLink({ projectId, cite, children }: { projectId: string; cite: string; children?: ReactNode }) {

- **Purpose:** Link to a source location in the code explorer.
- **Inputs:** { projectId, cite, children }: { projectId: string; cite: string; children?: ReactNode }
- **Process:** Contains 4 independent execution paths. Delegates to parseCite.
- **Outputs:** not declared
- **Dependencies:** parseCite
- **Used by:** src/app/p/[id]/architecture/page.tsx, src/app/p/[id]/ask/page.tsx, src/app/p/[id]/explain/page.tsx, src/app/p/[id]/files/page.tsx, src/app/p/[id]/map/page.tsx, src/app/p/[id]/review/page.tsx, ArchitecturePage, Cite, Origin, usePersisted
- **Important behavior:** Part of the module's public surface.

projectSummary (function)

src/lib/api.ts:41-49 · function projectSummary(p: ProjectRow) {

- **Purpose:** Function projectSummary (no doc comment; role inferred from name and relationships).
- **Inputs:** p: ProjectRow
- **Process:** Contains 3 independent execution paths. Delegates to getDb, reads jobs.
- **Outputs:** not declared
- **Dependencies:** getDb, reads jobs
- **Used by:** src/app/api/projects/[id]/analyze/route.ts, src/app/api/projects/[id]/overview/route.ts, src/app/api/projects/[id]/route.ts, src/app/api/projects/github/route.ts, src/app/api/projects/route.ts, src/app/api/projects/upload/route.ts, POST, GET
- **Important behavior:** Part of the module's public surface.

detectLanguage (function)

src/lib/ingest/languages.ts:84-94 · function detectLanguage(filePath: string): string {

- **Purpose:** Function detectLanguage (no doc comment; role inferred from name and relationships).
- **Inputs:** filePath: string
- **Process:** Contains 7 independent execution paths.
- **Outputs:** string
- **Dependencies:** none indexed
- **Used by:** src/lib/ingest/classify.ts, classifyPath
- **Important behavior:** Part of the module's public surface.

int (function)

src/lib/config.ts:7-12 · function int(name: string, fallback: number): number {

- **Purpose:** Function int (no doc comment; role inferred from name and relationships).
- **Inputs:** name: string, fallback: number
- **Process:** Contains 4 independent execution paths.
- **Outputs:** number
- **Dependencies:** none indexed
- **Used by:** config
- **Important behavior:** Internal to its module.

areaGraph (function)

src/lib/map/index.ts:214-240 · function areaGraph(projectId: string): Graph {

- **Purpose:** -----
- **Inputs:** projectId: string
- **Process:** Contains 30 independent execution paths. Delegates to loadModel, jobs.push.
- **Outputs:** Graph
- **Dependencies:** loadModel, jobs.push
- **Used by:** scripts/benchmark.mts, src/app/api/projects/[id]/graph/route.ts, src/lib/export/markdown.ts, tests/map.test.ts, GET, codeMapMarkdown, architectureMermaid
- **Important behavior:** Part of the module's public surface.

sanitizeRelativePath (function)

src/lib/ingest/paths.ts:9-23 · function sanitizeRelativePath(input: string): string | undefined {

- **Purpose:** Normalize an archive/upload path and reject anything that could escape the project root. Returns undefined for entries that should simply be skipped (directories, empty names).
- **Inputs:** input: string
- **Process:** Contains 10 independent execution paths. Delegates to AppError.
- **Outputs:** string | undefined
- **Dependencies:** AppError
- **Used by:** src/lib/ingest/fs.ts, src/lib/ingest/upload.ts, src/lib/ingest/zip.ts, tests/ingest.test.ts, readDirectory, ingestUpload, extractZip
- **Important behavior:** Part of the module's public surface.

exportUrl (function)

src/components/download.tsx:16-18 · function exportUrl(projectId: string, format: Format, scope: ReportScope, inline = false): string {

- **Purpose:** Function exportUrl (no doc comment; role inferred from name and relationships).

- **Inputs:** projectId: string, format: Format, scope: ReportScope, inline = false
- **Process:** Contains 2 independent execution paths.
- **Outputs:** string
- **Dependencies:** none indexed
- **Used by:** FormatRow
- **Important behavior:** Part of the module's public surface.

saveReport (function)

src/components/download.tsx:21-39 · async function saveReport(url: string, fallbackName: string): Promise<void> {

- **Purpose:** Fetch a report and save it, showing progress and surfacing server errors instead of navigating to a JSON error page.
- **Inputs:** url: string, fallbackName: string
- **Process:** Contains 7 independent execution paths. Delegates to json.
- **Outputs:** Promise
- **Dependencies:** json
- **Used by:** FormatRow
- **Important behavior:** Part of the module's public surface.

CreateProjectResult (interface)

src/lib/ingest/store.ts:73-77 · interface CreateProjectResult {

- **Purpose:** Interface CreateProjectResult (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** src/lib/ingest/upload.ts, IngestOutcome
- **Important behavior:** Part of the module's public surface.

LoadReportData (function)

src/lib/export/markdown.ts:16-24 · function loadReportData(projectId: string): ReportData {

- **Purpose:** Function loadReportData (no doc comment; role inferred from name and relationships).
- **Inputs:** projectId: string
- **Process:** Contains 5 independent execution paths. Delegates to getDb, reads projects, reads findings.
- **Outputs:** ReportData
- **Dependencies:** getDb, reads projects, reads findings
- **Used by:** src/lib/export/index.ts, buildHtml, buildJson, exportFile, buildMarkdown
- **Important behavior:** Part of the module's public surface.

ProviderStatus (interface)

src/lib/ai/index.ts:19-24 · interface ProviderStatus {

- **Purpose:** Interface ProviderStatus (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** ProviderHealth

- **Important behavior:** Part of the module's public surface.

SearchFilters (interface)

src/lib/retrieval/index.ts:24-33 · interface SearchFilters {

- **Purpose:** Interface SearchFilters (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** RetrievalOptions
- **Important behavior:** Part of the module's public surface.

Architecture (interface)

src/lib/discover/types.ts:129-152 · interface Architecture {

- **Purpose:** Interface Architecture (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** *src/app/api/projects/[id]/architecture/route.ts*, *src/app/api/projects/[id]/overview/route.ts*, *src/app/p/[id]/architecture/page.tsx*, *src/app/p/[id]/explain/page.tsx*, *src/app/p/[id]/page.tsx*, *src/lib/analysis/structural.ts*, *src/lib/ask/index.ts*, *src/lib/discover/index.ts*, *src/lib/docs/deterministic.ts*, *src/lib/docs/index.ts*
- **Important behavior:** Part of the module's public surface.

normalizeFiles (function)

src/lib/ingest/normalize.ts:15-106 · function normalizeFiles(raw: RawFile[]): { files: NormalizedFile[]; stats: IngestStats } {

- **Purpose:** Turn raw bytes into the normalized file model: gitignore filtering, binary detection, hashing, classification, duplicate detection, size flags. Files excluded by default are kept as metadata so users can inspect them.
- **Inputs:** raw: RawFile[]
- **Process:** Contains 39 independent execution paths. Delegates to *jobs.push*, *UsageMeter.add*, *classifyPath*, *sha256*, *extensionOf* and 1 more.
- **Outputs:** not declared
- **Dependencies:** *jobs.push*, *UsageMeter.add*, *classifyPath*, *sha256*, *extensionOf*, *countLines*, *writes files*
- **Used by:** *scripts/analyze.mts*, *scripts/benchmark.mts*, *scripts/verify-ai.mts*, *src/lib/ingest/upload.ts*, *src/lib/jobs/index.ts*, *tests/api.test.ts*, *tests/helpers/index.ts*, *tests/ingest.test.ts*, *tests/pipeline.test.ts*, *ingestUpload*
- **Important behavior:** Part of the module's public surface.

api (function)

src/lib/client.ts:16-31 · async function api<T>(path: string, init?: RequestInit): Promise<T> {

- **Purpose:** Function api (no doc comment; role inferred from name and relationships).
- **Inputs:** path: string, init?: RequestInit
- **Process:** Contains 9 independent execution paths. Delegates to *ApiError*, *text*.
- **Outputs:** Promise
- **Dependencies:** *ApiError*, *text*
- **Used by:** *src/app/p/[id]/ask/page.tsx*, *src/app/p/[id]/layout.tsx*, *src/app/p/[id]/map/page.tsx*, *src/app/page.tsx*, *src/components/project.tsx*, *Home*, *AnalysisProgress*

- **Important behavior:** Part of the module's public surface.

encryptSecret (function)

src/lib/ingest/credentials.ts:16-22 · function encryptSecret(plain: string): string {

- **Purpose:** Function encryptSecret (no doc comment; role inferred from name and relationships).
- **Inputs:** plain: string
- **Process:** Contains 1 independent execution paths. Delegates to key.
- **Outputs:** string
- **Dependencies:** key
- **Used by:** src/lib/ingest/store.ts, tests/secrets.test.ts, createPendingProject
- **Important behavior:** Part of the module's public surface.

createPendingProject (function)

src/lib/ingest/store.ts:10-22 · function createPendingProject(source: IngestSource, token?: string): string {

- **Purpose:** Insert an empty project row (used when file contents arrive later, e.g. a GitHub download inside a job).
- **Inputs:** source: IngestSource, token?: string
- **Process:** Contains 6 independent execution paths. Delegates to getDb, newId, encryptSecret.
- **Outputs:** string
- **Dependencies:** getDb, newId, encryptSecret
- **Used by:** src/app/api/projects/[id]/analyze/route.ts, src/app/api/projects/github/route.ts, POST, createProject
- **Important behavior:** Part of the module's public surface.

buildMarkdown (function)

src/lib/export/markdown.ts:35-174 · function buildMarkdown(projectId: string, opts: { scope?: ReportScope } = {}): string {

- **Purpose:** Function buildMarkdown (no doc comment; role inferred from name and relationships).
- **Inputs:** projectId: string, opts: { scope?: ReportScope } = {}
- **Process:** Contains 67 independent execution paths. Delegates to loadReportData, loadModel, jobs.push, para, cell and 1 more.
- **Outputs:** string
- **Dependencies:** loadReportData, loadModel, jobs.push, para, cell, cite, fence, apiTable, reviewSummary, findingMarkdown
- **Used by:** src/lib/export/index.ts, scripts/analyze.mts, scripts/benchmark.mts, buildHtml, exportFile, tests/exports.test.ts, tests/map.test.ts
- **Important behavior:** Part of the module's public surface.

intParam (function)

src/lib/api.ts:51-54 · function intParam(v: string | null, def: number, min = 1, max = 500): number {

- **Purpose:** Function intParam (no doc comment; role inferred from name and relationships).
- **Inputs:** v: string | null, def: number, min = 1, max = 500
- **Process:** Contains 4 independent execution paths.
- **Outputs:** number
- **Dependencies:** none indexed
- **Used by:** src/app/api/projects/[id]/findings/route.ts, src/app/api/projects/[id]/graph/route.ts, src/app/api/projects/[id]/search/route.ts, GET

- **Important behavior:** Part of the module's public surface.

ParsedFile (interface)

src/lib/parse/types.ts:53-68 · interface ParsedFile {

- **Purpose:** Interface ParsedFile (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** *src/lib/parse/cache.ts, src/lib/parse/extract.ts, src/lib/parse/index.ts, src/lib/parse/textfallback.ts*
- **Important behavior:** Part of the module's public surface.

parseGitHubUrl (function)

src/lib/ingest/github.ts:31-43 · function parseGitHubUrl(input: string): GitHubRef {

- **Purpose:** Function parseGitHubUrl (no doc comment; role inferred from name and relationships).
- **Inputs:** input: string
- **Process:** Contains 4 independent execution paths. Delegates to `AppError`.
- **Outputs:** `GitHubRef`
- **Dependencies:** `AppError`
- **Used by:** *src/app/api/github/validate/route.ts, src/app/api/projects/[id]/analyze/route.ts, src/app/api/projects/github/route.ts, src/lib/jobs/index.ts, tests/ingest.test.ts, POST, processJob*
- **Important behavior:** Part of the module's public surface.

extractZip (function)

src/lib/ingest/zip.ts:18-112 · async function extractZip(buffer: Buffer): Promise<ZipExtractResult> {

- **Purpose:** Extract a ZIP buffer defensively: entry count limits, per-file and total size limits, compression ratio checks (zip bombs), symlink rejection and path traversal rejection. Nothing is written to disk.
- **Inputs:** buffer: `Buffer`
- **Process:** Contains 21 independent execution paths. Delegates to `AppError`, `sanitizeRelativePath`, `jobs.push`, writes files, `stripCommonRoot`.
- **Outputs:** `Promise`
- **Dependencies:** `AppError`, `sanitizeRelativePath`, `jobs.push`, writes files, `stripCommonRoot`
- **Used by:** *src/lib/ingest/github.ts, src/lib/ingest/upload.ts, tests/exports.test.ts, tests/ingest.test.ts, downloadGitHubSnapshot, ingestUpload*
- **Important behavior:** Part of the module's public surface.

markdownToHtml (function)

src/lib/export/html.ts:12-62 · function markdownToHtml(md: string): string {

- **Purpose:** Converts the report's own Markdown subset (headings, tables, fences, lists, paragraphs) to HTML.
- **Inputs:** md: string
- **Process:** Contains 23 independent execution paths. Delegates to `jobs.push`, `escapeHtml`, `inline`.
- **Outputs:** string
- **Dependencies:** `jobs.push`, `escapeHtml`, `inline`
- **Used by:** *tests/map.test.ts, buildHtmlDocument*
- **Important behavior:** Part of the module's public surface.

nodeTypeForSymbol (function)

src/lib/map/index.ts:121-132 · function nodeTypeForSymbol(s: SymbolRow, file?: FileRow): NodeType {

- **Purpose:** Function nodeTypeForSymbol (no doc comment; role inferred from name and relationships).
- **Inputs:** s: SymbolRow, file?: FileRow
- **Process:** Contains 15 independent execution paths. Delegates to nodeTypeForFile.
- **Outputs:** NodeType
- **Dependencies:** nodeTypeForFile
- **Used by:** symbolGraph, changelImpact
- **Important behavior:** Part of the module's public surface.

changeImpact (function)

src/lib/map/index.ts:364-449 · function changeImpact(projectId: string, target: { type: "file" | "symbol"; id: string }, maxDepth = 4): ImpactResult | undefined {

- **Purpose:** Function changelImpact (no doc comment; role inferred from name and relationships).
- **Inputs:** projectId: string, target: { type: "file" | "symbol"; id: string }, maxDepth = 4
- **Process:** Contains 51 independent execution paths. Delegates to loadModel, jobs.push, nodeTypeForFile, nodeTypeForSymbol, UsageMeter.add.
- **Outputs:** ImpactResult | undefined
- **Dependencies:** loadModel, jobs.push, nodeTypeForFile, nodeTypeForSymbol, UsageMeter.add
- **Used by:** scripts/benchmark.mts, src/app/api/projects/[id]/files/content/route.ts, src/app/api/projects/[id]/impact/route.ts, src/app/api/projects/[id]/symbols/[symbolId]/route.ts, src/lib/export/markdown.ts, tests/map.test.ts, GET, codeMapMarkdown
- **Important behavior:** Part of the module's public surface.

LoadedFile (interface)

src/lib/graph/build.ts:22-24 · interface LoadedFile extends FileRow {

- **Purpose:** Interface LoadedFile (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Delegates to FileRow.
- **Outputs:** not declared
- **Dependencies:** FileRow
- **Used by:** src/lib/analysis/analyzers.ts, src/lib/analysis/structural.ts, src/lib/discover/detect.ts, src/lib/discover/index.ts, src/lib/docs/ai.ts, src/lib/docs/cache.ts, src/lib/docs/deterministic.ts, src/lib/review/ai.ts, src/lib/review/verify.ts, tests/ai.test.ts
- **Important behavior:** Part of the module's public surface.

createProject (function)

src/lib/ingest/store.ts:79-84 · function createProject(source: IngestSource, files: NormalizedFile[], stats: IngestStats, token?: string): CreateProjectResult {

- **Purpose:** Function createProject (no doc comment; role inferred from name and relationships).
- **Inputs:** source: IngestSource, files: NormalizedFile[], stats: IngestStats, token?: string
- **Process:** Contains 1 independent execution paths. Delegates to createPendingProject, attachFiles, getDb.
- **Outputs:** CreateProjectResult
- **Dependencies:** createPendingProject, attachFiles, getDb
- **Used by:** scripts/analyze.mts, scripts/benchmark.mts, scripts/verify-ai.mts, src/lib/ingest/upload.ts, tests/api.test.ts, tests/helpers/index.ts, tests/pipeline.test.ts, ingestUpload, analyze
- **Important behavior:** Part of the module's public surface.

stripCommonRoot (function)

src/lib/ingest/paths.ts:26-36 · function stripCommonRoot(paths: string[]): (p: string) => string {

- **Purpose:** Remove a single shared top-level directory (GitHub zipballs wrap everything in owner-repo-sha/).
- **Inputs:** paths: string[]
- **Process:** Contains 6 independent execution paths.
- **Outputs:** (p: string)
- **Dependencies:** none indexed
- **Used by:** src/lib/ingest/upload.ts, src/lib/ingest/zip.ts, ingestUpload, extractZip
- **Important behavior:** Part of the module's public surface.

requestCancel (function)

src/lib/jobs/index.ts:44-55 · function requestCancel(jobId: string): JobRow | undefined {

- **Purpose:** Function requestCancel (no doc comment; role inferred from name and relationships).
- **Inputs:** jobId: string
- **Process:** Contains 4 independent execution paths. Delegates to getDb, getJob.
- **Outputs:** JobRow | undefined
- **Dependencies:** getDb, getJob
- **Used by:** src/app/api/jobs/[id]/cancel/route.ts, src/app/api/projects/[id]/route.ts, tests/pipeline.test.ts, POST, DELETE
- **Important behavior:** Part of the module's public surface.

queryTerms (function)

src/lib/retrieval/index.ts:164-169 · function queryTerms(query: string): string[] {

- **Purpose:** Function queryTerms (no doc comment; role inferred from name and relationships).
- **Inputs:** query: string
- **Process:** Contains 8 independent execution paths. Delegates to tokenize, UsageMeter.add.
- **Outputs:** string[]
- **Dependencies:** tokenize, UsageMeter.add
- **Used by:** src/lib/ask/index.ts, askRepository, retrievalOnly, search
- **Important behavior:** Part of the module's public surface.

explainAIError (function)

src/lib/ai/errors.ts:2-14 · function explainAIError(message: string): { summary: string; hint: string } {

- **Purpose:** Turn provider failures into a short summary plus an actionable hint. Never includes credentials.
- **Inputs:** message: string
- **Process:** Contains 28 independent execution paths.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** src/lib/ai/index.ts, tests/provider.test.ts, GET, checkProvider, processJob
- **Important behavior:** Part of the module's public surface.

fetchGitHubMetadata (function)

src/lib/ingest/github.ts:76-108 · async function fetchGitHubMetadata(ref: GitHubRef, token?: string): Promise<GitHubMetadata> {

- **Purpose:** Function fetchGitHubMetadata (no doc comment; role inferred from name and relationships).
- **Inputs:** ref: GitHubRef, token?: string

- **Process:** Contains 7 independent execution paths. Delegates to AppError.
- **Outputs:** Promise
- **Dependencies:** AppError
- **Used by:** src/app/api/github/validate/route.ts, src/app/api/projects/[id]/analyze/route.ts, src/app/api/projects/github/route.ts, tests/ingest.test.ts, POST
- **Important behavior:** Part of the module's public surface.

JobStage (interface)

src/lib/db/schema.ts:224-231 · interface JobStage {

- **Purpose:** Interface JobStage (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** src/components/project.tsx, src/lib/jobs/index.ts, src/lib/jobs/stages.ts
- **Important behavior:** Part of the module's public surface.

DocReport (interface)

src/lib/docs/types.ts:89-110 · interface DocReport {

- **Purpose:** Interface DocReport (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** scripts/verify-ai.mts, src/app/api/projects/[id]/files/content/route.ts, src/app/api/projects/[id]/overview/route.ts, src/app/api/projects/[id]/report/route.ts, src/app/api/projects/[id]/symbols/[symbolId]/route.ts, src/app/p/[id]/explain/page.tsx, src/lib/docs/ai.ts, src/lib/docs/deterministic.ts, src/lib/docs/index.ts, src/lib/export/markdown.ts
- **Important behavior:** Part of the module's public surface.

attachFiles (function)

src/lib/ingest/store.ts:25-71 · function attachFiles(projectId: string, source: IngestSource, files: NormalizedFile[], stats: IngestStats): IncrementalStats {

- **Purpose:** Store normalized files for a project and compute the incremental diff against the previous analysis of the same source.
- **Inputs:** projectId: string, source: IngestSource, files: NormalizedFile[], stats: IngestStats
- **Process:** Contains 17 independent execution paths. Delegates to getDb, sha256, reads projects, reads files, UsageMeter.add and 1 more.
- **Outputs:** IncrementalStats
- **Dependencies:** getDb, sha256, reads projects, reads files, UsageMeter.add, newId
- **Used by:** src/lib/jobs/index.ts, createProject, processJob
- **Important behavior:** Part of the module's public surface.

blankLiterals (function)

src/lib/analysis/rules.ts:114-151 · function blankLiterals(line: string): string {

- **Purpose:** Blank the contents of string, template and regex literals and drop trailing // comments so code rules only see code. Line based and deliberately simple: an unterminated template continues on the next line as plain text, which errs towards fewer matches.

- **Inputs:** line: string
- **Process:** Contains 32 independent execution paths.
- **Outputs:** string
- **Dependencies:** none indexed
- **Used by:** tests/precision.test.ts, scanText
- **Important behavior:** Part of the module's public surface.

FindingDraft (interface)

src/lib/review/types.ts:11-32 · interface FindingDraft {

- **Purpose:** Interface FindingDraft (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** src/lib/analysis/analyzers.ts, src/lib/analysis/rules.ts, src/lib/analysis/structural.ts, src/lib/review/ai.ts, src/lib/review/index.ts, src/lib/review/verify.ts, tests/review.test.ts
- **Important behavior:** Part of the module's public surface.

isDefaultExcludedPath (function)

src/lib/ingest/classify.ts:56-63 · function isDefaultExcludedPath(path: string): string | undefined {

- **Purpose:** Function isDefaultExcludedPath (no doc comment; role inferred from name and relationships).
- **Inputs:** path: string
- **Process:** Contains 5 independent execution paths.
- **Outputs:** string | undefined
- **Dependencies:** none indexed
- **Used by:** classifyPath
- **Important behavior:** Part of the module's public surface.

tryAnalyze (function)

src/lib/ai/index.ts:59-69 · async function tryAnalyze<T>(provider: AIProvider | null, meter: UsageMeter, req: AnalysisRequest<T>): Promise<T | undefined> {

- **Purpose:** Run a structured request and swallow provider failures into the meter so one failed pass never aborts the whole job.
- **Inputs:** provider: AIProvider | null, meter: UsageMeter, req: AnalysisRequest
- **Process:** Contains 4 independent execution paths. Delegates to AnthropicProvider.analyze, UsageMeter.add, jobs.push.
- **Outputs:** Promise<T | undefined>
- **Dependencies:** AnthropicProvider.analyze, UsageMeter.add, jobs.push
- **Used by:** src/lib/ask/index.ts, src/lib/docs/ai.ts, src/lib/review/ai.ts, src/lib/review/verify.ts, askRepository, enhanceWithAI, runAiReview, verifyWithAI
- **Important behavior:** Part of the module's public surface.

closeDatabase (function)

src/lib/db/client.ts:52-57 · function closeDatabase(): void {

- **Purpose:** Function closeDatabase (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Contains 1 independent execution paths.
- **Outputs:** void

- **Dependencies:** none indexed
- **Used by:** scripts/benchmark.mts, scripts/verify-ai.mts, tests/helpers/index.ts, freshDb
- **Important behavior:** Part of the module's public surface.

RawFile (interface)

src/lib/ingest/types.ts:1-5 · interface RawFile {

- **Purpose:** Interface RawFile (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** src/lib/ingest/fs.ts, src/lib/ingest/github.ts, src/lib/ingest/normalize.ts, src/lib/ingest/upload.ts, src/lib/ingest/zip.ts
- **Important behavior:** Part of the module's public surface.

parseFile (function)

src/lib/parse/index.ts:14-36 · async function parseFile(filePath: string, language: string, source: string): Promise<ParsedFile> {

- **Purpose:** Parse one file into symbols, imports, calls and relationships. AST parsing is used when a grammar exists; otherwise a documented text fallback runs.
- **Inputs:** filePath: string, language: string, source: string
- **Process:** Contains 9 independent execution paths. Delegates to emptyParse, grammarKeyFor, parseWithTreeSitter, extractFromTree, fallback.
- **Outputs:** Promise
- **Dependencies:** emptyParse, grammarKeyFor, parseWithTreeSitter, extractFromTree, fallback
- **Used by:** src/lib/graph/build.ts, tests/parse.test.ts, buildGraph
- **Important behavior:** Part of the module's public surface.

setAIPProvider (function)

src/lib/ai/index.ts:15-17 · function setAIPProvider(p: AIPProvider | null | undefined): void {

- **Purpose:** Inject a provider (tests, or embedding applications). Pass undefined to restore configuration.
- **Inputs:** p: AIPProvider | null | undefined
- **Process:** Contains 1 independent execution paths.
- **Outputs:** void
- **Dependencies:** none indexed
- **Used by:** tests/helpers/index.ts, tests/ai.test.ts, tests/analyzers.test.ts, freshDb
- **Important behavior:** Part of the module's public surface.

SeverityBadge (component)

src/components/ui.tsx:14-21 · function SeverityBadge({ severity }: { severity: string }) {

- **Purpose:** Component SeverityBadge (no doc comment; role inferred from name and relationships).
- **Inputs:** { severity }: { severity: string }
- **Process:** Contains 2 independent execution paths.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** src/app/p/[id]/files/page.tsx, src/app/p/[id]/page.tsx, src/app/p/[id]/review/page.tsx, usePersisted, OverviewPage, Diff
- **Important behavior:** Part of the module's public surface.

serializeParsed (function)

src/lib/parse/cache.ts:19-22 · function serializeParsed(p: ParsedFile): string {

- **Purpose:** Function serializeParsed (no doc comment; role inferred from name and relationships).
- **Inputs:** p: ParsedFile
- **Process:** Contains 1 independent execution paths.
- **Outputs:** string
- **Dependencies:** none indexed
- **Used by:** putCachedParse
- **Important behavior:** Part of the module's public surface.

buildHtmlDocument (function)

src/lib/export/html.ts:67-89 · function buildHtmlDocument(title: string, markdown: string, opts: { print?: boolean } = {}): string {

- **Purpose:** Function buildHtmlDocument (no doc comment; role inferred from name and relationships).
- **Inputs:** title: string, markdown: string, opts: { print?: boolean } = {}
- **Process:** Contains 2 independent execution paths. Delegates to markdownToHtml, escapeHtml.
- **Outputs:** string
- **Dependencies:** markdownToHtml, escapeHtml
- **Used by:** src/lib/export/index.ts, tests/map.test.ts, buildHtml
- **Important behavior:** Part of the module's public surface.

UsageMeter (class)

src/lib/ai/index.ts:48-56 · class UsageMeter {

- **Purpose:** Tracks usage across a whole analysis job.
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** src/lib/ask/index.ts, src/lib/docs/ai.ts, src/lib/docs/index.ts, src/lib/jobs/index.ts, src/lib/review/ai.ts, src/lib/review/index.ts, src/lib/review/verify.ts, askRepository, processJob
- **Important behavior:** Part of the module's public surface.

renderDocx (function)

src/lib/export/docx.ts:82-118 · async function renderDocx(doc: ReportDocument, meta: { footer: string; author?: string }): Promise<Buffer> {

- **Purpose:** Render the report as a Word document.
- **Inputs:** doc: ReportDocument, meta: { footer: string; author?: string }
- **Process:** Contains 5 independent execution paths. Delegates to outline, render.
- **Outputs:** Promise
- **Dependencies:** outline, render
- **Used by:** scripts/convert.mts, src/lib/export/index.ts, exportFile
- **Important behavior:** Part of the module's public surface.

renderPdf (function)

src/lib/export/pdf.ts:279-284 · async function renderPdf(doc: ReportDocument, meta: { footer: string; author?: string }): Promise<Buffer> {

- **Purpose:** Render the report as a PDF with cover, contents with page numbers, bookmarks and a running footer.

- **Inputs:** doc: ReportDocument, meta: { footer: string; author?: string }
- **Process:** Contains 1 independent execution paths. Delegates to build.
- **Outputs:** Promise
- **Dependencies:** build
- **Used by:** scripts/convert.mts, src/lib/export/index.ts, exportFile
- **Important behavior:** Part of the module's public surface.

repositoryTree (function)

src/lib/map/index.ts:152-189 · function repositoryTree(projectId: string, opts: { includeExcluded?: boolean } = {}): TreeNode {

- **Purpose:** Function repositoryTree (no doc comment; role inferred from name and relationships).
- **Inputs:** projectId: string, opts: { includeExcluded?: boolean } = {}
- **Process:** Contains 29 independent execution paths. Delegates to getDb, loadModel, reads files, jobs.push.
- **Outputs:** TreeNode
- **Dependencies:** getDb, loadModel, reads files, jobs.push
- **Used by:** src/app/api/projects/[id]/files/route.ts, src/lib/export/markdown.ts, src/lib/jobs/index.ts, tests/map.test.ts, GET, codeMapMarkdown, processJob
- **Important behavior:** Part of the module's public surface.

treeToText (function)

src/lib/map/index.ts:191-209 · function treeToText(root: TreeNode, opts: { depth?: number; maxChildren?: number; name?: string } = {}): string {

- **Purpose:** Function treeToText (no doc comment; role inferred from name and relationships).
- **Inputs:** root: TreeNode, opts: { depth?: number; maxChildren?: number; name?: string } = {}
- **Process:** Contains 16 independent execution paths. Delegates to riskGlyph, jobs.push, walk.
- **Outputs:** string
- **Dependencies:** riskGlyph, jobs.push, walk
- **Used by:** src/app/api/projects/[id]/files/route.ts, src/lib/export/markdown.ts, src/lib/jobs/index.ts, tests/map.test.ts, GET, codeMapMarkdown, processJob
- **Important behavior:** Part of the module's public surface.

loadProjectFiles (function)

src/lib/graph/build.ts:54-60 · function loadProjectFiles(projectId: string, opts: { includeExcluded?: boolean } = {}): LoadedFile[] {

- **Purpose:** Load included, text-bearing files for a project along with content.
- **Inputs:** projectId: string, opts: { includeExcluded?: boolean } = {}
- **Process:** Contains 5 independent execution paths. Delegates to getDb, reads files, getFileContents.
- **Outputs:** LoadedFile[]
- **Dependencies:** getDb, reads files, getFileContents
- **Used by:** src/lib/discover/index.ts, src/lib/docs/index.ts, src/lib/review/index.ts, discoverArchitecture, generateDocs, buildGraph, runStaticReview, runReview
- **Important behavior:** Part of the module's public surface.

drainQueue (function)

src/lib/jobs/worker.ts:37-46 · async function drainQueue(): Promise<number> {

- **Purpose:** Run queued jobs in the foreground until the queue is empty (used by tests and the CLI).
- **Inputs:** none declared
- **Process:** Contains 2 independent execution paths. Delegates to claimNextJob, processJob.

- **Outputs:** Promise
- **Dependencies:** claimNextJob, processJob
- **Used by:** scripts/analyze.mts, scripts/benchmark.mts, scripts/verify-ai.mts, tests/helpers/index.ts, tests/pipeline.test.ts, analyze
- **Important behavior:** Part of the module's public surface.

architectureDiagramText (function)

src/lib/map/index.ts:462-488 · function architectureDiagramText(projectId: string): string {

- **Purpose:** Generate the system architecture map from detected layers, routes, models and services.
- **Inputs:** projectId: string
- **Process:** Contains 18 independent execution paths. Delegates to loadModel, reads files, jobs.push, box, glyphFor and 1 more.
- **Outputs:** string
- **Dependencies:** loadModel, reads files, jobs.push, box, glyphFor, center
- **Used by:** src/app/api/projects/[id]/architecture/route.ts, src/lib/export/markdown.ts, src/lib/jobs/index.ts, tests/map.test.ts, GET, codeMapMarkdown, processJob
- **Important behavior:** Part of the module's public surface.

architectureMermaid (function)

src/lib/map/index.ts:508-510 · function architectureMermaid(projectId: string): string {

- **Purpose:** Function architectureMermaid (no doc comment; role inferred from name and relationships).
- **Inputs:** projectId: string
- **Process:** Contains 1 independent execution paths. Delegates to graphToMermaid, areaGraph.
- **Outputs:** string
- **Dependencies:** graphToMermaid, areaGraph
- **Used by:** src/app/api/projects/[id]/architecture/route.ts, src/lib/export/markdown.ts, src/lib/jobs/index.ts, tests/map.test.ts, GET, codeMapMarkdown, processJob
- **Important behavior:** Part of the module's public surface.

erMermaid (function)

src/lib/map/index.ts:512-527 · function erMermaid(projectId: string): string | undefined {

- **Purpose:** Function erMermaid (no doc comment; role inferred from name and relationships).
- **Inputs:** projectId: string
- **Process:** Contains 9 independent execution paths. Delegates to loadModel, jobs.push.
- **Outputs:** string | undefined
- **Dependencies:** loadModel, jobs.push
- **Used by:** src/app/api/projects/[id]/architecture/route.ts, src/lib/export/markdown.ts, src/lib/jobs/index.ts, tests/map.test.ts, GET, codeMapMarkdown, processJob
- **Important behavior:** Part of the module's public surface.

exportFile (function)

src/lib/export/index.ts:65-92 · async function exportFile(projectId: string, format: ExportFormat, scope: ReportScope = "full"): Promise<ExportResult> {

- **Purpose:** Produce one report file. PDF and Word are generated from the same Markdown the app shows, so all formats agree.
- **Inputs:** projectId: string, format: ExportFormat, scope: ReportScope = "full"
- **Process:** Contains 11 independent execution paths. Delegates to loadReportData, safe, buildJson, buildMarkdown, buildHtml and 1 more.

- **Outputs:** Promise
- **Dependencies:** loadReportData, safe, buildJson, buildMarkdown, buildHtml, parseReport, renderPdf, renderDocx
- **Used by:** scripts/analyze.mts, scripts/benchmark.mts, src/app/api/projects/[id]/export/route.ts, tests/exports.test.ts, GET
- **Important behavior:** Part of the module's public surface.

fmtDuration (function)

src/lib/client.ts:78-81 · function fmtDuration(ms: number): string {

- **Purpose:** Function fmtDuration (no doc comment; role inferred from name and relationships).
- **Inputs:** ms: number
- **Process:** Contains 2 independent execution paths.
- **Outputs:** string
- **Dependencies:** none indexed
- **Used by:** src/components/project.tsx, AnalysisProgress
- **Important behavior:** Part of the module's public surface.

renderEvidence (function)

src/lib/ai/prompt.ts:42-60 · function renderEvidence(items: EvidenceItem[], charBudget: number): { text: string; included: EvidenceItem[] } {

- **Purpose:** Render numbered source excerpts for a model. Secrets are redacted before leaving the process.
- **Inputs:** items: EvidenceItem[], charBudget: number
- **Process:** Contains 6 independent execution paths. Delegates to neutralizeDelimiters, redactSecrets, jobs.push.
- **Outputs:** not declared
- **Dependencies:** neutralizeDelimiters, redactSecrets, jobs.push
- **Used by:** askRepository, enhanceWithAI, runAiReview, verifyWithAI
- **Important behavior:** Part of the module's public surface.

key (function)

src/lib/ingest/credentials.ts:11-14 · function key(): Buffer {

- **Purpose:** Function key (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Contains 2 independent execution paths.
- **Outputs:** Buffer
- **Dependencies:** none indexed
- **Used by:** encryptSecret, decryptSecret, symbolNeighborhood
- **Important behavior:** Internal to its module.

AnalysisRequest (interface)

src/lib/ai/types.ts:3-11 · interface AnalysisRequest<T> {

- **Purpose:** Interface AnalysisRequest (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** src/lib/ai/anthropic.ts, src/lib/ai/index.ts, src/lib/ai/openai.ts

- **Important behavior:** Part of the module's public surface.

AnalysisResult (interface)

src/lib/ai/types.ts:19-24 · interface AnalysisResult<T> {

- **Purpose:** Interface AnalysisResult (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** *src/lib/ai/anthropic.ts*, *src/lib/ai/index.ts*, *src/lib/ai/openai.ts*
- **Important behavior:** Part of the module's public surface.

AIUnavailableError (class)

src/lib/ai/types.ts:38-43 · class AIUnavailableError extends Error {

- **Purpose:** Class AIUnavailableError (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** *src/lib/ai/index.ts*, *requireProvider*
- **Important behavior:** Part of the module's public surface.

mapLimit (function)

src/lib/util/concurrency.ts:2-13 · async function mapLimit<T, R>(items: T[], limit: number, fn: (item: T, index: number) => Promise<R>): Promise<R[]> {

- **Purpose:** Run tasks with bounded concurrency, preserving result order.
- **Inputs:** items: T[], limit: number, fn: (item: T, index: number)
- **Process:** Contains 2 independent execution paths.
- **Outputs:** Promise): Promise<R[]>
- **Dependencies:** none indexed
- **Used by:** *src/lib/docs/ai.ts*, *src/lib/review/ai.ts*, *src/lib/review/verify.ts*, *enhanceWithAI*, *runAiReview*, *verifyWithAI*
- **Important behavior:** Part of the module's public surface.

decryptSecret (function)

src/lib/ingest/credentials.ts:24-33 · function decryptSecret(payload: string): string | undefined {

- **Purpose:** Function decryptSecret (no doc comment; role inferred from name and relationships).
- **Inputs:** payload: string
- **Process:** Contains 2 independent execution paths. Delegates to key.
- **Outputs:** string | undefined
- **Dependencies:** key
- **Used by:** *src/app/api/projects/[id]/analyze/route.ts*, *src/lib/jobs/index.ts*, *tests/secrets.test.ts*, POST, *processJob*
- **Important behavior:** Part of the module's public surface.

relevantHits (function)

src/lib/ask/index.ts:183-190 · function relevantHits(hits: SearchHit[], terms: string[]): SearchHit[] {

- **Purpose:** Keep only hits whose text covers a meaningful share of the question's terms.
- **Inputs:** hits: SearchHit[], terms: string[]
- **Process:** Contains 6 independent execution paths.
- **Outputs:** SearchHit[]
- **Dependencies:** none indexed
- **Used by:** askRepository, retrievalOnly
- **Important behavior:** Part of the module's public surface.

LegendText (function)

src/lib/map/legend.ts:43-54 · function legendText(): string {

- **Purpose:** Function legendText (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Contains 1 independent execution paths.
- **Outputs:** string
- **Dependencies:** none indexed
- **Used by:** GET, buildMarkdown, codeMapMarkdown, processJob, tests/map.test.ts
- **Important behavior:** Part of the module's public surface.

symbolNeighborhood (function)

src/lib/map/index.ts:275-316 · function symbolNeighborhood(projectId: string, symbolId: string): SymbolNeighborhood | undefined {

- **Purpose:** Function symbolNeighborhood (no doc comment; role inferred from name and relationships).
- **Inputs:** projectId: string, symbolId: string
- **Process:** Contains 51 independent execution paths. Delegates to loadModel, UsageMeter.add, jobs.push, key, reads findings.
- **Outputs:** SymbolNeighborhood | undefined
- **Dependencies:** loadModel, UsageMeter.add, jobs.push, key, reads findings
- **Used by:** src/app/api/projects/[id]/symbols/[symbolId]/route.ts, src/lib/ask/index.ts, tests/map.test.ts, GET, graphAnswer
- **Important behavior:** Part of the module's public surface.

Run (interface)

src/lib/export/document.ts:4-10 · interface Run {

- **Purpose:** A styled run of inline text.
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared
- **Dependencies:** none indexed
- **Used by:** src/lib/export/docx.ts, src/lib/export/pdf.ts
- **Important behavior:** Part of the module's public surface.

ReportDocument (interface)

src/lib/export/document.ts:22-26 · interface ReportDocument {

- **Purpose:** Interface ReportDocument (no doc comment; role inferred from name and relationships).
- **Inputs:** none declared
- **Process:** Leaf logic with no calls to other indexed symbols.
- **Outputs:** not declared

- **Dependencies:** none indexed
- **Used by:** src/lib/export/docx.ts, src/lib/export/pdf.ts
- **Important behavior:** Part of the module's public surface.

extractFromTree (function)

src/lib/parse/extract.ts:778-798 · function extractFromTree(language: string, grammarKey: string, tree: Tree, src: string, filePath: string): ParsedFile {

- **Purpose:** Function extractFromTree (no doc comment; role inferred from name and relationships).
- **Inputs:** language: string, grammarKey: string, tree: Tree, src: string, filePath: string
- **Process:** Contains 15 independent execution paths. Delegates to emptyParse, extractTsJs, extractPython, extractGo, extractJavaLike and 1 more.
- **Outputs:** ParsedFile
- **Dependencies:** emptyParse, extractTsJs, extractPython, extractGo, extractJavaLike, extractRuby, extractRust, extractPhp, extractCss, extractHtml
- **Used by:** src/lib/parse/index.ts, parseFile
- **Important behavior:** Part of the module's public surface.

grammarKeyFor (function)

src/lib/parse/treesitter.ts:50-54 · function grammarKeyFor(language: string, filePath: string): string | undefined {

- **Purpose:** Function grammarKeyFor (no doc comment; role inferred from name and relationships).
- **Inputs:** language: string, filePath: string
- **Process:** Contains 6 independent execution paths.
- **Outputs:** string | undefined
- **Dependencies:** none indexed
- **Used by:** src/lib/parse/index.ts, parseFile
- **Important behavior:** Part of the module's public surface.

17. Recommendations

1. Add tests for src/lib/client.ts, which many files depend on. (src/lib/client.ts)
2. Add tests for src/lib/util/text.ts, which many files depend on. (src/lib/util/text.ts)

18. Detailed Code Map

1. Repository Tree

```
brody/
├── .github/ # build, deployment and operations (1 files)
│   ├── workflows/ # infrastructure (1 files)
│   │   └── ci.yml
├── docs/ # project documentation (4 files)
│   ├── validation/
│   │   ├── VALIDATION_REPORT.docx
│   │   ├── VALIDATION_REPORT.md
│   │   └── VALIDATION_REPORT.pdf
│   └── ARCHITECTURE.md
├── drizzle/
│   ├── meta/ # configuration (2 files)
│   │   ├── _journal.json
│   │   └── 0000_snapshot.json
│   └── 0000_robust_hawkeye.sql
├── e2e/ # automated tests, fixtures and mocks (3 files)
└── performance.spec.ts
```

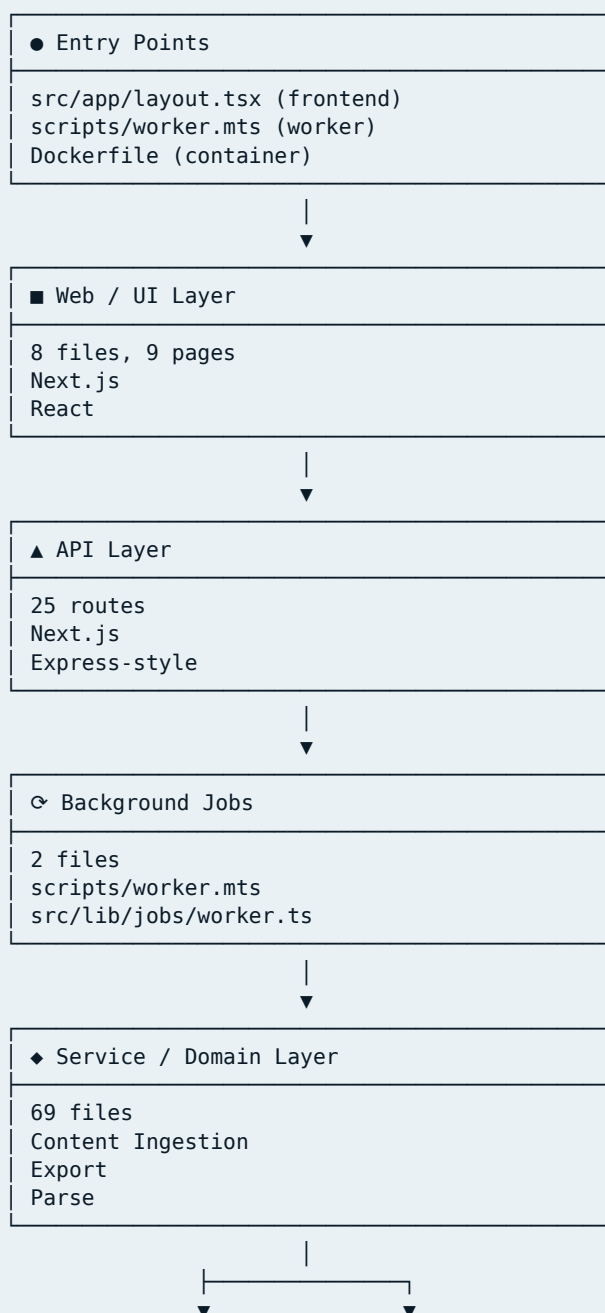
- └─ styling.spec.ts
- └─ workflow.spec.ts
- └─ fixtures/ ○ # automated tests, fixtures and mocks (17 files)
 - └─ sample-shop/ ○ # tests (17 files)
 - └─ migrations/ # schemas, models, migrations and data access (1 files)
 - └─ src/ ○ # tests (9 files)
 - └─ tests/ # automated tests, fixtures and mocks (1 files)
 - └─ worker/ # asynchronous and scheduled work (1 files)
 - └─ .env.example
 - └─ .gitignore
 - └─ Dockerfile
 - └─ package.json
 - └─ README.md
- └─ public/ # unknown (5 files)
 - └─ file.svg
 - └─ globe.svg
 - └─ next.svg
 - └─ vercel.svg
 - └─ window.svg
- └─ scripts/ # build, deployment and operations (5 files)
 - └─ analyze.mts
 - └─ benchmark.mts
 - └─ convert.mts
 - └─ verify-ai.mts
 - └─ worker.mts
- └─ src/ ●
 - └─ app/ ● # API layer (34 files)
 - └─ api/ ○ # http/rpc endpoints and request handling (21 files)
 - └─ p/ ● # API layer (9 files)
 - └─ favicon.ico
 - └─ globals.css
 - └─ layout.tsx
 - └─ page.tsx ●
 - └─ components/ # pages, components and client-side state (6 files)
 - └─ download.tsx
 - └─ graph.tsx
 - └─ legend.tsx
 - └─ mermaid.tsx
 - └─ project.tsx
 - └─ ui.tsx
 - └─ lib/ ● # reusable helpers and shared code (67 files)
 - └─ ai/ # language-model prompts, agents and inference (6 files)
 - └─ analysis/ # services (3 files)
 - └─ ask/ # services (1 files)
 - └─ db/ # schemas, models, migrations and data access (2 files)
 - └─ discover/ ●
 - └─ docs/ ● # services (5 files)
 - └─ export/ # services (7 files)
 - └─ graph/ ● # services (2 files)
 - └─ ingest/ # collecting and importing data from external sources (13 files)
 - └─ jobs/ # asynchronous and scheduled work (3 files)
 - └─ map/ # services (2 files)
 - └─ parse/ ● # services (6 files)
 - └─ retrieval/ # indexing and querying content (1 files)
 - └─ review/ # services (5 files)
 - └─ util/ # reusable helpers and shared code (4 files)
 - └─ api.ts
 - └─ ... 2 more
 - └─ instrumentation.ts
 - └─ proxy.ts
- └─ tests/ ○ # automated tests, fixtures and mocks (14 files)
 - └─ helpers/ # reusable helpers and shared code (2 files)
 - └─ index.ts
 - └─ zip.ts
 - └─ ai.test.ts ○
 - └─ analyzers.test.ts
 - └─ api.test.ts ○

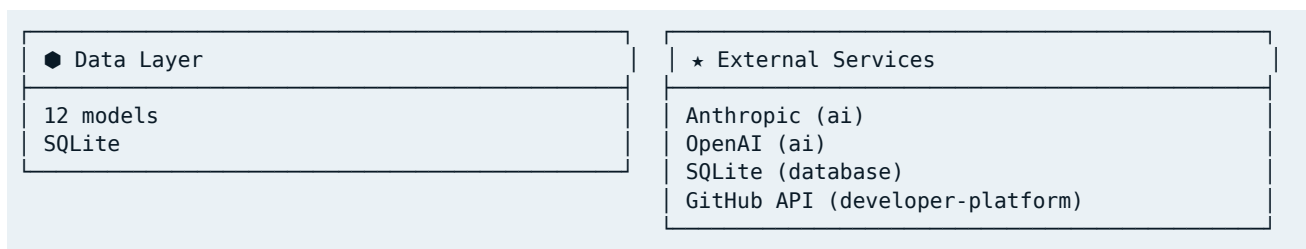
```

├── exports.test.ts
├── ingest.test.ts
├── map.test.ts
├── parse.test.ts
├── pipeline.test.ts ○
├── precision.test.ts
├── provider.test.ts
├── review.test.ts
├── secrets.test.ts ○
├── .dockerignore
├── .env.example
├── .gitignore
├── AGENTS.md
├── CLAUDE.md
├── docker-compose.yml
├── Dockerfile
├── ... 10 more

```

2. Functional Component Map





Functional-area dependencies:

- API Layer → Shared Utilities (142 references)
- API Layer → User Interface (128 references)
- Testing → Content Ingestion (57 references)
- API Layer → Map (40 references)
- Testing → Data Layer (40 references)
- API Layer → Content Ingestion (34 references)
- Content Ingestion → Util (30 references)
- API Layer → Data Layer (29 references)
- Background Jobs → Data Layer (28 references)
- Testing → Background Jobs (28 references)
- Infrastructure & Deployment → Background Jobs (27 references)
- File & Media Storage → Shared Utilities (26 references)
- Testing → Map (26 references)
- Infrastructure & Deployment → Data Layer (25 references)
- Export → Map (25 references)
- Search & Retrieval → Data Layer (25 references)
- Infrastructure & Deployment → Content Ingestion (24 references)
- API Layer → Util (24 references)
- Testing → AI Processing (22 references)
- Infrastructure & Deployment → Export (21 references)
- Review → Data Layer (21 references)
- Export → Data Layer (20 references)
- File & Media Storage → Data Layer (19 references)
- Discover → Data Layer (19 references)
- Map → Data Layer (18 references)
- Parse → Infrastructure & Deployment (18 references)
- Testing → Analysis (18 references)
- Content Ingestion → Data Layer (17 references)
- Background Jobs → Content Ingestion (17 references)
- Review → Analysis (17 references)

Diagram omitted here. Open the Code Map in the app, or the Markdown export, for the Mermaid source.

3. Runtime / Data Flow Map

```

GET /path
  ○ GET /path (src/lib/discover/detect.ts:250)
    ↓
  ○ lineAt (src/lib/discover/detect.ts:6)
    ↓
  ○ callArgs (src/lib/discover/detect.ts:218)
    ↓

```

```
○ isInlineFn (src/lib/discover/detect.ts:248)
↓
● reads symbols (src/lib/db/schema.ts:82)
↓
○ jobs.push (.github/workflows/ci.yml:3)
```

```
POST /api/github/validate
○ POST /api/github/validate (src/app/api/github/validate/route.ts:10)
↓
○ guard (src/lib/api.ts:20)
↓
○ json (src/lib/api.ts:8)
↓
○ AppError (src/lib/util/errors.ts:2)
↓
○ parseGitHubUrl (src/lib/ingest/github.ts:31)
↓
○ fetchGitHubMetadata (src/lib/ingest/github.ts:76)
↓
○ fail (src/lib/api.ts:12)
↓
○ isAppError (src/lib/util/errors.ts:14)
↓
○ scrubToken (src/lib/ingest/credentials.ts:36)
```

```
GET /api/jobs/:id
○ GET /api/jobs/:id (src/app/api/jobs/[id]/route.ts:8)
↓
○ guard (src/lib/api.ts:20)
↓
○ getJob (src/lib/jobs/index.ts:36)
↓
○ AppError (src/lib/util/errors.ts:2)
↓
○ json (src/lib/api.ts:8)
↓
○ fail (src/lib/api.ts:12)
↓
○ getDb (src/lib/db/client.ts:48)
↓
● reads jobs (src/lib/db/schema.ts:159)
↓
○ isAppError (src/lib/util/errors.ts:14)
↓
○ scrubToken (src/lib/ingest/credentials.ts:36)
↓
○ openDatabase (src/lib/db/client.ts:32)
```

```
POST /api/jobs/:id/cancel
○ POST /api/jobs/:id/cancel (src/app/api/jobs/[id]/cancel/route.ts:8)
↓
○ guard (src/lib/api.ts:20)
↓
○ requestCancel (src/lib/jobs/index.ts:44)
↓
○ AppError (src/lib/util/errors.ts:2)
↓
○ json (src/lib/api.ts:8)
↓
○ fail (src/lib/api.ts:12)
↓
○ getDb (src/lib/db/client.ts:48)
↓
○ getJob (src/lib/jobs/index.ts:36)
↓
○ isAppError (src/lib/util/errors.ts:14)
↓
```

```
○ scrubToken (src/lib/ingest/credentials.ts:36)
  ↓
○ openDatabase (src/lib/db/client.ts:32)
  ↓
● reads jobs (src/lib/db/schema.ts:159)
```

```
GET /api/projects
○ GET /api/projects (src/app/api/projects/route.ts:7)
  ↓
○ guard (src/lib/api.ts:20)
  ↓
○ getDb (src/lib/db/client.ts:48)
  ↓
○ json (src/lib/api.ts:8)
  ↓
● reads projects (src/lib/db/schema.ts:4)
  ↓
○ fail (src/lib/api.ts:12)
  ↓
○ openDatabase (src/lib/db/client.ts:32)
  ↓
○ isAppError (src/lib/util/errors.ts:14)
  ↓
○ scrubToken (src/lib/ingest/credentials.ts:36)
  ↓
○ migrationsFolder (src/lib/db/client.ts:20)
```

```
DELETE /api/projects/:id
○ DELETE /api/projects/:id (src/app/api/projects/[id]/route.ts:15)
  ↓
○ guard (src/lib/api.ts:20)
  ↓
○ getProject (src/lib/api.ts:28)
  ↓
○ listJobs (src/lib/jobs/index.ts:271)
  ↓
○ requestCancel (src/lib/jobs/index.ts:44)
  ↓
○ deleteProject (src/lib/ingest/store.ts:102)
  ↓
○ json (src/lib/api.ts:8)
  ↓
○ fail (src/lib/api.ts:12)
  ↓
○ getDb (src/lib/db/client.ts:48)
  ↓
○ AppError (src/lib/util/errors.ts:2)
  ↓
● reads projects (src/lib/db/schema.ts:4)
  ↓
● reads jobs (src/lib/db/schema.ts:159)
  ↓
○ getJob (src/lib/jobs/index.ts:36)
  ↓
○ isAppError (src/lib/util/errors.ts:14)
```

```
GET /api/projects/:id
○ GET /api/projects/:id (src/app/api/projects/[id]/route.ts:8)
  ↓
○ guard (src/lib/api.ts:20)
  ↓
○ json (src/lib/api.ts:8)
  ↓
○ projectSummary (src/lib/api.ts:41)
  ↓
○ getProject (src/lib/api.ts:28)
  ↓
```

```

○ fail (src/lib/api.ts:12)
↓
○ getDb (src/lib/db/client.ts:48)
↓
● reads jobs (src/lib/db/schema.ts:159)
↓
○ AppError (src/lib/util/errors.ts:2)
↓
● reads projects (src/lib/db/schema.ts:4)
↓
○ isAppError (src/lib/util/errors.ts:14)
↓
○ scrubToken (src/lib/ingest/credentials.ts:36)
↓
○ openDatabase (src/lib/db/client.ts:32)

```

```

POST /api/projects/:id/analyze
○ POST /api/projects/:id/analyze (src/app/api/projects/[id]/analyze/route.ts:17)
↓
○ guard (src/lib/api.ts:20)
↓
○ getProject (src/lib/api.ts:28)
↓
○ getDb (src/lib/db/client.ts:48)
↓
○ decryptSecret (src/lib/ingest/credentials.ts:24)
↓
○ parseGitHubUrl (src/lib/ingest/github.ts:31)
↓
○ fetchGitHubMetadata (src/lib/ingest/github.ts:76)
↓
○ createPendingProject (src/lib/ingest/store.ts:10)
↓
○ enqueueAnalysis (src/lib/jobs/index.ts:24)
↓
○ json (src/lib/api.ts:8)
↓
○ projectSummary (src/lib/api.ts:41)
↓
● reads credentials (src/lib/db/schema.ts:29)
↓
● reads projects (src/lib/db/schema.ts:4)
↓
○ fail (src/lib/api.ts:12)

```

4. Dependency Map

Top 35 of 125 source files by importance (drill down by area in the application).

Diagram omitted here. Open the Code Map in the app, or the Markdown export, for the Mermaid source.

5. API Map

Method	Route	Handler	Authentication	Kind	Location
GET	/	uploadWithProgress	Not visible	page · Next.js	src/app/page.tsx:15
POST	/api/github/validate	POST (route handler)	Not visible	api · Next.js	src/app/api/github/validate/route.ts:10
GET	/api/jobs/:id	GET (route handler)	Not visible	api · Next.js	src/app/api/jobs/[id]/route.ts:8
POST	/api/jobs/:id/cancel	POST (route handler)	Not visible	api · Next.js	src/app/api/jobs/[id]/cancel/route.ts:8
GET	/api/projects	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/route.ts:7

Method	Route	Handler	Authentication	Kind	Location
DELETE	/api/projects/:id	DELETE (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/route.ts:15
GET	/api/projects/:id	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/route.ts:8
POST	/api/projects/:id/analyze	POST (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/analyze/route.ts:17
GET	/api/projects/:id/architecture	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/architecture/route.ts:8
GET	/api/projects/:id/ask	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/ask/route.ts:11
POST	/api/projects/:id/ask	POST (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/ask/route.ts:19
GET	/api/projects/:id/export	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/export/route.ts:13
GET	/api/projects/:id/files	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/files/route.ts:9
GET	/api/projects/:id/files/content	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/files/content/route.ts:13
GET	/api/projects/:id/findings	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/findings/route.ts:9
GET	/api/projects/:id/graph	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/graph/route.ts:9
GET	/api/projects/:id/impact	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/impact/route.ts:8
GET	/api/projects/:id/overview	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/overview/route.ts:11
GET	/api/projects/:id/report	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/report/route.ts:8
GET	/api/projects/:id/search	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/search/route.ts:8
GET	/api/projects/:id/symbols/:symbolId	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/symbols/[symbolId]/route.ts:13
POST	/api/projects/github	POST (route handler)	Not visible	api · Next.js	src/app/api/projects/github/route.ts:16
POST	/api/projects/upload	POST (route handler)	Not visible	api · Next.js	src/app/api/projects/upload/route.ts:14
GET	/api/status	GET (route handler)	Not visible	api · Next.js	src/app/api/status/route.ts:30
GET	/p/:id	OverviewPage	Not visible	page · Next.js	src/app/p/[id]/page.tsx:26
GET	/p/:id/architecture	ArchitecturePage	Not visible	page · Next.js	src/app/p/[id]/architecture/page.tsx:16
GET	/p/:id/ask	Cite	Not visible	page · Next.js	src/app/p/[id]/ask/page.tsx:20
GET	/p/:id/explain	Origin	Not visible	page · Next.js	src/app/p/[id]/explain/page.tsx:11
GET	/p/:id/files	Splitter	Not visible	page · Next.js	src/app/p/[id]/files/page.tsx:36

Method	Route	Handler	Authentication	Kind	Location
GET	/p/:id/map	ImpactPanel	Not visible	page · Next.js	src/app/p/[id]/map/page.tsx:18
GET	/p/:id/reports	ReportsPage	Not visible	page · Next.js	src/app/p/[id]/reports/page.tsx:16
GET	/p/:id/review	Diff	Not visible	page · Next.js	src/app/p/[id]/review/page.tsx:14
GET	/path	handler	Not visible	api · Express-style	src/lib/discover/detect.ts:315
GET	/x	GET /x (inline)	Not visible	api · Express-style	src/lib/parse/extract.ts:306

6. Data Model Map

Model	ORM	Fields	References	Location
projects	Drizzle + SQL table projects (drizzle/0000_robust_hawkeye.sql:117)	id, name, sourceType, sourceUrl, owner, branch, commit, sourceHash		src/lib/db/schema.ts:4
credentials	Drizzle + SQL table credentials (drizzle/0000_robust_hawkeye.sql:7)	projectId, encrypted, createdAt, project_id, created_at		src/lib/db/schema.ts:29
blobs	Drizzle + SQL table blobs (drizzle/0000_robust_hawkeye.sql:1)	hash, content, size		src/lib/db/schema.ts:36
files	Drizzle + SQL table files (drizzle/0000_robust_hawkeye.sql:20)	id, projectId, path, name, directory, extension, language, size		src/lib/db/schema.ts:42
symbols	Drizzle + SQL table symbols (drizzle/0000_robust_hawkeye.sql:165)	id, projectId, fileId, filePath, name, qualifiedName, kind, startLine		src/lib/db/schema.ts:82
relationships	Drizzle + SQL table relationships (drizzle/0000_robust_hawkeye.sql:148)	id, projectId, kind, sourceType, sourceId, targetType, targetId, filePath		src/lib/db/schema.ts:110
findings	Drizzle + SQL table findings (drizzle/0000_robust_hawkeye.sql:54)	id, projectId, code, title, category, severity, confidence, origin		src/lib/db/schema.ts:128
jobs	Drizzle + SQL table jobs (drizzle/0000_robust_hawkeye.sql:93)	id, projectId, status, stages, currentStage, error, log, createdAt		src/lib/db/schema.ts:159
parseCache	Drizzle + SQL table parse_cache (drizzle/0000_robust_hawkeye.sql:111)	key, payload, createdAt, created_at		src/lib/db/schema.ts:182
explainCache	Drizzle + SQL table explain_cache (drizzle/0000_robust_hawkeye.sql:13)	key, kind, payload, createdAt, created_at		src/lib/db/schema.ts:189
indexEntries	Drizzle + SQL table index_entries (drizzle/0000_robust_hawkeye.sql:80)	projectId, kind, refId, filePath, title, terms, text, embedding		src/lib/db/schema.ts:197
questions	Drizzle + SQL table questions (drizzle/0000_robust_hawkeye.sql:139)	id, projectId, question, answer, createdAt, project_id, created_at		src/lib/db/schema.ts:212

Diagram omitted here. Open the Code Map in the app, or the Markdown export, for the Mermaid source.

7. Test Map

Test file	Kind	Framework	Exercises
e2e/performance.spec.ts	e2e	Playwright	.github/workflows/ci.yml, src/lib/jobs/index.ts
e2e/styling.spec.ts	e2e	Playwright	.github/workflows/ci.yml, src/lib/parse/extract.ts
e2e/workflow.spec.ts	e2e	Playwright	src/lib/parse/extract.ts
fixtures/sample-shop/.env.example	fixture		.env.example

Test file	Kind	Framework	Exercises
fixtures/sample-shop/.gitignore	fixture		.dockerignore
fixtures/sample-shop/Dockerfile	fixture		Dockerfile
fixtures/sample-shop/README.md	fixture		README.md
fixtures/sample-shop/migrations/001_init.sql	fixture		not resolved
fixtures/sample-shop/package.json	fixture		package.json
fixtures/sample-shop/src/config.ts	fixture		src/lib/config.ts
fixtures/sample-shop/src/db/client.ts	fixture		not resolved
fixtures/sample-shop/src/db/models.ts	fixture		not resolved
fixtures/sample-shop/src/routes/auth.ts	fixture		src/lib/api.ts, src/lib/ai/openai.ts
fixtures/sample-shop/src/routes/orders.ts	fixture		src/lib/ai/openai.ts, src/lib/api.ts
fixtures/sample-shop/src/server.ts	fixture		src/lib/api.ts, src/lib/jobs/index.ts
fixtures/sample-shop/src/services/orderService.ts	fixture		not resolved
fixtures/sample-shop/src/services/paymentService.ts	fixture		not resolved
fixtures/sample-shop/src/utils/pricing.ts	fixture		not resolved
fixtures/sample-shop/tests/pricing.test.ts	fixture	Vitest	not resolved
fixtures/sample-shop/worker/tasks.py	fixture		not resolved
tests/ai.test.ts	unit	Vitest	src/lib/db/client.ts, src/lib/docs/ai.ts, src/lib/docs/types.ts, src/lib/graph/build.ts
tests/analyzers.test.ts	unit	Vitest	src/lib/analysis/analyzers.ts, src/lib/graph/build.ts, src/lib/retrieval/index.ts, src/lib/ai/index.ts
tests/api.test.ts	integration	Vitest	src/lib/db/client.ts, src/lib/ingest/normalize.ts, src/lib/ingest/store.ts, src/lib/jobs/index.ts
tests/exports.test.ts	unit	Vitest	src/lib/ask/index.ts, src/lib/export/document.ts, src/lib/export/index.ts, src/lib/ingest/zip.ts
tests/helpers/index.ts	unit	Jest-style	src/lib/db/client.ts, src/lib/ai/index.ts, src/lib/ingest/normalize.ts, src/lib/ingest/store.ts
tests/helpers/zip.ts	unit		.github/workflows/ci.yml, src/lib/ingest/zip.ts
tests/ingest.test.ts	unit	Vitest	src/lib/config.ts, src/lib/db/client.ts, src/lib/ingest/github.ts, src/lib/ingest/fs.ts
tests/map.test.ts	unit	Vitest	src/lib/map/index.ts, src/lib/ask/index.ts, src/lib/retrieval/index.ts, src/lib/export/html.ts
tests/parse.test.ts	unit	Vitest	src/lib/parse/index.ts
tests/pipeline.test.ts	unit	Vitest	src/lib/db/client.ts, src/lib/discover/types.ts, src/lib/jobs/index.ts, src/lib/jobs/worker.ts
tests/precision.test.ts	unit	Vitest	src/lib/analysis/rules.ts, src/lib/discover/types.ts, src/lib/db/client.ts
tests/provider.test.ts	unit	Vitest	src/lib/ai/anthropic.ts, src/lib/ai/errors.ts, src/lib/ai/openai.ts, src/lib/ai/index.ts
tests/review.test.ts	unit	Vitest	src/lib/analysis/rules.ts, src/lib/review/ai.ts, src/lib/review/patch.ts, src/lib/review/verify.ts

Test file	Kind	Framework	Exercises
tests/secrets.test.ts	unit	Vitest	src/lib/ingest/secrets.ts, src/lib/ingest/credentials.ts

Important components with no tests:

- src/lib/client.ts — high-importance ui file with no test referencing it
- src/lib/util/text.ts — high-importance util file with no test referencing it
- src/components/ui.tsx — high-importance ui file with no test referencing it

8. External Integration Map

```

★ Anthropic (ai)
  ↳ package.json, src/lib/ai/anthropic.ts:1, src/lib/config.ts:35
★ OpenAI (ai)
  ↳ src/lib/config.ts:39, src/lib/config.ts:38, .env.example:12
★ SQLite (database)
  ↳ package.json, src/lib/db/client.ts:1
★ GitHub API (developer-platform)
  ↳ src/lib/config.ts:57, src/lib/config.ts:56

```

9. High-Risk Component Map

Risk	File	Role	Importance	Findings
● medium	src/lib/parse/extract.ts	api	0.33	3
● medium	src/lib/graph/build.ts	service	0.23	2
● medium	src/lib/docs/ai.ts	service	0.07	2
● medium	src/lib/discover/detect.ts	api	0.05	3
● medium	src/app/p/[id]/files/page.tsx	api	0.03	1
● medium	src/app/page.tsx	api	0.03	2
○ low	src/lib/client.ts	ui	0.19	1
○ low	src/app/api/github/validate/route.ts	api	0.06	1
○ low	src/lib/discover/index.ts	service	0.05	1

10. Change Impact Relationships

text (src/lib/parse/extract.ts:20) — Changing text can affect 201 upstream components, including 14 routes, and depends on 0 downstream components. 12 test file(s) exercise this code.

```

text
  ↓ used by
api

text
  ↓ used by
extractTsJs

text
  ↓ used by
extractPython

text
  ↓ used by
extractGo

text
  ↓ used by
extractJavaLike

```

getDb (src/lib/db/client.ts:48) — Changing getDb can affect 182 upstream components, including 21 routes, and

depends on 2 downstream components. 10 test file(s) exercise this code.

```
getDb
  ↓ used by
POST

getDb
  ↓ used by
GET

getDb
  ↓ used by
GET

getDb
  ↓ used by
GET

getDb
  ↓ used by
GET
```

openDatabase (src/lib/db/client.ts:32) — Changing openDatabase can affect 178 upstream components, including 21 routes, and depends on 1 downstream component. 9 test file(s) exercise this code.

```
openDatabase
  ↓ used by
getDb
  ↓ used by
POST

openDatabase
  ↓ used by
getDb
  ↓ used by
GET

openDatabase
  ↓ used by
getDb
  ↓ used by
GET

openDatabase
  ↓ used by
getDb
  ↓ used by
GET

openDatabase
  ↓ used by
getDb
  ↓ used by
GET

openDatabase
  ↓ used by
getDb
  ↓ used by
GET
```

migrationsFolder (src/lib/db/client.ts:20) — Changing migrationsFolder can affect 156 upstream components, including 16 routes, and depends on 0 downstream components. 9 test file(s) exercise this code.

```
migrationsFolder
  ↓ used by
openDatabase
  ↓ used by
getDb
  ↓ used by
POST

migrationsFolder
```

```

    ↓ used by
openDatabase
    ↓ used by
getDb
    ↓ used by
GET

migrationsFolder
    ↓ used by
openDatabase
    ↓ used by
getDb
    ↓ used by
GET

migrationsFolder
    ↓ used by
openDatabase
    ↓ used by
getDb
    ↓ used by
GET

migrationsFolder
    ↓ used by
openDatabase
    ↓ used by
getDb
    ↓ used by
GET

```

json (src/lib/api.ts:8) — Changing json can affect 93 upstream components, including 25 routes, and depends on 0 downstream components. 5 test file(s) exercise this code.

```

json
    ↓ used by
POST /api/login

json
    ↓ used by
POST /api/calc

json
    ↓ used by
POST /api/orders

json
    ↓ used by
GET /api/orders/:id

json
    ↓ used by
GET /api/orders

```

AppError (src/lib/util/errors.ts:2) — Changing AppError can affect 101 upstream components, including 21 routes, and depends on 1 downstream component. 5 test file(s) exercise this code.

```

AppError
    ↓ used by
POST

AppError
    ↓ used by
POST

AppError
    ↓ used by

```

GET

AppError

↓ used by

POST

AppError

↓ used by

GET

11. Legend

LEGEND

- Application Entry Point
- ◆ Service
- UI Component
- ▲ API Endpoint
- Data Model
- Utility
- ★ External Service
- T Test
- C Configuration
- ⌚ Background Job
- Infrastructure

- Calls
- Imports
- Data flow
- ↳ Writes to
- ↵ Reads from
- ... Optional dependency

Risk:

- ! Critical
- ▲ High
- Medium
- Low

19. Legend

LEGEND

- Application Entry Point
- ◆ Service
- UI Component
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- Calls
- Imports
- Data flow
- ↳ Writes to
- ↵ Reads from
- ... Optional dependency

Risk:

- ! Critical
- ▲ High

- Medium
- Low